C6	98	1.607	0.782
C9	78	1.560	0.395
I3	165	1.897	0.487

Date: 20260801
Title: Collect B. sub supernatant and pellet of J
Purpose: To prepare supernatant and pellet for analyses
Procedure:
J: Pveg-EEBD-RBS0-csn:ACCD-B0015

1. Transfer culture of J1, J2, and J3 to 1.5 mL tube.
2. Centrifuge at 16000 g for 1 minute.
3. Collect supernatants in 1.4 mL tubes.
4. Repeat steps 1–3 until all cultures are collected.

Done by: Sarah, Naomi, Sophie H.

Date: 20260801
Title: Make acrylamide gel
Purpose: create a fine-pored matrix that separates proteins or small nucleic acids by size during electrophoresis.
Procedure:

1. Assemble holder, glass plates, comb.
2. Prepare resolver solution:
    a. Resolver A — 3 mL
    b. Resolver B — 3 mL
    c. 10 % APS — 30 µL
    d. TEMED — 3 µL
3. Fill to 1.5 - 1 cm below the bottom of comb teeth with a resolver solution (~ 6 mL).
4. Prepare stacker solution:
    a. Resolver A — 1 mL
    b. Resolver B — 1 mL
    c. 10 % APS — 10 µL
    d. TEMED — 2 µL
5. Fill the top with a stacker solution (~2 mL).
6. Insert comb & allow gel to polymerize.

Done by: Sophie C, Chloe

Date: 20260801
Title: Make 0.8 % agarose gel (x3)
Purpose: To prepare gel for enzyme digestion
Procedure:

1. Weigh 0.48 g of agarose powder and 60 mL 1x TAE buffer.
2. Add the components into a conical flask.

3. Assemble mold, tray, and comb.
4. Microwave for 30 sec after plastic wrap & holes poking. Agitate. Repeat until there are no sediments.
5. Wait until no more smoke, add 3 µL of safety dye.
6. Pour the melted gel into the three assembled molds.
7. Put the comb and wait for it to solidify.

Done by: Sophie H.

Date: 20260801
Title: Enzyme Digestion
Purpose: To confirm plasmid size for each samples before moving on to golden gate assembly
Procedure:
Samples:

- A1, A2, A3: pSTK-1A - PtagC + RBS - Ho1:pcyA - L3S2P21
- C6, C9: pSTK-1B - PlepA - RBS0 - ccaSm3 - B0015
- I3: pSTK-1D - PcpcG2-veg - EEBD-RBS0 – csn:ACCD - B0015

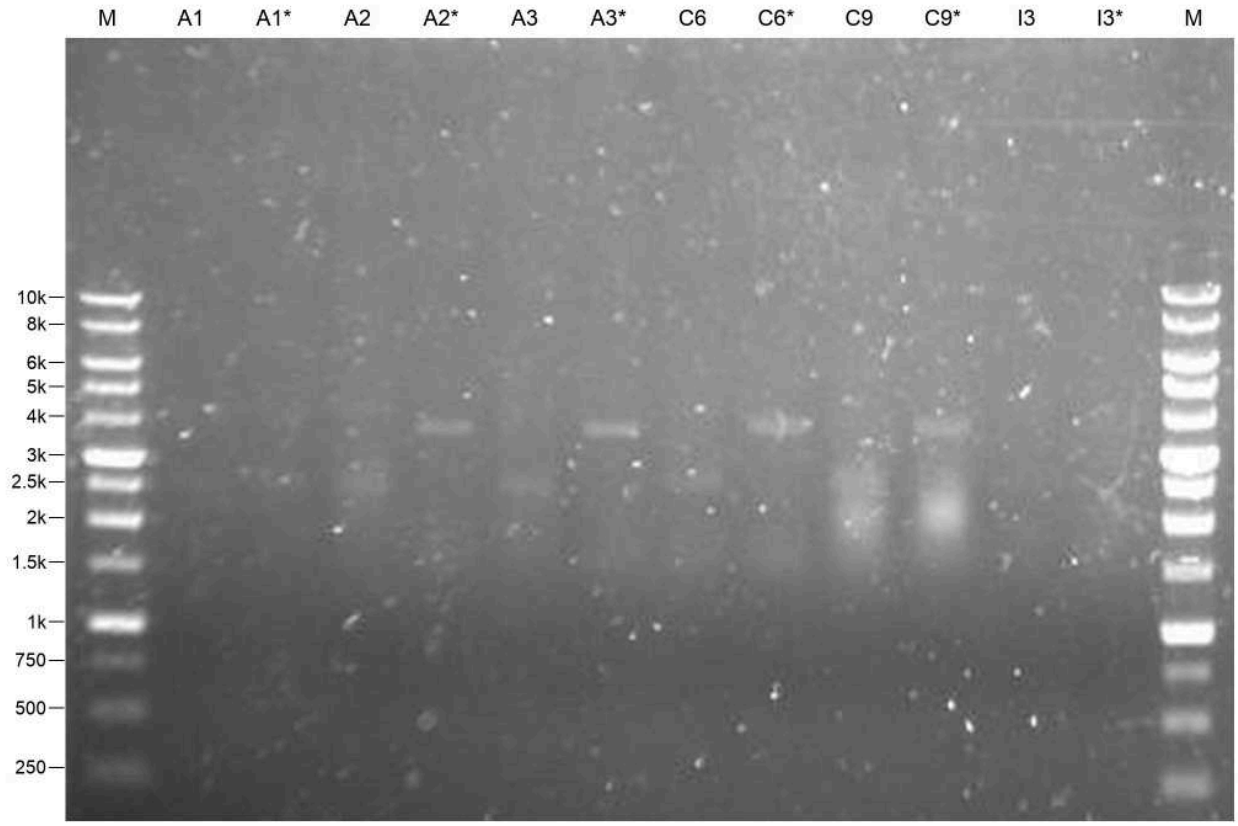
1. Prepare following reaction:
    a. A1 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 4.67 µL
        iii. ddH_2O: 12.84 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    b. A2 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.62 µL
        iii. ddH_2O: 11.88 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    c. A3 (base pair 3754)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.32 µL
        iii. ddH_2O: 12.18 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    d. C6 (base pair 3508)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 5.1 µL
        iii. ddH_2O: 12.4 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    e. C9 (base pair 3828)

        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 6.41 µL
        iii. ddH_2O: 11.09 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
    f. I3 (base pair 3757)
        i. 10× Cutsmart buffer: 2 µL
        ii. DNA: 3.03 µL
        iii. ddH_2O: 14.47 µL
        iv. EcoRI (1 site): 0.5 µL
        v. Total volume: 20 µL
2. For digested samples, add 0.5 µL of restriction enzyme EcoRI.
3. For undigested samples, add 2 µL more of ddH_2O.
4. Incubate the mixture at 37 °C for 1 hr.
5. After finishing incubation, run gel electrophoresis.

Done by: Sarah, Naomi, Sophie H.

Date: 20260801
Title: Electrophoresis for Enzyme Digestion
Purpose: To separate fragments and ensure plasmid size for each samples from 20260801 enzyme digestion
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Sample	Plasmid Construct	Enzyme	Expected Size (bp)
A1~3	PtagC+RBS-Ho1:pcyA-L3S2P21	EcoRI	3754
C6,9	PlepA-RSS0-ccaSm3-B0015	EcoRI	3828
I3	PcpcG2-Veg-EEBD-RBSO-csn:ACCD-B0015	EcoRI	3857
Note: *: digested; Enzyme: EcoRI (for all)			

Done by: Sarah, Naomi, Sophie H.

Date: 20260801
Title: Colony PCR
Purpose: screen for successful cloning after transformation

Procedure:

Samples:

1. Prepare the following reaction (3 master mix) and dispense solutions into PCR tubes

G: PcpcG2-ver-EEBD-RBS0-Csn:LEA-bkdBcoldD

bp: 1375

	1 reaction	13 reactions
Forward primer (CsnF)	1μL	13 μL
CDS primer (LEA R)	1μL	13 μL
Q-amp master mix	10 μL	130 μL
ddH_2O	8 μL	104 μL

M: PcpcG2-veg-EEBD-RBS0-GanA-Gala-L3S2P21

bp: 2837

	1 reaction	13 reactions
Forward primer (Biobrick Prefix)	1 μL	13 μL
CDS Reverse primer (STK seq rv)	1 μL	13 μL
Q-amp master mix	10 μL	130 μL
ddH_2O	8 μL	104 μL

L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21

bp: 2411

	1 reaction	17 reactions
Forward primer (pSTK-Exp-seq-rv)	1 μL	17 μL
CDS reverse primer (Exp-seq4-rv)	1 μL	17 μL
Q-amp master mix	10 μL	170 μL
ddH_2O	8 μL	136 μL

2. Circle 12 colonies from plate G and M, 16 colonies from L
3. Pick up colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid

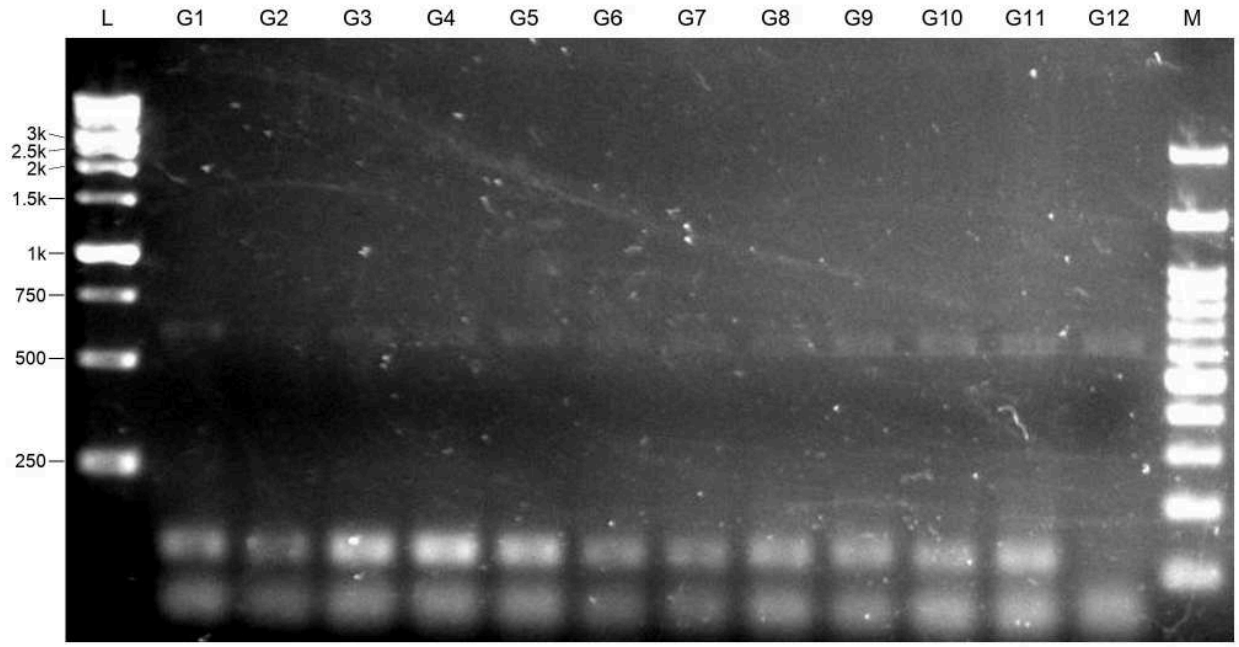
6. Place into PCR machine for thermal cycle

95 °C	2 min	1 cycle
95 °C	30 sec	30 cycles
65 °C	40 sec	
72 °C	3 min	
72 °C	1 min	1 cycle

Done by Ryan Y, Olivia L, Abby T

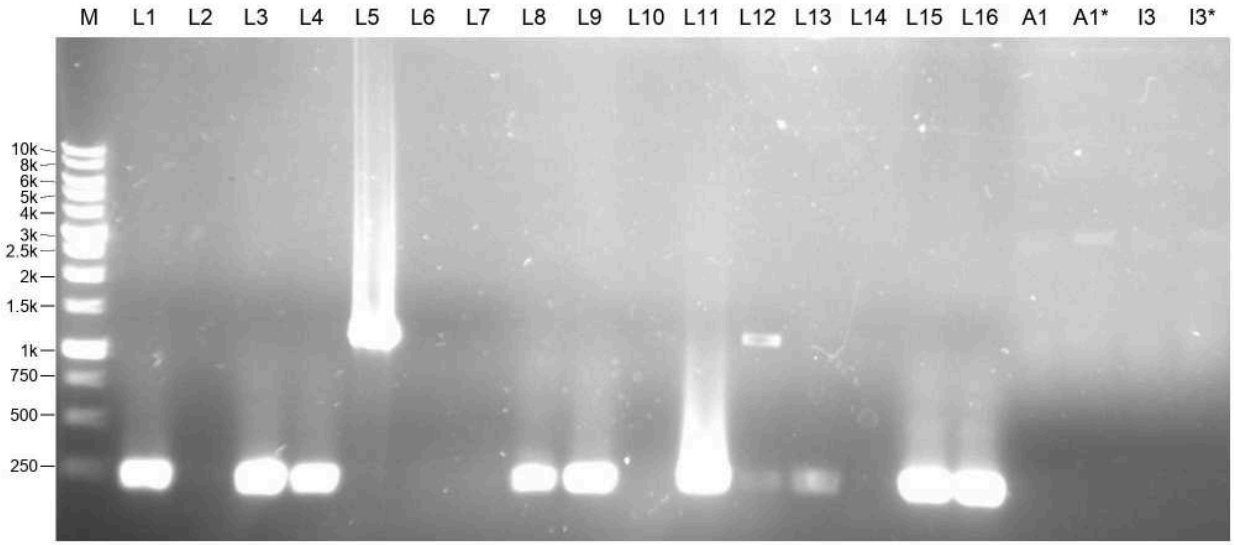
Date: 20260801
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



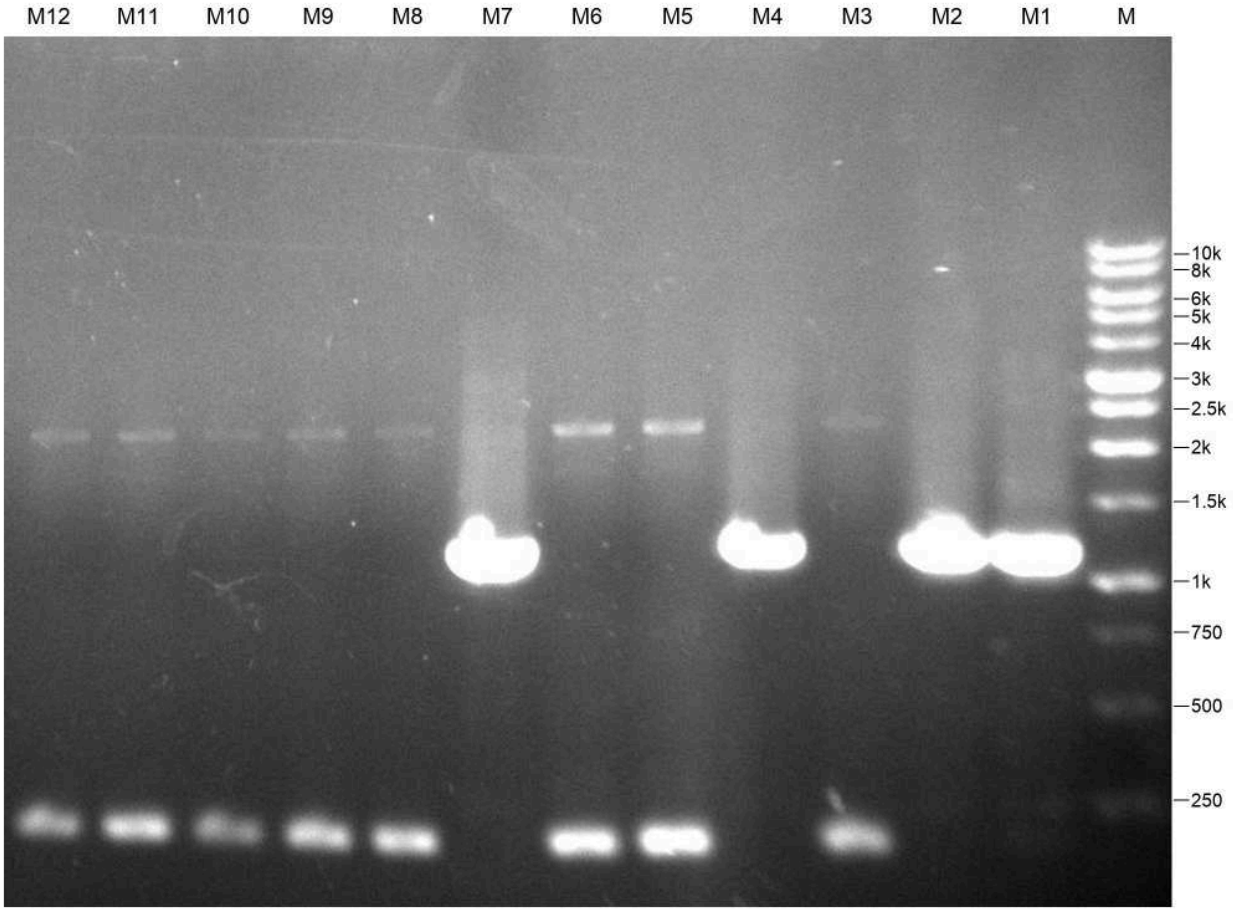
L: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G: PcpcG2-veg-EEBD-RBS0-Csn-LEA-bkdB Cold (12)	Csn F	LEA R	585

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Sample	Plasmid Construct	Primers / Enzyme	Expected Size (bp)
L1-L16	L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21	pSTK-Exp-seq-fw / Exp-seq4-rv	
A1, A1*	PtagC+RBS-Ho1:pcyA-L3S2P21	EcoRI	3754
I3, I3*	PcpcG2-Veg-EEBD-RBSO-csn:ACCD-B0015	EcoRI	3857
Note: *: digested			

5. Load the gel in the layout shown below (5 µL each):
6. Resolve at 100 v for 30 min



Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
M: PcpcG2-veg-EEBD-RBS0-GanA-Gala-L3S2P21 (12)	Biobrick prefix	STK rv	2837

Done by Olivia Lin, Abby Tsai,Ryan Y

Date: 20260801
Title: Liquid culture
Purpose: bacteria culture from successful colony PCR product
Procedure:
Samples from 0801 colony PCR

- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G10)
- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G11)
- PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold(G12)
- Pveg-EEBD-RBS0-csn:AccD-B0015(J1)
- Exp1-Pveg-SamyQ-TAT-HA-AmilCP(#8)

1. Label 5 liquid culture tube
2. Turn on bunsen burner
3. Add 3 mL of TB medium and 3 µL of ampicillin to each of the G10,G11,G12 tube
4. Add 1 mL of TB medium and 5 µL of erythromycin to J1 and #8 tubes
5. Dip a pipette tip into the colonies
6. Dispense tips in their respective tubes
7. Place the tubes in the incubator, slant around 45 %

Done by Olivia Lin, Ryan Y
Supervised by Gabriel

Date: 20260801
Title: Level 2 Golden Gate Assembly
Purpose: assemble level 1 constructs from different modules
Procedure:

1. Prepare the reaction as follows

(A) pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21
(B) pSTK-1B-PliaG-RBS0-ccaSm3-B0015
(C) pSTK-1B-PlepA-RBS0-ccaSm3-B0015
(D) pSTK-1C-Pveg-RBS0-ccaR-B0015
(E) pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015
(I) pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015
A, I, C-> liquid culture

A+B+D+I

Part	Original (µL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871(1/5 dilution)
(A) pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - concentration:107 - Bp: 3754 1953	0.4560934579	1.36828037(1/3dilution)
(B) pSTK-1B-PliaG-RBS0-ccaSm3-B0015 - concentration: 83ng/ul - Bp: 3792	0.5939277108	1.78178313 (1/3 dilution)

(D) pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134 ng/μL - Bp: 3243 bp 1442	0.314619403	1.25847761 (1/4 dilution)
(I) pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015 - Concentration: 165 ng/μL - Bp: 3757	0.2960060606	1.4800303 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.09375988 μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+C+D+E

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - bp: 5104	0.2035337423	1.01766871 (1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 107 ng/μL - bp: 3754	0.4560934579	1.36828037 (1/3 dilution)

(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/μL - Bp: 3828 bp	0.5077959184	1.52338776 (1/3 dilution)
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134 ng/μL - Bp: 3243 bp	0.314619403	1.25847761 (1/4 dilution)
(E)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015 - Concentration:168 ng/μL - Bp: 3485 bp	0.269672619	1.3483631 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.48382245 μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+B+D+E

Part	Original (μL)	Dilution
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871(1/5 dilution)
(A) pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21	0.4560934579	1.36828037(1/3dilution)

- Concentration: 107 ng/μL - bp: 3754		
(B) pSTK-1B-PliaG-RBS0-ccaSm3-B0015 - concentration: 83 ng/μL - bp: 3792	0.5939277108	1.78178313 (1/3 dilution)
(D) pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134 ng/μL - bp: 3243 bp	0.314619403	1.25847761 (1/4 dilution)
(E) pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015 - Concentration:168 ng/μL - bp: 3485 bp	0.269672619	1.3483631 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.22542708 μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

A+C+D+I

Part	Original (μL)	Dilution

Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	0.2035337423	1.01766871 (1/5 dilution)
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 107 ng/μL - Bp: 3754	0.4560934579	1.36828037(1/3dilution)
(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/μL - Bp: 3828 bp	0.5077959184	1.52338776(1/3dilution)
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134 ng/μL - Bp: 3243 bp	0.314619403	1.25847761 (1/4 dilution)
(I)pSTK-1D-PcpcG2-veg-EEBD-MF001-csn:ACCD-B0015 - Concentration: 165 ng/μL - Bp: 3757	0.2960060606	1.4800303 (1/5 dilution)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	9.35215525 μL
T4 DNA ligase	1 μL	1 μL
BsmB1	1 μL	1 μL
Total	20 μL	20 μL

2. Run thermal cycle

55 °C	5 min	45 cycles
16 °C	5 min	
55 °C	10 min	1 cycle
80 °C	20 min	

Done by Abby T, Audrey H

Date: 20260801
Title: Level 1 Golden Gate Assembly
Purpose: Assemble parts
Procedure:

1. Prepare the reaction as follows

L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21

Part	Original (μL)	Dilution (μL)
Vector (shuttle vector): pSTK-EXP-1-sfGFP concentration: 162 ng/ μl 6929 bp	0.2780154321	1.4 (dilute 1/5)
Insert promoter(3P) concentration: 106 ng/ μl 2909bp	0.3567641509	1.43 (dilute 1/4)
Insert RBS (EEBD-RBS0) concentration: 151 2392 bp	0.2059337748	1.03 (dilute 1/5)
Insert CDS(GanA-Gala) concentration: 177 4148 bp	0.3046553672	1.22 (dilute 1/4)
Insert Terminator (L3S2P21) concentration: 170 2145 bp	0.1640294118	1.64 (dilute 1/10)
T4 DNA ligase buffer	2	2
Sterile water	9.28	

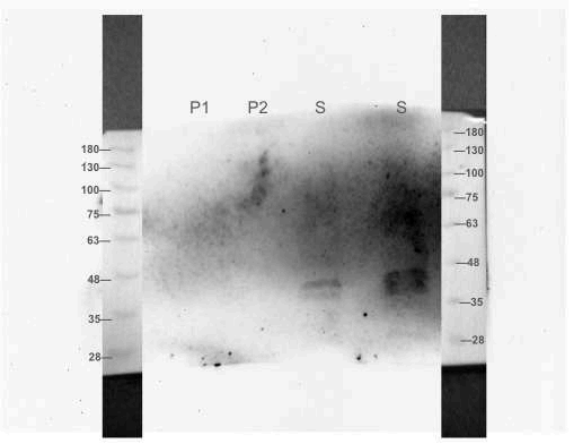
T4 DNA ligase	1	1
BsaI-HFv2	1	1
Total	20	20

Done by Sophie C, Abigail

Date: 20260801
Title: Western blot
Purpose: detect specific protein within a complex biological sample
Procedure:

1. Collecting *Bacillus subtilis* by centrifuge at 16000 g 30 sec
2. Transfer supernatant to the new 15 mL tube. Add 1 mL cellytic B lysis buffer and resuspend
3. Add 10 ng/mL lysozyme 20 mL and incubate at RT for 15 min
4. Transfer 125 mL lysate to the new microtube, add 25 mL dumpling buffer and incubate at 95 °C for 5 min as P1
5. Transfer 125 mL medium to the new microtube, add 25 mL sampling buffer and incubate at 95°C for 5 min as S.
6. Centrifuge the lysate at 16000 g for 30 sec. Transfer 125 mL supernatant to the new microtube. Add 25 mL sampling buffer and incubate at 95 °C for 5 min as P2
7. Resolve samples in SDS-PAGE(10 %) by the following layout:
   ladder P1 P2 S S ladder
   5 mL 50 mL 50 mL 50 mL 50 mL 2 mL
8. Transfer to PVDF membrane
9. Blocking in 5 % milk in TBST for 2 hr
10. Replace blocking solution by anti His-tag mouse m/tb (1:1000) in 5 % Milk. Incubate on the orbital shaker at 4 °C for overnight

Results:



P1: Bacillus total pellet
P2: Bacillus soluble pellet
S: Supernatant

Exposure: 6 min

Done by Chloe, Sophie C, Chars

Date: 20260818
Title: Electroporation competent cell preparation
Purpose: To prepare electroporation competent cell for electroporation
Procedure:

1. Overnight culture *Bacillus subtilis* WB800
2. Dilute the overnight culture of WB800 *Bacillus subtilis* in 16 fold in growth medium and grow at 37 ℃ to OD_{600} of 0.907
3. Cool the cells on ice with shake
4. Collect the cells in 2 mL microtube at 4 ℃ and 5000 g for 5 min
5. Wash the cells with 1 mL cold electroporation medium 2 times
6. Suspend the cells in 1/50 of the culture volume of the electroporation medium
7. The cells stored at -80 ℃

Done by Gabriel

Date: 20260818
Title: Electroporation
Purpose: Introduce DNA sequence into competent cells
Procedure:

1. Following the protocol in 0811
2. Mix 2.4 µL pveg-AccD(J) plasmid (410ng/µL) in the 100 µL EPC from 0817
3. Electroporate the cell at 2.3 kV 300st 35F
4. Recovery with recovery medium (1mL) for 3 hr
5. Centrifuge at 5000 g for 5 min
6. Spread at LBE plate

Done by Gabriel

Date: 20260818
Title: Colony PCR (QA, QB, QC, QD)
Purpose: Amplify the number of targeted gene
Procedure:
QA: pSTK-OC-SamyQ-LEA-2.1
QB: pSTK-OC-SamyQ-AccD-v3t1
QC: pSTK-OC-SamyQ-LEA-2.2
QD: pSTK-OC-SamyQ-AccD-2.1

1. Prepare the following master mix x9

Fw primer Samy Q F	9 µL
Rv primer sSTK-seq-rv	9 µL
Q-amp master mix	90 µL
ddH_2O	72 µL

2. Distribute 20μL to each PCR tube
3. Pick up colonies with tip (QA1, QA3, QC3,QCs,QB,QB3,QD2,QD11)
4. Dispense tip in PCR tube. Take out tip
5. Close the lid
6. Thermal cycle in PCR machine

95 ℃	5 min	x1
95 ℃	20 sec	35 cycles
60 ℃	20 sec	
72 ℃	1 min 30 sec	
72 ℃	5 min	x1

Done by Sophie C

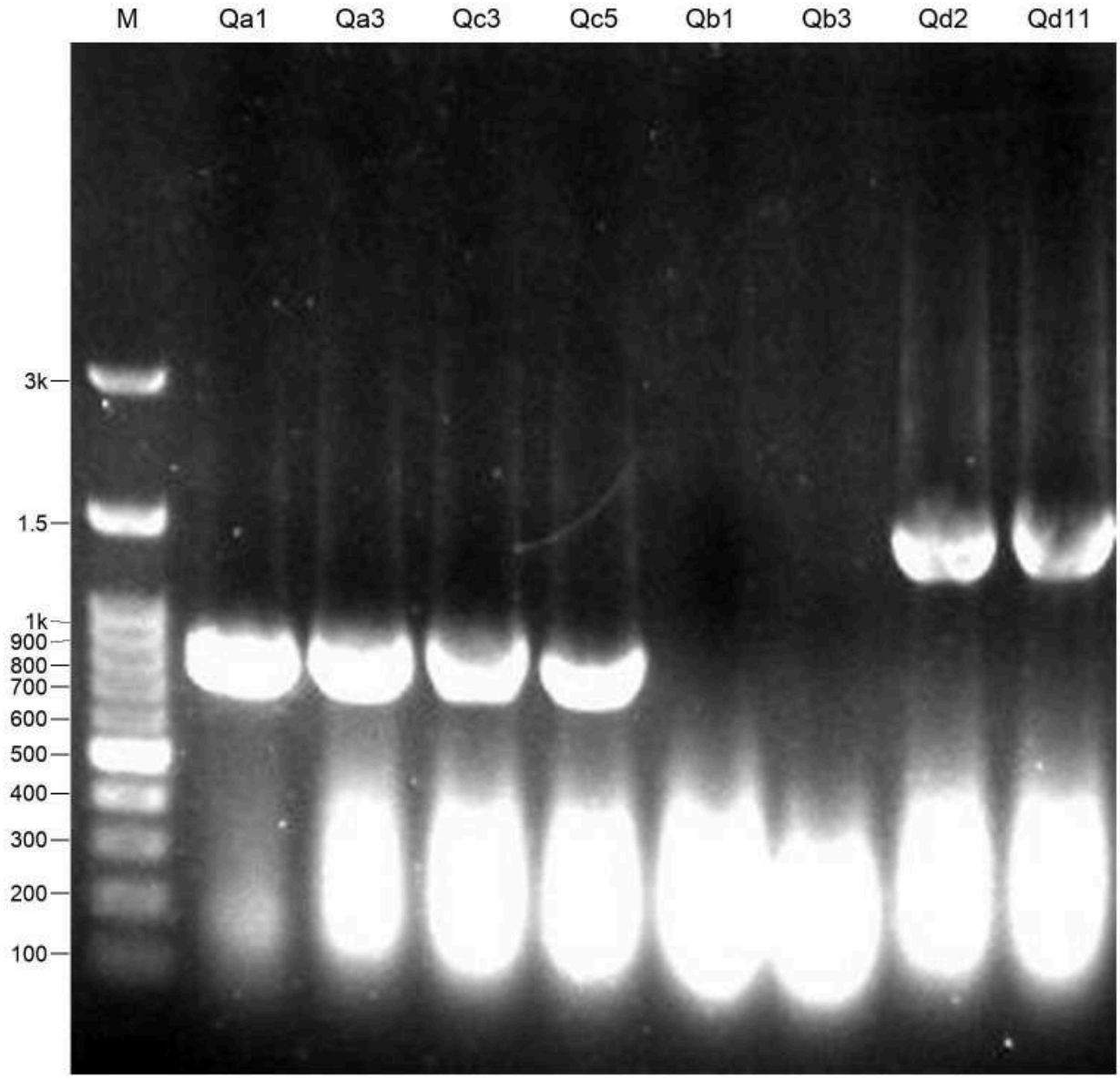
Date: 20260818
Title: Make 1 % Gel x2
Purpose:prepare gel for electrophoresis
Procedure:

1. Assemble mold, tray, comb
2. Weigh 0.2 g agarose gel, 20 mL 1XTAE buffer (for 1)
3. Add all in conical flask
4. Cover with plastic wrap&puncture holes
5. Microwave for 30 sec agitate
6. Repeat until no sediments
7. When no smoke, add 1μL safety dye agitate
8. Pour melted gel into mold
9. Let it solidify at RT

Done by Sophie C

Date: 20260818
Title: Gel electrophoresis
Purpose: To validate PCR product
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Qa1, 3 (LEA 2.1)	SamyQ F	STK Rv	770
Qc3, 5 (LEA 2.2)	SamyQ F	STK Rv	770
Qb1, 3 (ACCD v3t1)	SamyQ F	STK Rv	1331
Qd2, 11 (ACCD 2.1)	SamyQ F	STK Rv	1331

Done by Abby K

Date: 20260818
Title: Colony PCR (L)
Purpose: to screen for successful cloning after transformation
Procedure:
L: 3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

1. Prepare the following reactions in PCR tube

L1:

Fw primer Biobrick prefix	1 μL
Rv primer CsnR	1 μL
Q-amp master mix	10 μL
ddH_2O	8 μL

L2:

Fw primer pSTK-EXP-fw	1 μL
Rv primer CsnR	1 μL
Q-amp master mix	10 μL
ddH_2O	8 μL

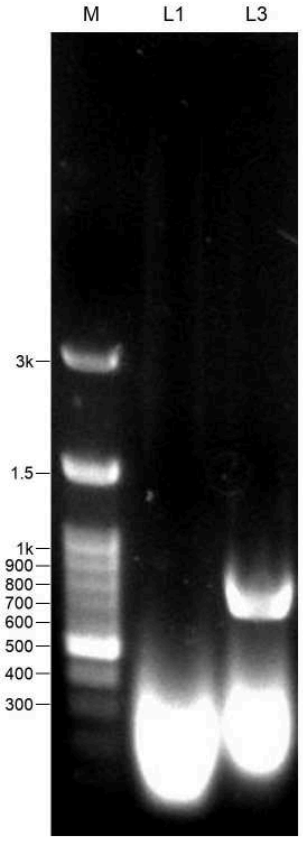
2. Pick up colony with tip
3. Dispense tip in pcr tube
4. Take out tip, close lid
5. PCR machine for thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	1 min 30 sec	
25 ℃	5 min	x1

Done by Sophie C

Date: 20260818
Title: Gel electrophoresis (L)
Purpose:
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L1, L3: 3P-EEBD-RBS0-csn:GanA-Gala-L3S2P21)	pSTK-EXP F	csn	1.3k

Done by Sophie C

Date: 20260818

Title: Golden gate assembly (level 2)
Purpose: to assemble parts to level 2
Procedure:
Samples:

- X: ACDI
- V: ABDI

V:A+B+D+I

V	
Part	Amount
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
B(83ng/μL, 3792bp)	1.484μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3787bp)	0.75μL
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.84 μl
T4 DNA ligase	1μL
BsmB1	1μL

X: A+C+D+I (1:1)

Part	Original (pmol=0.05)
Vector: pSTK-EXP-2-sfGFP - Concentration: 163 - Bp: 5104	1.017 μL
(A)pSTK-1A-PtagC+RBS-Ho1:pcyA-L3S2P21 - Concentration: 109 ng/μl - Bp: 3754	1.119 μL
(C)pSTK-1B-PlepA-RBS0-ccaSm3-B0015 - Concentration: 98 ng/μl - Bp: 3828 bp	1.27 μL
(D)pSTK-1C-Pveg-RBS0-ccaR-B0015 - Concentration: 134ng/μl - Bp: 3243 bp	0.786 μL

(I)pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015 - Concentration: 165 ng/µl - Bp: 3857	0.75 µL
T4 DNA ligase buffer	2 µL
Sterile water	11.05 µL
T4 DNA ligase	1 µL
BsmB1	1 µL
Total	20 µL

Done by Sophie C.

Date: 20260818

Title: Add NaCl + CH peptide to contreatment (exp set #7 trial 3)

Purpose: contreatment test

Procedure:

NaCl: 7.5mM, 100mM

CH peptide: pretreat, contreat, aftertreat

1. Dilute protectant by 1/500, NaCk from 3M NaCl master mix
    - 75 mM (total volume: 15 mL 375 µL (3 M NaCl)+ 14.625 mL (sterilized water)
    - 100 mM (total volume : 15 mL 500 µL (3 M NaCl)+14.5 mL (sterilized water)
2. Under the BSC, use a 200 µL pipette to remove the excess water
3. Add 3 mL of CH protectant to the agar plates
4. Move it around the plate until the entire surface is covered
5. Wait for 1 min and remove 2 mL of CH peptide and discard it
6. Mix 1 mL of CH peptide (diluted) with 1 mL of 75 mM NaCl in a 15 mL centrifuge tube by pipetting and label (CH peptide, 75mM NaCl)
7. Mix 1 mL of CH peptide (diluted) with 1 mL of 100 mM NaCl in another 15 mL centrifuge tube by pipetting and label (CH peptide, 100 mM NaCl)
8. Pipette 2 mL of protectant + NaCl master mix to each of the corresponding agar groups and more the plates until surface covered entirely
9. Pipette 1 mL NaCl to 75mM-> (control. 75 mM NaCl) plate / 100 mM -> (control, 1–mM NaCl) plate and move the plate until surface covered entirely
10. Take pics with phone (wrapped in plastic wrap) to upload daily observation
11. Use breath able tape to tape out and seal edges of plates
12. Place them back in the growth chamber

Done by Abigail

Date: 20260802
Title: Miniprep
Purpose: Isolate plasmid DNA
Procedure:
G10, G11, G12: PcpcG2-veg-EEBD-RBS0-Csn:LEA-bleb-cold double GanA-GalA Level 0

1. Transfer 1000 µL liquid culture to 1.5 ml microtube
2. Centrifuge at 16,000 g for 1min, discard supernatant
3. Repeat step 1&2 until all medium are collected
4. Add 200 µL of MX1 buffer
5. Resuspend by pipetting or vortexting
6. Add 250 µL of MX2 buffer
7. Invert slowly 8-10 times (don't pipette or vortex!), and incubate at RT for 2 min
8. Add 350 µL of MX3 buffer
9. Centrifuge at 23444 g for 5 min
10. Transfer 800 µL of supernatant to column
11. Centrifuge at 7000 g for 30 sec and discard the flow through
12. Add 500 µL of WN buffer, centrifuge at 7000 g for 30 sec
13. Centrifuge at 7000 g for 30 sec and discard the flow through
14. Centrifuge empty column for 8min at 10000 g and place column into 1.5 mL tube
15. Add 50 µL EB buffer at the center of the column
16. Centrifuge at 23444 g for 5 min

	Concentration (ng/µL)	A260/230	A260/280
G10	48.0	1.655	1.920
G11	57.0	1.676	1.966
G12	62.0	1.879	1.958
Gana-GalA level 0	278	2.122	1.9021

Done by Olivia L, Ryan Y

Date: 20260802
Title: PCR
Purpose: Amplify specific fragment of DNA
Procedure:
A:MF001
B:MF002
C:L3S1P32-L3S2P56

1. Prepare the reaction as follow:

	MF001 5.2 rxn	MF002 5.2 rxn	L3S1P32-L3S2P56 5.2 rxn
1.0x Ex taq buffer	13 µL	13 µL	13 µL
2.5uM dNTP Mix	10.4 µL	10.4 µL	10.4 µL
Forward Reverse MF001(each)	2.6 µL	x	x
Forward Reverse MF002(each)	x	2.6 µL	x
Forward Reverse L3S1P32-L3S2P56	x	x	2.6 µL
Sterile water	98.15 µL	98.15 µL	98.15 µL

2. Distribute 24.875µL of the 5rxn mixture into each tube for the corresponding part
3. Add 0.125µL of Ex Tax polymerase to each tube
4. Place in PCR machine applying the thermal cycle below:

98°C	30 sec	1 cycle
98°C	10 sec	25 cycle
55°C	30 sec	
72°C	15 sec	
72°C	5 min	1 cycle
25°C	∞	

Done by Ryan Y, Sophia Y
Redone by Ryan Y

Date: 20260802
Title: Prepare 2 % agarose gel to verify the size of the PCR products
Purpose: validate PCR results
Procedure:

1. Weight 1.6 g of agarose powder and measure 80 mL of 1X TAE buffer
2. Add the ingredients to a flask and cover the flask with a plastic wrap punctured with holes
3. Microwave for 30 secs and agitate it. Repeat until there are no sediments.
4. Assemble the trays and the combs

5. Wait until no smoke then add 4 µL of safety dye and agitate it
6. Pour melted gel to the molds
7. Wait until the gel solidifies

Done by Naomi

Date: 20260802
Title: Prepare 1 % agarose gel to verify the size of the PCR products
Purpose: validate PCR results
Purpose: validate PCR results
Procedure:

1. Weigh 0.2 g of agarose powder and measure 20 mL of 1X TAE buffer
2. Add the ingredients into the flask, cover the flask with plastic wrap with punctured holes.
3. Microwave for 30 sec and agitate it. Repeat until no particles are present
4. Assemble the tray and the comb
5. Add 1 µL of safety dye to the gel
6. Pour the gel into the mold
7. Wait until the gel solidifies

   Done by Naomi

Date: 20260802
Title: Enzyme Digestion
Purpose: Check plasmid size before DNA sequencing
Procedure:

1. Make sample for each vector to be digested and undigested using:

G10: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2 µL
DNA plasmid	10.916 µL
Nuclear free water	7.984 µL

G11: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2 µL
DNA plasmid	8.7719 µL
Nuclear free water	9.2281 µL

G12: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double

10x restriction buffer	2 µL
DNA plasmid	8.0645 µL

Nuclear free water	9.9355 µL

GanA-GalA level 0

10x restriction buffer	2 µL
DNA plasmid	1.7985 µL
Nuclear free water	10.2015 µL

2. For digested sample , add 0.5 µL of EcoR1
3. For undigested sample , add 2µL of ddH_2O
4. Incubate mixture at 37 ℃

Done by Ryan Y, Olivia Lin

Date: 20260802
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
ABDI:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PliaG-RBS0-CcaSm3-B0015)B+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD–RBS0-csn:AccD-B0015)I
ACDE:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PlepA-RBS0-CcaSm3-B0015)C+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD-RBS0-csn:AmilcP-B0015)E
ABDE:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PliaG-RBS0-CcaSm3-B0015)B+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD-RBS0-csn:AmilcP-B0015)E
ACDI:(PtagC+RBS-Ho1:PcyA-L3S2P21)A+(PlepA-RBS0-CcaSm3-B0015)C+(Pveg-RBS0-ccaR-B0015)D+(PcpcG2-veg-EEBD–RBS0-csn:AccD-B0015)I
L:3P-EEBD-RBS0-GanaA-Gala-L3S2P21

1. Set the water bath to 42 °C and prepare ice
2. Get 1 LBA plate and 4 LBC plates
3. Thaw competent cells until they’re 50 % thawed and add 5 µL of plasmid
4. Incubate on ice for 20 min
5. Heat shock at 42 °C for 45 sec
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium
8. Incubate for 1 hr at 37 °C
9. Centrifuge at 5000 g for 5 min
10. Remove 1 mL of supernatant then vertex
11. Spread on their respective plates
12. Incubate at 37°C for 16 hr

Done by Alex
Supervised by Gina

Date: 20260803
Title: Gel extraction of Mf001, Mf002, L3S2P56
Purpose: Extract targeted DNA
Procedure:

1. Place the tubes in a dry bath on 60 ℃ to dissolve the gel slices from 20260802
2. Load no more than 700 µL of melted gel into the collection tube
3. Centrifuge for 30 sec at 7000 g, discard the flow through. Repeat step 2 and 3 until all the for rest of the mixture
4. Wash the column once with 0.5 mL of WN buffer by centrifuging for 30 sec.Discard FT
5. Wash the column once with 0.5 mL of WS buffer by centrifuging for 30 sec. Discard FT
6. Centrifuge the column at full speed 23444 g for 3 min. To remove residual ethanol
7. Place column into new 1.5 mL microtube. Add 15 µL of EB buffer onto the center of the membrane
8. Stand column for 2 min. Centrifuge at 23444 g for 2 min to elute DNA

Done by Sophie C

Biodrop Results:

part	Concentration (ng/µL)	A230/260	A260/180
MF001	15	1.071	1.875
MF002	19	1.583	1.900
L3S1PS2-L3S2P56	14	0.875	1.750

Date: 20260803
Title: Colony PCR
Purpose: Amplify the number of targeted gene
Procedure:

1. Prepare the reaction as followed:

	1 rxn	26 rxn
Fr premier (Exp)	1 µL	26 µL
Rv promier (Exp)	1 µL	26 µL
Q-amp master mix	10 µL	260 µL
dd H_2O	8 µL	208 µL

ABDI: ACDI: 6486 bp

2. Separate 20 µL of the master mix into each PCR tubes (24 PCR tubes in total)
3. Pick the total of 24 colonies from the three plates under UV light

4. Pick up colonies using tip after turning on the bunsen burner
5. Point the tip(with colony) on another plate (labeled) and slightly point it for 3 times
6. Place the tip in PCR tubes and let it incubate for 5 min, gently mix by shaking horizontally
7. Put it in the PCR machine

Thermal cycle:

98°C	2 min	1 cycle
98°C	30 secs	30 cycles
55°C	4 secs	
72°C	2 min	
25°C	1 min	1 cycle

Done by Sophie H, Sophia Y.

Date: 20260803
Title: Golden gate assembly
Purpose: Assemble level 0 parts
Procedure:

A. MF001

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/μL): 346 ● bp: 3090	0.05804913295 μL	1.16 μL (dilute 1/20)
Insert MF001 ● concentration(ng/μL): 15 ● bp: 82	0.07106666667 μL	1.14 μL (dilute 1/16)
T4 DNA ligase buffer	2 μL	
Sterile water	13.7 μL	
T4 DNA ligase	1 μL	
BbsI-HF	1 μL	

Total	20 µL	

B. MF002

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert MF002 ● concentration(ng/µL): 19 ● bp: 82	0.056 µL	1.12 µL (dilute 1/20)
T4 DNA ligase buffer	2 µL	
Sterile water	13.72 µL	
T4 DNA ligase	1 µL	
BbsI-HF	1 µL	
Total	20 µL	

C. L3S1P52:L3S2P56

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● concentration(ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20)
Insert L3S1P52:L3S2P56 ● concentration(ng/µL): 14 ● bp: 175	0.1625 µL	1.14 µL(dilute 1/7)
T4 DNA ligase buffer	2 µL	

Sterile water	13.7 µL	
T4 DNA ligase	1 µL	
BbsI-HF	1 µL	
Total	20 µL	

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 min	25-45 cycles
16 °C (Ligation)	3 min	
37 °C (Restriction enzyme digestion)	5 min	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 min	1 cycle

Done by Sophie C.

Date: 20260803
Title: Make 0.8 % gel
Purpose: To prepare gel for gel electrophoresis
Procedure:

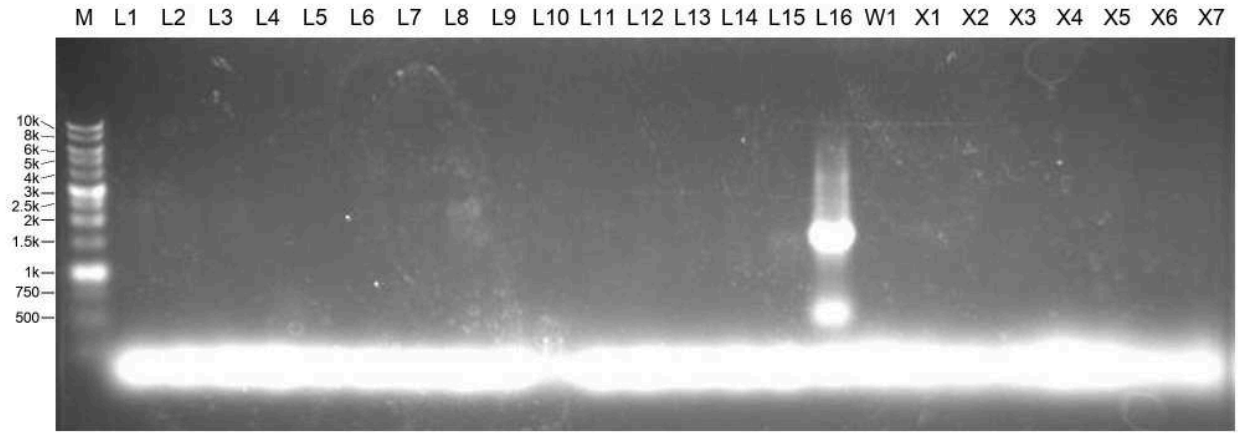
1. Assemble comb, tray mold
2. Add 0.64 g of agarose powder and 80 mL 1XTAE buffer into conical flask
3. Cover flask with plastic wrap & puncture holes
4. Microwave for 30 sec. Agitate. Repeat fill no sediments
5. When no smoke, add 4 µL safety dye.Agitate
6. Pour melted gel into the mold
7. Wait until gel solidifies

Done by Sophie C/H,Naomi

Date: 20260803
Title:Gel electrophoresis

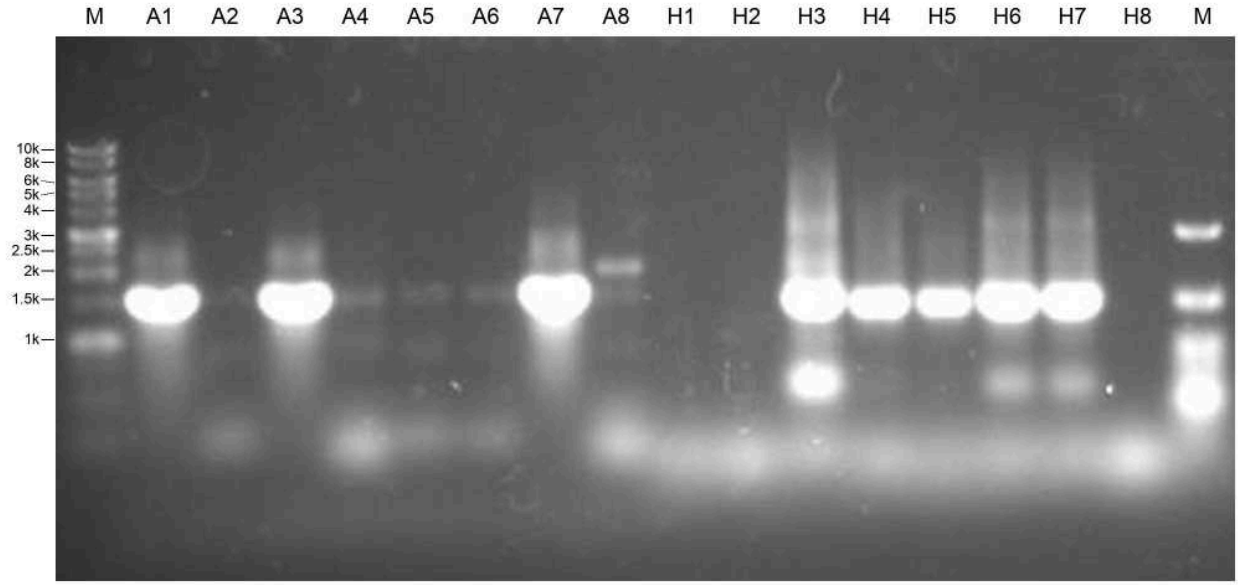
Purpose: ensure plasmid size for each samples
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L: 3P-EEBD-RBS0-GanA-Gala-L3 S2P21	EXP-seq-fw	EXP-seq-rv	3448
W: ABDE	EXP-seq-fw	AmilCP R	6298
X: ACDI	EXP-seq-fw	ACCD R	6363

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg-RBS0-Ho1:pcyA-B0015	BioBrick fw	STK rv	1513
H: Pveg-EEBD-RBS0-csn:LEA-dkdB Cold	EXP-seq-fw	EXP-seq-rv	6968

Done by Ethan C, Sophie C

Date: 20260803
Title: Liquid Culture
Purpose: bacteria culture
Procedure:
J: Pveg-EEBD-RBS0-csn:AccD-B0015

1. Label culture tube
2. Turn on bunsen burner
3. Add 5 mL LB medium and 25 µL of erythromycin to the tube
4. Dip a pipette in J1
5. Dispense tip into tube
6. Place tube in the incubator(shape)

Done by Sophie C

Date: 20260804
Title: 250mL TB medium preparation
Purpose: Prepare TB medium for Arabidopsis planting on plates
Procedure:

1. Weight 11.9 g of broth powder, 250 mL ddH_2O, and 2 mL trehalose
2. Add the ingredient above in order (broth powder→ddH_2O→trehalose) in 250 mL flask and add a stirring bar
3. Open stirring machine and let it stir until all the broth powder was fully dissolved
4. Cover the flask with aluminum foil and stick an autoclave tape on top
5. Autoclave for 20 min at 127 °C.

Done by Sophie H, Naomi

Date:20260804
Title: Miniprep of A8, H4, and H5
Purpose: Isolate plasmid DNA
Procedure:
A8: Pveg-Ho1:PcyA (did to step 5 because of too obvious blue color through observation)
H4: Pveg-Csn:LEA
H5: Pveg-Csn:LEA

1. Transfer 1.5 mL Tb medium to a 2 mL microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Repeat step 2 and 3 until all the liquid culture were collected
5. Add 200 µL of MX1 buffer
6. Resuspend the pellet by vortexing or pipetting
7. Add 250 µL of MX2 buffer
8. Gently mix the buffer well by inverting the microtube 8-10 times
9. Incubate at Rt for 2 min
10. Add 350 mL of MX3 buffer
11. Gently mix via inversion 4~5 times
12. Centrifuge at 23444 g for 5 min
13. Assemble the column and the tube while waiting for centrifuge
14. Transfer the supernatant (approximately 800 µL) to the column
15. Centrifuge the column at 7000 g for 30 sec
16. Discard the flow through
17. Add 500 µL WN buffer
18. Centrifuge the column at 7000 g for 30 sec
19. Discard the flowthrough
20. Add 700 µL of WS buffer
21. Centrifuge the column at 7000 g for 30 sec
22. Discard the flow through
23. Centrifuge the empty column 10,000 g for 3 min
24. Place the column into new 1.5 mL microtube

25. Add 50 µL of EB buffer to the center of the membrane
26. Incubate at RT for 3 min
27. Centrifuge at 10,000 g for 3 min

Done by Sarah, Naomi

Biodrop Result:

	Concentration (ng/µL)	A260/230	A260/280
H5	131.0	1.899	1.048
H4	132.0	1.913	1.048

Date: 20260804
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence G,L
Procedure:
G: PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb cold double
L:3P-EEBD-RBS0-GanaA-Gala-L3S2P21

1. Prepare the following reaction:

G:

	1 rxn	26 rxn
Csnfw	1 µL	26 µL
pSTK-seq rv	1 µL	26 µL
Q-amp master mix	10 µL	260 µL
ddH_2O	8 µL	208 µL

L:

	1 rxn	26 rxn
pStK-Exp-seq fw	1 µL	26 µL
pSTK-Exp-seq rv	1 µL	26 µL
Q-amp master mix	10 µL	260 µL
ddH_2O	8 µL	208 µL

2. Pick up colonies with tip and streak on grid
3. Wait for 5 mins and pick up tips
4. Put in thermal cycle

95 °C	5 min	1 cycle
95 °C	20 sec	35 cycles
65 °C	20 sec	
72 °C	3 min 40 sec	
72 °C	5 min	1 cycle

Done by Alex, Sophie C, Chloe

Date:20260804
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence W,ACDI,ACDE
Procedure:

1. Prepare the following reaction

W:

	1 rxn	3 rxn
CcaR Fw	1 µL	3 µL
PcpcG2-veg rv	1 µL	3 µL
Q-amp master mix	10 µL	30 µL
ddH_2O	8 µL	24 µL

X:

	1 rxn	12 rxn
CcaR Fw	1 µL	12 µL
PcpcG2-veg rv	1 µL	12 µL
Q-amp master mix	10 µL	120 µL
ddH_2O	8 µL	96 µL

V:

	1 rxn	4 rxn
CcaR Fw	1 µL	4 µL
PcpcG2-veg rv	1 µL	4 µL

Q-amp master mix	10 µL	40 µL
ddH_2O	8 µL	32 µL

2. Pick up colonies with tip & streak on grid
3. Wait 5 min and take out tips
4. Put in thermal cycle(same as previous page)

Done by Chloe & Sophie C

Date:20260804
Title: make 0.8 % gel
Purpose:prepare gel for electrophoresis
Procedure:

1. Add 0.64 g agarose powder and 80 mL of 1XTAE buffer into a conical flask
2. Cover it with a plastic wrap and puncture holes
3. Assemble tray, mold, comb
4. Microwave flask for 30 sec. Agitate. Repeat until no sediments
5. When no smoke, add 4 µL safety dye. Agitate
6. Pour melted gel into mold
7. Let it solidify at RT

Done by Sophie C, Chloe

Date:20260804
Title: make 1 % gel
Purpose:prepare gel for electrophoresis
Procedure:

1. Add 0.4 g agarose powder and 40 mL 1XTAE buffer into conical flask
2. Cover it with a plastic wrap and puncture holes
3. Microwave flask for 30 sec Agitate. Repeat until no sediments
4. Assemble mold, fray, comb
5. When there is no smoke, add 2 µL safety dye. Agitate
6. Pour melted gel into mold
7. Let it solidify at RT

Done by Sophie C, Chloe

Date:20260804
Title: Transformation
Purpose: Amplification MF001, MF002,L3S1PS2-L3S2P56, V:ABDI, X:ACDI
Procedure:

1. Set the water bath to 42 °C and prepare ice
2. Get 1 LBA and 5 LBC plate
3. Thaw competent cells until they’re 50 % thawed and add 10 µL plasmid to 100 µL competent cell
4. Incubate on ice for 20 min

5. Heat shock at 42 °C for 45 sec
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium
8. Incubator for 1 hr at 37 °C
9. Centrifuge at 5000 g for 5 min
10. Remove 1 mL of supernatant & vortex
11. Spread on their respective plates
12. Incubate at 37 °C for 16 hr

Done by Sophie C, Chloe

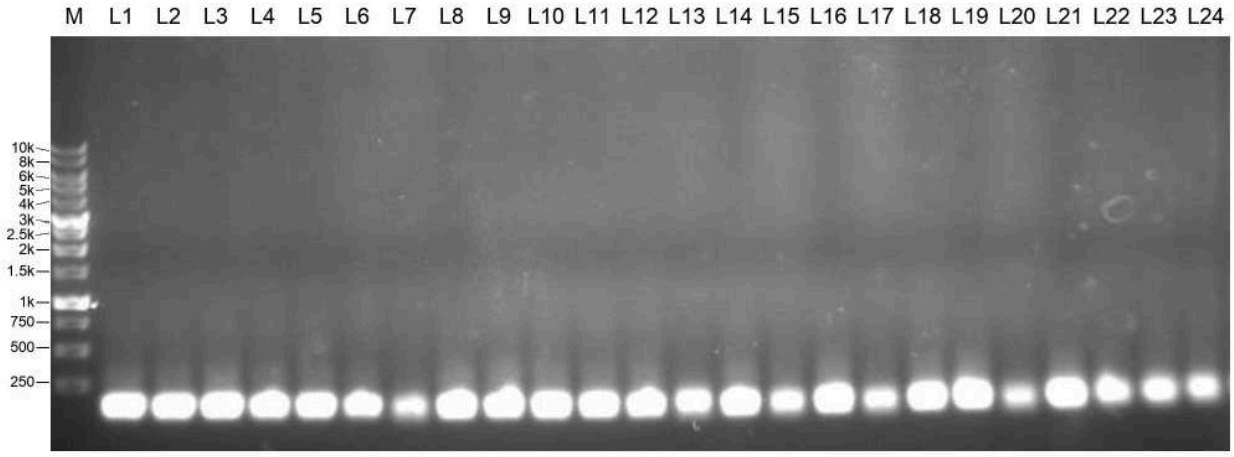
Date:20260804
Title: Liquid culture
Purpose: bacteria culture from successful colony PCR product
Procedure:
A:PtagC+RBS-Ho1-PcyA-L3S2P21
G7~9:pcpcG2-veg-eEBD-RBS0-csn:LEA-bkbd cold

1. Label 4 liquid culture tubes
2. Add 3 mL TB medium and 3 µL ampicillin in each tube
3. Dip a pipette into colonies (each)
4. Dispense tips in tubes(each)
5. Place tubes in incubator(shake)

Done by Sophie C

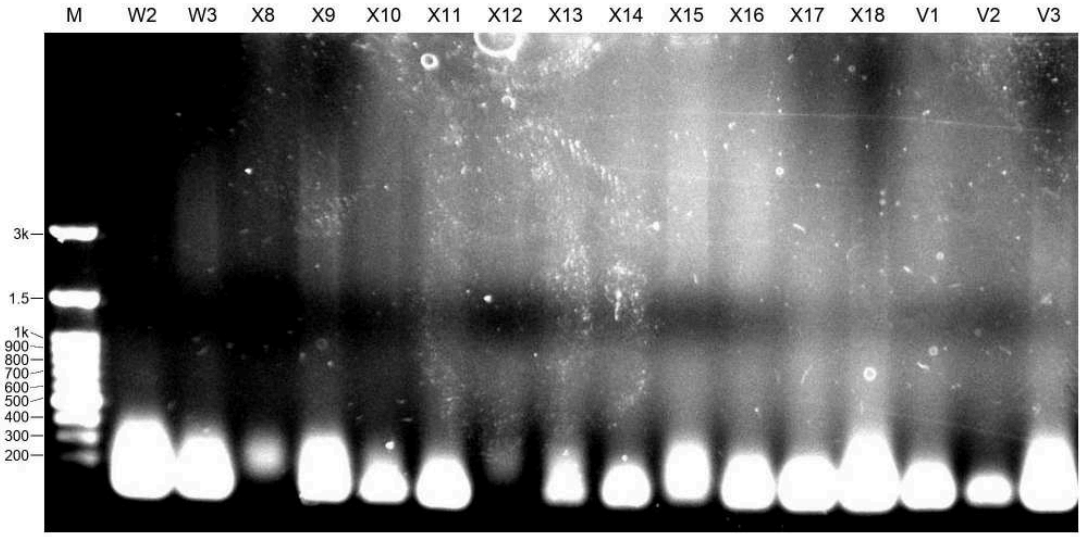
Date:20260804
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L: 3P-EEBD-RBS0-GanA-Gala-L3 S2P21	EXP-seq fw	EXP-seq rv	3516
Note: All fail -> ReGolden Gate			

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



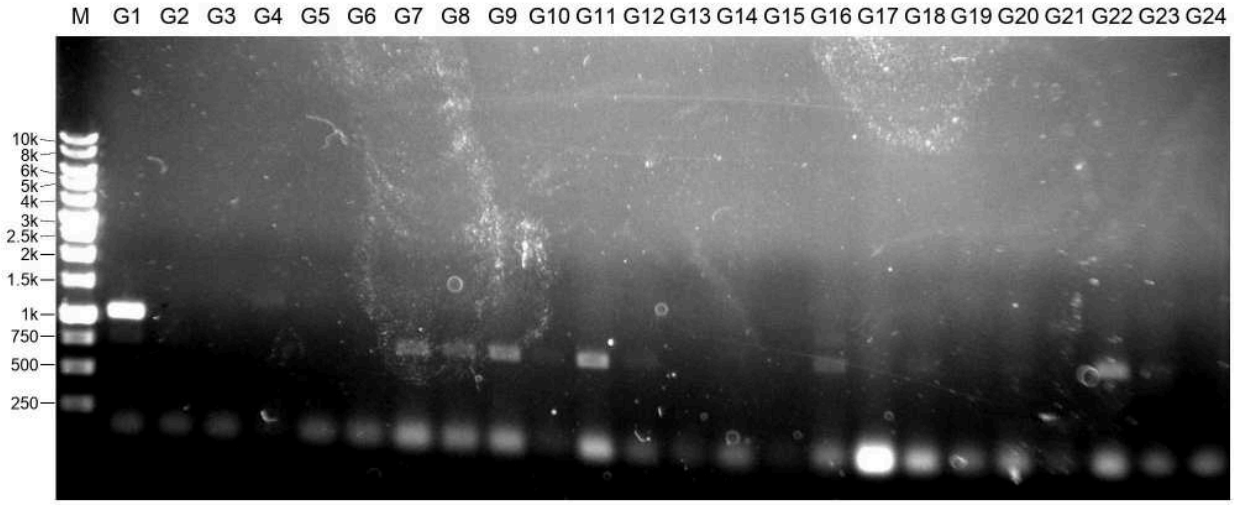
M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
W: ABDE	CcaR fw	PcpcG2-veg rv	1045
X: ACDI	CcaR fw	PcpcG2-veg rv	1045
V: ACDE	CcaR fw	PcpcG2-veg rv	1045
Note: All fail -> Retransform 0804			

Done by Sophie C, Chloe

Date:20260804
Title: Gel electrophoresis
Purpose: validate the bps of pcr results
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G: PcpcG2-veg-EEBD-RBS0-Csn: Lea-bkdB-Cold	Csn fw	STK-seq rv	787
Note: G7, G8, G9 -> liquid culture 0804			

Done by Sophie C

Date:20260804
Title: Add 正瀚protectant to exp7. Circular plates (pretreatment group: 0, 正瀚/ 100 NaCl, 正瀚)
Purpose: to test when to add the protectant is the best
Procedure:

1. Dilute 正瀚protectant by 1/500 (add 6986 µL water + 14 µL protectant) into the 15 mL tube in the BSC
2. Add 1mL of the diluted protectant into the 2 agar plates (0, 正瀚/100mM NaCl, 正瀚)
3. Evenly distribute the liquid protectant by filling the plate sideways until all the plants are covered with a thin layer protectant
4. Seal the plates with tape and label the pretreatment date

Done by Sophia Y.
Supervised by Audrey H, Abby T., Abigail

Date: 20260804
Title: Agar exp set 6 fresh weight
Purpose: compare different groups' growth results
Procedure:

1. Using a tweezer, lift and pick all 9 arabidopsis simultaneously from the agar plate in a single motion
2. Careful not to hurt the plants by squeezing the tweezer
3. Lift from below the leaf area
4. Put the 9 plants on a tissue paper, lightly press to remove the agar sticking to the plants
5. Careful not to press the leaf areas too hard
6. Place all 9 plants in a weighing plate, weight the fresh weight of arabidopsis (a plate at a time)
7. Pick them up and put them(plants) in a 2 mL microtube, take care that the entire plant and the roots need to go in too
8. Label the group concentration as well as the respective fresh weight on the microtube
9. Repeat these steps for as many times as the sets of arabidopsis (8 sets)

Done by Audrey H
Supervised by Abby T

Result:

Group	Weight(g)
150 mM NaCl	0.0175
Control	0.01756

100 mM NaCl 5 mM Trehalose	0.0517
75 mM NaCl 5 mM Trehalose	0.0655
75 mM NaCl	0.0655
5 mM Trehalose	0.1755
100 mM NaCl	0.0298
150 mM NaCl 5mM Trehalose	0.0187

Date: 20260805
Title: Miniprep
Purpose: Isolate plasmid DNA
Procedure:
G7, G8, G9: PcpcG2-veg-EEBD-RBS0-Csn:LEA-bdlB-cold
A1: PtagC+RBS-Ho1-pcyA-L3S2P2

1. Transfer 1.8 mL of TB medium to a 2 mL microtube
2. Centrifuge for 30 sec 16000 g, then discard the supernatant
3. Repeat set 1&2 until all the liquid culture were collected
4. Add 200 µL MX1 buffer into each tube
5. Resuspend the pellet by vortexing or pipetting
6. Add 250 µL of MX2 buffer
7. Let it incubate in RT for 2 min after inverting for 8~10times
8. Add 350 µL of MX3 buffer, then slowly invert it 4~5times
9. Centrifuge at 23444 g for 5 min
10. Assemble the column and the tube
11. Transfer about 400 µL of supernatant into the column
12. Centrifuge the column at 7000 g for 3 min
13. Discard the flow through, then add 500 µL WN buffer
14. Centrifuge the column at 7000 g for 3 min
15. Discard the flow through, then add 500 µL of WS buffer
16. Centrifuge the column at 7000 g for 3 min
17. Discard the flow through
18. Centrifuge the empty column at 1000 g for 3 min and place it into new 1.5 mL microtube
19. Add 50 µL EB buffer into the column (right in the center)
20. Incubate at RT for 3 min
21. Centrifuge at 10000g for 3 min
22. Using biodrop testing the concentration

	Concentration(ng/µL)	A260/280	A260/230
A1	109.0	1.912	0.940
G7	115.0	1.917	0.966
G8	145.9	2.014	1.280
G9	147.0	1.934	1.114

Done by Naomi, Sophie H

Date: 20260805
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence

Procedure:
MF: MF002
LL:L3S1PS2-L3S2P5G
X:XACDI

1. Prepare the following reactions and distribute into PCR tube

LLx12rxn

Fw primer L3S1PS2 fw	12 µL
Rv primer L3S1PS2 rv	12 µL
Q-amp master mix	120 µL
ddH_2O	96 µL

MFx13rxn

Fw primer MF002 fw	13 µL
Rv primer MF002 rv	13 µL
Q-amp master mix	130 µL
ddH_2O	104 µL

Lx26rxns

Fw primer Exp-seq-fw	26 µL
Rv primer Exp-seq4-rv	26 µL
Q-amp master mix	260 µL
ddH_2O	208 µL

Xx2rxn

Fw primer CcaR fw	2 µL
Rv primer PcpcG2-veg rv	2 µL

Q-amp master mix	20 μL
ddH_2O	16 μL

2. Pick up colonies with tip and streak on grid
3. Wait 5min&take out tips
4. Put in thermal cycle:

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	4 min	
72 ℃	5 min	x1

Done by Sophie C, Abby T

Date: 20260805
Title: Make 1 % & 0.8 % gel
Purpose: make gel for electrophoresis of colony PCR
Procedure:

1. Add the ingredient as follow in to two 250ml flask
2. 1 %-> 0.4 g agarose powder / 40 ml TAE buffer
   0.8 %->0.36 g agarose powder / 40 ml TAE buffer
3. Put the flask into microwave and heat until no powder in the flask
4. Wait for it to stop smoking and add 2 μL safety dye into each flask
5. Pour the gel into two assembled tray, use tip to push the air out and pop bubbles
6. Assemble the comb
7. Wait for the two tray to solidify

Done by Sophie H

Date: 20260805
Title: Protein Purification & extraction
Purpose: Isolating a specific protein based on binding affinity(his-tag)
Procedure:

1. Take 24 2 mL centrifuge tubes and add 1800 μL of overnight liquid culture(*Bacillus subtilis*)
2. Centrifuge 16000 g for 30 seconds
3. Take tubes out of microcentrifuge and pour the supernatant into a 50mL centrifuge tube
4. Repeat the steps 1-3 until the 50 mL centrifuge tube is filled
5. Pour the supernatant into a syringe connected to 0.22 μm filter and filter the supernatant into a 50 mL centrifuge tube

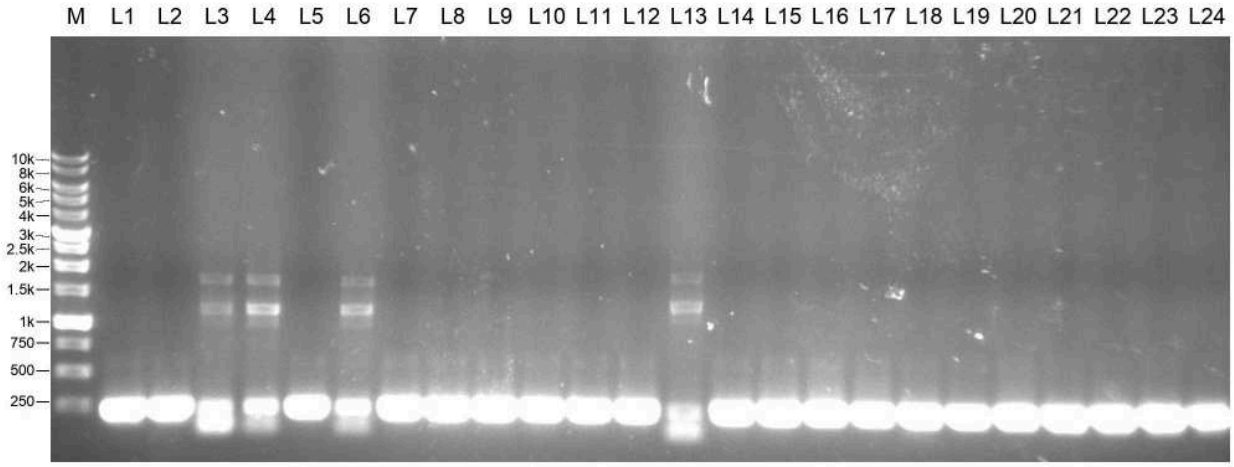
6. Repeat the previous steps until all culture is filtered
7. Centrifuge the Ni-NTA column at 175 g for 2 min to remove the storage buffer
8. Add 6mL of equilibrate buffer into the column by dispensing the buffer along the walls of the tube
9. Add 237.5 µL of imidazole into supernatant
10. Add all supernatant from 1 tube into the column
11. Add 6 mL of wash buffer and remove the flow-through repeat this step 3 times
12. Centrifuge at 175 g for 2 min
13. Collect the flow through from the last wash buffer step
14. Add 3 mL of elute buffer
15. Centrifuge at 175 g for 2 min
16. Collect the flow-through as the purified protein

Note: only did one tube, others filtered store in refrigerator for 0806 purification

Done by Abigail
Supervised by Gabriel

Date: 20260805
Title: Gel Electrophoresis
Procedure:

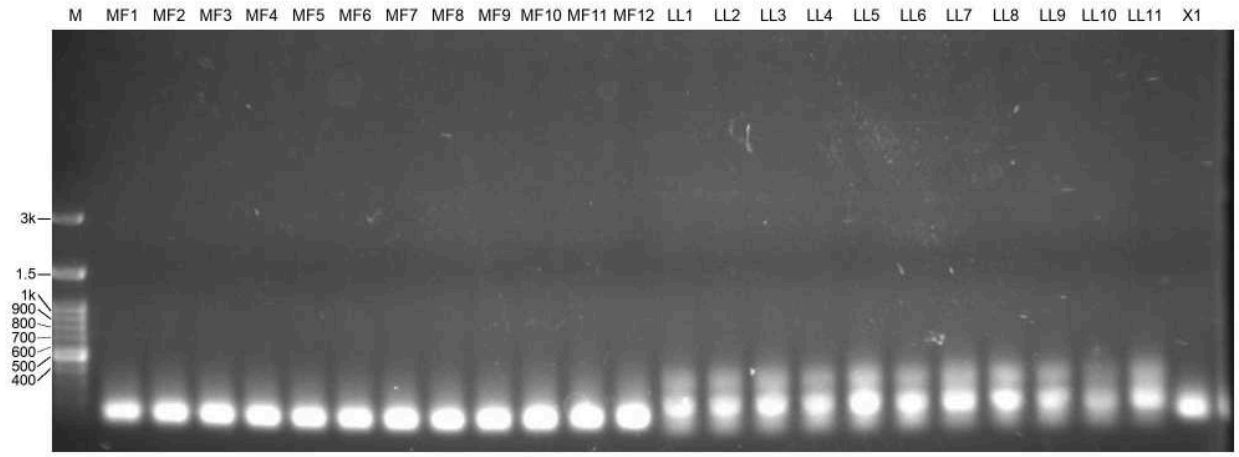
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

L: 3P-EEBD-RBS0-GanA-Gala-B0015	EXP-seq-fw	EXP-seq4-rv	3626

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
MF: MF002	MF002 fw	MF002 rv	46
LL: L3S1PS2-L3S2P56	L3S1PS2-L3S2P56 fw	L3S1PS2-L3S2P56 rv	139

Done by Sophie C

Date: 20260805
Title: Golden Gate Assembly
Purpose: assemble level 1 parts to level 2
Procedure:
A: pSTK-1A-Ptag+RBS-Ho1:pcayA-L3S2P21
B:psTK-1B-Pllaq-RBS0-CcaSm3-B0015
C:pSTK-1B-PcepA-RBS0-CcaSM3-B0015
D:pSTK-1C-Pveg-RBS0-CcaR-B0015
I:pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015
C:pStK-Exp1-3P-EEBD-RBS0-GanA-Gala-L3S2P21

✓ABDI

Part	Original	Dilution
pSTK-Exp2-SfGfP (163 ng/μL 5104 bp)	0.2 μL	1.02 μL (1/5)
A (109 ng/μL 3754 bp)	0.46 μL	1.37 μL (1/3)
B (83 ng/μL 3793 bp)	0.59 μL	1.78 μL (1/3)
D (134 ng/μL 3243 bp)	0.31 μL	0.26 μL (1/3)
I (165 ng/μL 3757 bp)	0.29 μL	1.48 μL (1/5)
T4 DNA ligase buffer	2 μL	
T4 DNA ligase	1 μL	
BsmBI-v2	1 μL	
ddH_2O	9.05 μL	

x:ACD1

Part	Original	dilution
pSTK-Exp2-SfGfP (163 ng/μL 5104 bp)	0.2 μL	1.02 μL (1/5)
A	0.46 μL	1.37 μL (1/3)
C (98 ng/μL 3744 bp)	0.31 μL	1.26 μL (1/4)
I	0.29 μL	1.48 μL (1/5)
T4 DNA liq buff	2 μL	
T4 DNA liq	1 μL	
BsmBI-v2	1 μL	
ddH_2O	9.35 μL	

L

Part	Original	dilution

pSTK-Exp1	0.28 µL	1.4 µL (1/5)
3P	0.36 µL	1.43 µL (1/4)
EEBD-RBS0	0.21 µL	1.05 µL (1/5)
GanaGala	0.3 µL	1.22 µL (1/4)
L3S2P21	0.16 µL	1.64 µL (1/10)
pSTK-ocx-spacer	0.11 µL	1.07 µL (1/10)
pSTK-DC2-nissx	0.12 µL	1.16 µL (1/10)
T4 DNA liquid buffer	2 µL	
T4 DNA liq	1 µL	
BsaI-HFv2	1 µL	
ddH_2O	7.05 µL	

Done by Chloe, Sophic C

Date: 20260805
Title: Make acrylamide gel
Purpose:separates proteins or small nucleic acids by size during electrophoresis.
Procedure:

1. Assemble holder, plates, comb
2. Prepare resolves solution

Resolver A	3 mL
Resolver B	3 mL
10 % APS	30 µL
TEMED	3 µL

3. Full to 1.5-1cm below bottom of comb teeth w resolver solution(~6mL)
4. Prepare stacker solution

Stacker A	1 ml
Stacker B	1 mL
10 % APS	10 µL

TEMED	2μL

5. Fill to top with stacker solution(~2 mc)
6. Insert comb and allow the gel to polymerize

Done by Sophie C

Date: 20260805

Title: agar (exp set #6) chlorophyll extraction

Purpose: compare different groups' growth results

Procedure:

1. Take the microtubes from 0804
2. Dilute acetone to 80 %, add 4 mL to each tube
3. Add 10 metal bits into each tube
4. Wait for the chlorophyll to extract, and use a vortex to crush the plants to speed up the process (done when the plants are no longer green)
5. Add the liquid to the 96 well, 200 μL in each well, 3 wells for each sample
6. ELISA Reader, insert the 96 well, run the results through the equation
7. Make sure to select the correct wavelength
    a. A633 -> used the wrong wavelength this time, it should be 663

Done by Audrey H, Phoebe, Sarah C

Date: 20260805

Title: Seed planting on agar

Purpose: grow the seeds on agar

Procedure:

1. Open the BSC
2. Use the seed template (circular plates) to plant 25 seeds on each agar plate
3. Use breathable tape to seal each of the agar plates
4. Label the vernalization date, incubator date, and seed type
5. Wrap the agar plates with aluminium foil
6. Vernalize in the 4°C refrigerator for 4 days and avoid light

Done by Audrey H, Naomi, Sophie H

Date: 20260805

Title: Add CH protectant and NaCl on horizontal agar plates(exp set #7)

Purpose: to test when to add protectant is the best (Co-treatment)

Procedure:

1. Use the following amount to apply for each plate

3M NaCl	1667 μL

CH biopeptide	1000 µL (dilution rate 1/500 -> 6986 µL sterile water + 14 µL peptide)

2. Apply NaCl for the following plates

Co-treatment	100mM NaCl, CH biopeptide (Day 0)
Pre-treatment	100mM NaCl, CH biopeptide (Day -1)
After-treatment	100mM NaCl, CH biopeptide (Day 1)

3. Apply CH biopeptide to the following plates

Co-treatment	100 mM NaCl, CH biopeptide	(Day 0)
	0 mM NaCl, CH biopeptide	

Done by Audrey H

Date: 20260806
Title: Colony PCR of G, H
Purpose: Amplify desired DNA sequence
Procedure:

1. Prepare the master mix as below:

	1 rxn	34 rxn
Fr Primer (biobrick prefix)	1 µL	34 µL
Rv primer (LEA reverse)	1 µL	34 µL
ddH_2O	8 µL	272 µL
Qamp master mix	10 µL	340 µL

	1 rxn	30 rxn
Fr Primer (pSTK Exp1 Fw)	1 µL	30 µL
Rv Primer (pSTK Exp1 Rv)	1 µL	30 µL
ddH_2O	8 µL	240 µL
Qamp master mix	10 µL	300 µL

2. Mix the master mix by vortexing
3. Distribute the master mix into PCR tubes(remember to label the PCr tubes first, each PCR tube should have 20 µL master mix）
4. Centrifuge PCR tubes before starting to pick up colonies
5. Open bunsen burner, and use pipette tip to pick up labelled single colony
6. Dip/draw a few lines with the tip on another labeled LBA plate
7. Dip the tip into respective PCR tubes and dispense it inside
8. After all the labeled colonies were collected, let the PCR tubes incubate at RT for 5 min, gently shake the tip at the same time
9. Remove pipette tips and put PCr tubes in the PCR machine
10. Set up thermal cycle as below:

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	

72°C	2.5 min	
72°C	1 min	1 cycle

Done by Sophie H, Sarah

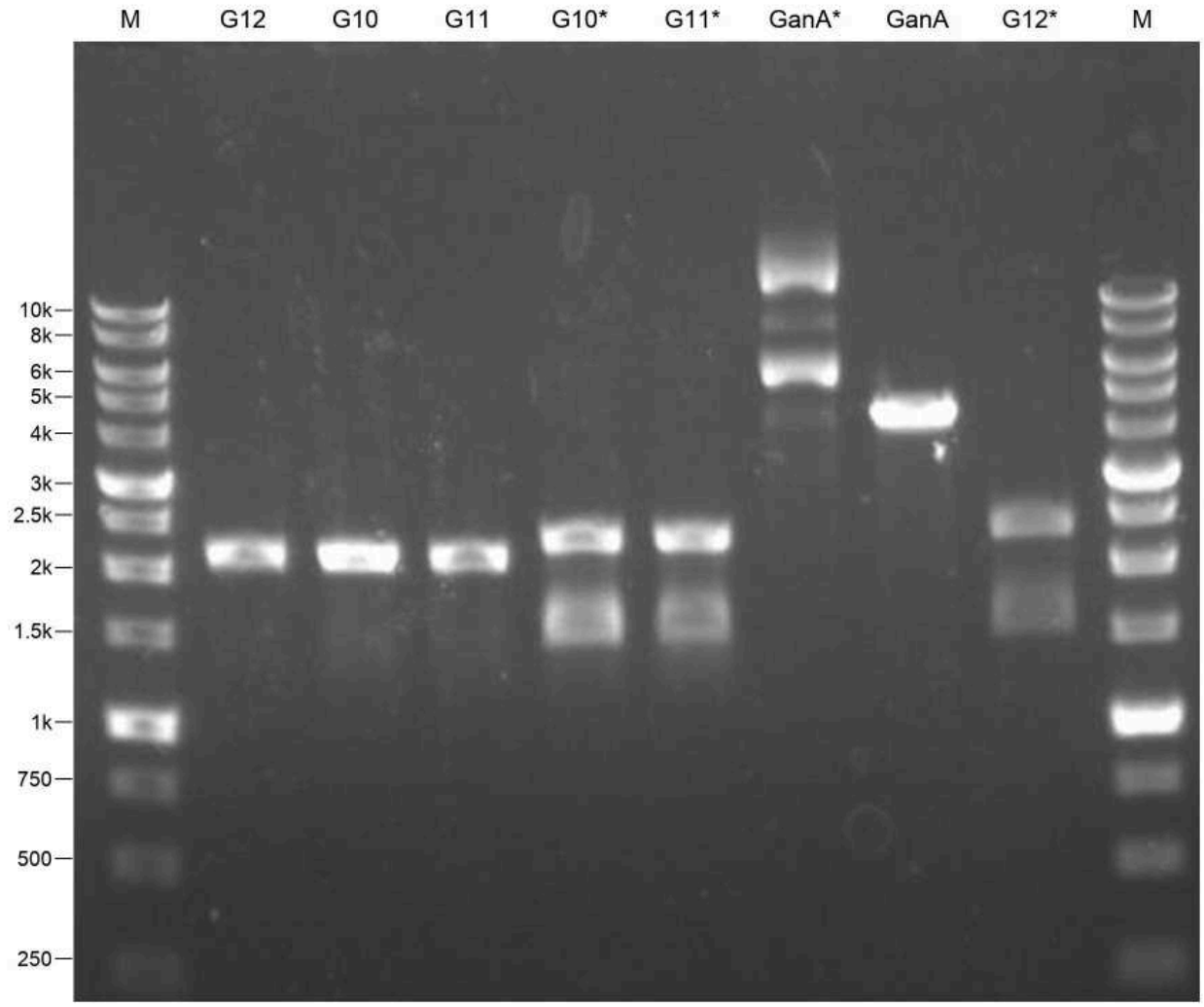
Date: 20260806
Title: Make 0.8 % gel for enzyme digestion in 8/2
Purpose: Make gel for electrophoresis
Procedure:

1. Weight 0.32 g of Agarose powder and 40 mL TEB buffer
2. Add the ingredients above in 250 mL flask and cover the top with double layer plastic wrap
3. Poke some holes on plastic wrap with pipette tip
4. Microwave the gel until the gel is clear
5. Remove the plastic warp and wait until the gel stops smoking but is still hot
6. Add 2 µL of diluted safety dye into liquid gel and mix it by shaking the flask gently
7. Pour the gel into assembled tray to the black line in the tray
8. Use pipette tip to squeeze the air out and pop bubbles
9. Wait for gel to solidify about 30min)

Done by Naomi

Date: 20260806
Title: Gel electrophoresis of 0802 enzyme digestion
Purpose: check the result of 0802 enzyme digestion

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Sample	Plasmid Construct	Enzyme	Expected Size (bp)
G10~12	PcpcG2-veg-EEBD-RBS0-csn:LEA-bkdb-cold double	EcoR1	3234 (undigest)
GanA	GanA-GalA level 0	EcoR1	4148 (undigest)
Note: *= undigested			

Done by Sarah , Naomi

Date: 20260806

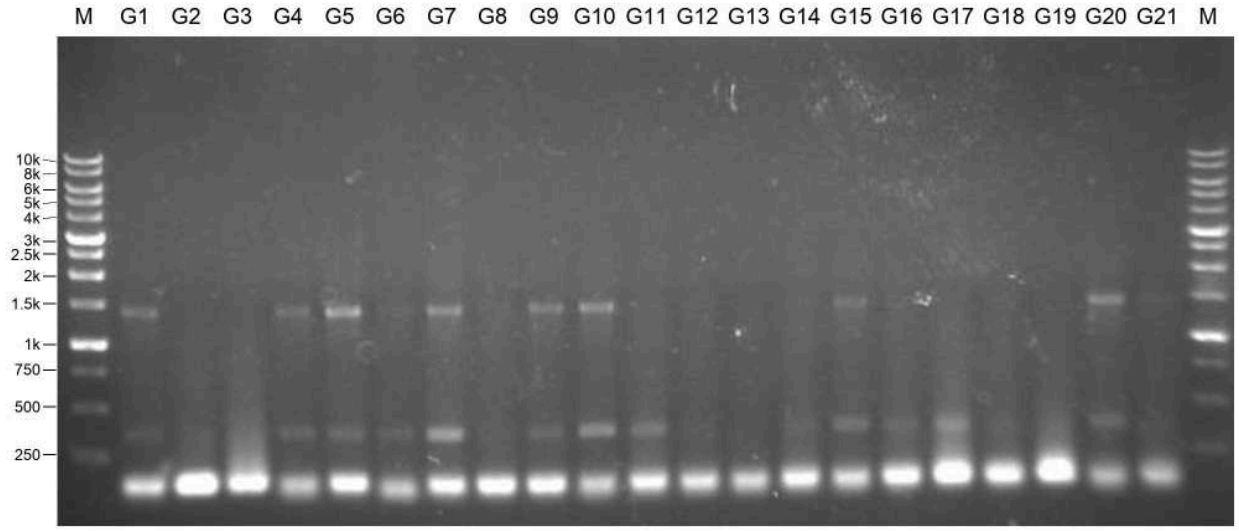
Title: Make 1 % gel for colony PCR(25 well*3)
Purpose: Make gel for electrophoresis
Procedure:

1. Measure 1.2 g of Agarose powder and 120 mL of TEB buffer
2. Add the ingredient above into 250 mL flask and cover the flask with double layer
3. Pop a few holes in plastic wrap with pipette tip
4. Microwave liquid gel until no powder can be seen
5. Take out plastic wrap and wait until no smoke
6. Add 6μL of safety dye into liquid gel
7. Pour the gel into 3 assembled tray
8. Wait for them to solidify

Done by Sarah Chou, Sophie H

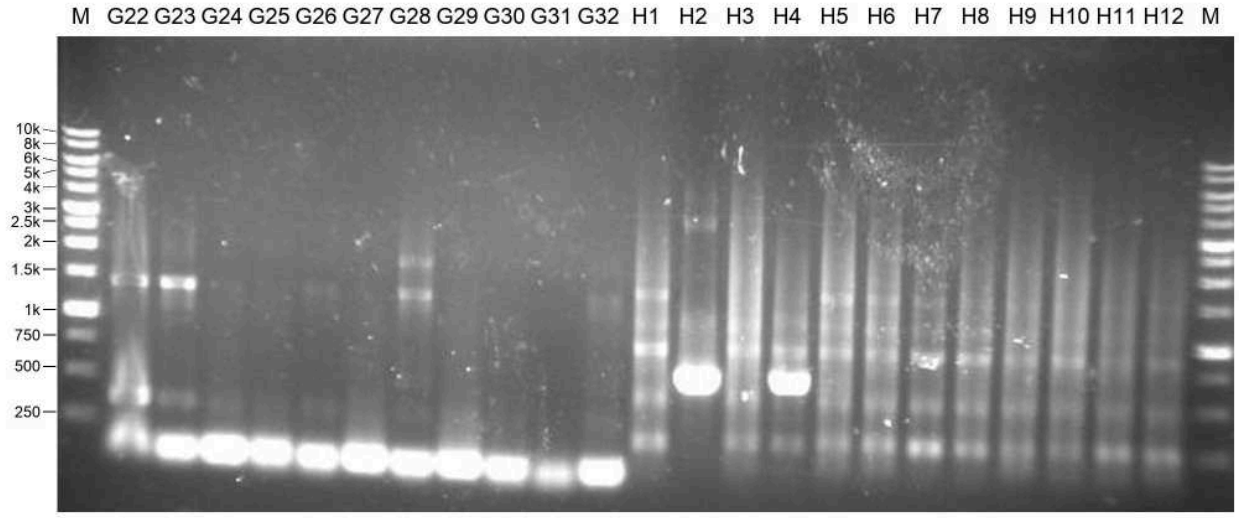
Date: 20260806
Title: Load gel for colony PCR
Purpose: Load gel for colony PCR product of G and H
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



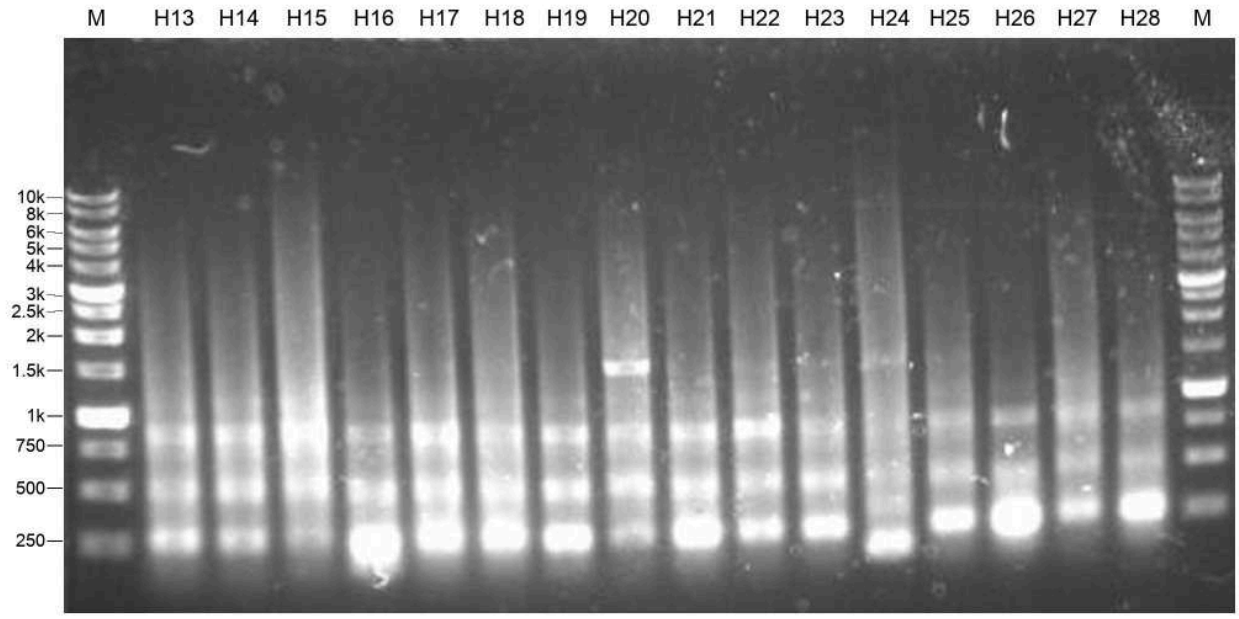
M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G1~21: PcpcG2-veg-EEBD-RBS0-csn: LEA-bkdb-cold double	biobrick prefix	LEA reverse	3.1 K

3. Load the gel in the layout shown below (5 μL each):
4. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G22~28: PcpcG2-veg-EEBD-RBS0-csn: LEA-bkdb-cold double	biobrick prefix	LEA reverse	
H1~H12: 3P-EEBD-RBS0-Csn:LEA-B0015	pSTK Exp1 Fw	pSTK Exp1 Rv	3.1 K

5. Load the gel in the layout shown below (5 μL each):
6. Resolve at 100 v for 30 min



M: 10kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
H13~H28: 3P-EEBD-RBS0-Csn:LEA-B0015	pSTK Exp1 Fw	pSTK Exp1 Rv	3.1 K

Date: 20260806
Title: make LBA plate with glycerol
Purpose: prepare LBA plate
Procedure:

1. Weight 250 mL ddH_2O and 6.25 agar powder*2
2. Add 1mL glycerol and put a stirring bar in
3. Stir the liquid mixture until no powder on glycerol can be seen
4. Cover the flask with double-layer aluminum foil
5. Autoclave for 20 min at 127 °C.
6. Add 250 µL ampicillin in the BSC while it's still a bit hot
7. Pour 20 mL into each plate and wait for it to solidify

Done by Sophie H

Date: 20260806
Title: Protein purification(take from 0805 filtered tubes)
Purpose: isolated ACCD protein
Procedure:

1. Use the following buffer for the procedures:(formula for 50mL and buffers are prepared/stocked beforehand)

Buffer	Imidazole final concentration (mM)	10xPBS (mL)	2M imidazole (pM)	Water (mL)
Equilibrate buffer	10	5	250	44.75
Wash buffer	25	5	625	44.375
Elute buffer	150	5	3750	41.25

2. Centrifuge the Ni-NTA column at 175 g for 2 min to remove storage buffer
3. Add 6mL of equilibrate buffer
4. Before adding supernatant, add and pipette 237.5 µl of imidazole to the tube of supernatant with 47.5mL volume and add 62.5 µl to the tube of supernatant with 12.5 mL volume
5. Adding all supernatant into the column by washing the walls(sides) of the column
6. Add 6mL of wash buffer and remove the flow-through. Repeat this step 3 times. *At the 3rd time, collect the flowthrough to load gel
7. Centrifuge at 175 g for 2 min
8. Collect the flow-through from the last wash buffer step
9. Add 3 mL of elute buffer
10. Centrifuge at 175 g for 2 min
11. Collect flow-through as purified protein

Note 1: did 2 tubes with volumes of 47.5 mL and 12.5 mL and store others in 4 °C
Note 2: method to store Ni-NTA column after use

1. Add 30 mL of Ni-NTA regenerate buffer(10x of resin's volume(3mL)) to the column to wash
2. Add 30 mL ddH_2O (10x) of resin's volume) to the column to wash
3. Add 20 % of ethanol in the column
4. Store at 4°C

Done by Abigail and Sophie C
Supervised by Gabriel

Date: 20260806
Title: SDS-PAGE electrophoresis
Purpose: separate proteins by size
Procedure:

1. Assemble casting frames with two 1.5 mm glass plates
2. Prepare solutions for the stacking(top) and resolving(bottom) gels

components	Stacker (top gel)	Resolver (bottom gel)
Resolver A	x	4.5 mL
Resolver B	x	4.5 mL
Stacker A	1.5 mL	x
Stacker B	1.5 mL	x
Total Volume	3mL	9 mL
TEMED	3 µl	4.5 µl
10 % APS	15 µl	45 µl

3. Transfer resolving gel between glass plates using a 1000 µl pipette
4. Add the stacking gel to the top of the green part(line) of the assembled rack ~6mL
5. Insert comb to form wells and allow it to set for ~10-15 min
6. Remove the comb and mount the gel into the electrophoresis tank
7. Mix 50 µl of protein sample (25mM and 150mM) with 10 µl of 6x Laemml: Sample Buffer(final volume's 1/6)
8. Heat samples at 95°C for 5 minutes
9. Briefly spin down
10. Fill the inner and outer chambers of the gel tank with 1x running buffer
11. Load the protein ladder in the first and last lane
12. Load prepared protein samples into remaining wells

Gel one(purpose: western blot):

	(0805)	(0805)	(0805)	(0805)	(0806)	(0806)	
Ladder	25 mM	10 mM	150 mM	250 mM	25 mM	150 mM	Ladder
10 µl	25 µl	25 µl	25 µl	25 µl	25 µl	25 µl	3 µl

Gel two(purpose: coomassie staining):

	(0805)	(0805)	(0805)	(0805)	(0806)	(0806)	
Ladder	25 mM	10 mM	150 mM	250 mM	25 mM	150 mM	Ladder
3 µl	25 µl	25 µl	25 µl	25 µl	25 µl	25 µl	10 µl

13. Close the lid and connect the electrodes
14. Run at constant voltage of 200 V for 10 min until the dye reaches the bottom of the stacking gel then continue electrophoresis at a constant voltage (100-150 V) for 40 min until the dye reaches near the bottom of the gel
15. Carefully disassemble the glass plates and collect the gel for subsequent analysis

Done by Abigail and Sophie C
Supervised by Gabriel

Date: 20260806
Title: Coomassie Blue Staining
Purpose: Visualize proteins in polyacrylamide gels after electrophoresis
Procedure:

1. Make the solutions and buffer according to the recipe below:

Coomassie blue solution recipe (40mL)	
methanol	50 %
Acetic acid	10 %

Coomassie blue destain buffer(40mL)	
methanol	30 %
Acetic acid	10 %

2. Transfer gel 2 to a staining tray
3. Wash gel with ddH_2O for 5 minutes, repeat 3 times
4. Stain the gel with 0.05 % Coomassie Blue R250 for 1 hour on the shaker at 50 rpm
5. Destain the gel by soaking in Coomassie Blue destain buffer for at least 1 hour until protein bonds are clearly visible against a transparent background
6. Document the gel image and compare the protein bands to the ladder

Done by Abigail and Sophie C
Supervised by Gabriel

Results

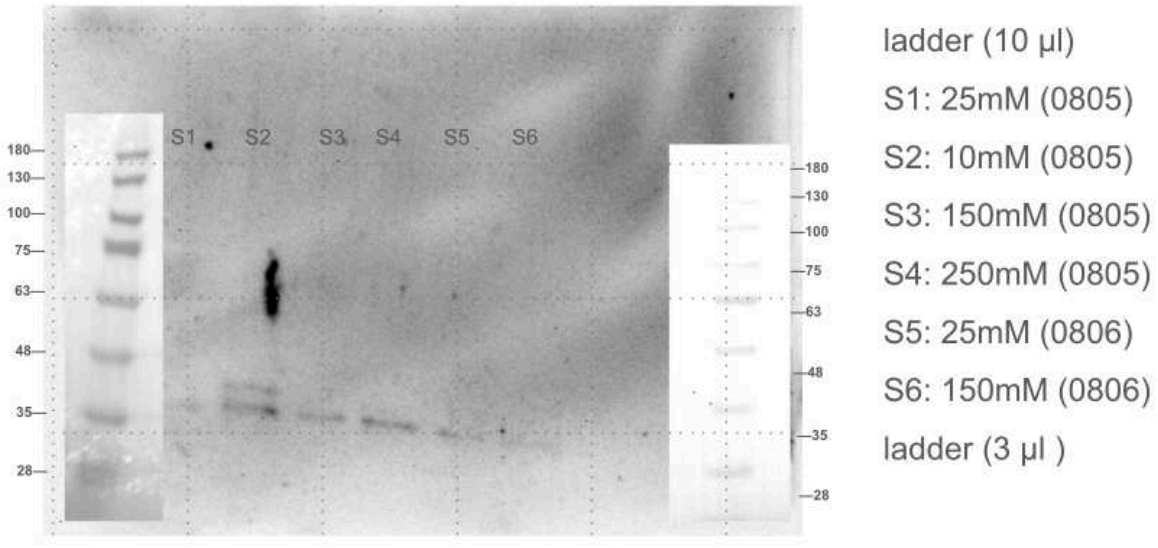


Date: 20260806
Title: Western Blotting
Purpose: to identify specific protein within a complex sample mixture using gel separation and antibody binding
Procedure:

1. After SDS-PAGE, carefully collect gel 1 for transfer
2. Soak the PVDF membrane in methanol for 1 minute.
2. Equilibrate the membrane in 1X transfer buffer for 3 minutes
3. Equilibrate the gel in 1X transfer buffer for 5 minutes
4. Wet 3 M papers in 1X transfer buffer
5. Assemble the transfer sandwich in the cassette in the following order: sponge pad → filter paper → gel → PVDF membrane → filter paper → sponge pad. Ensure there are no air bubbles between the gel and membrane
6. Place the cassette into the tank filled with cold 1X transfer buffer and an ice block
7. Transfer at 100 V for 90 minutes
8. Block in 5 % non-fat milk in TBST for 2 h at room temperature
9. Replace blocking solution by anti- His-tag mouse mAb (1:1000) In 5 % milk
10. Incubate on the orbital shaker at 4 °C overnight

Done by Sophie C and Abigail
Supervised by Gabriel

Results



Date: 20260806
Title: Protein Dialysis
Purpose: Remove imidazole which may interfere with future experiments
Procedure:

1. Dilute 10x PBS: Mix 160mL of 10x PBS and 1.6 L of deionized water
2. Adjust pH to 8
3. Top up H_2O to 1.6L and wrap opening of beaker with plastic wrap
4. Prepare three beakers of 1.6L PBS
5. Open the dialysis cassette by gently twisting the cap counterclockwise
6. Add the protein lysate to the cassette, slowly withdrawing the pipette while dispensing. Don't overload the cassette
7. Remove the excess air in the cassette by simultaneously pressing the membrane gently on both sides using gloved thumb and forefinger and insert the cap
8. Insert cap and lock by gently twisting it clockwise until the arrows align and a firm stop is felt
9. Place dialysis cassette in 2 L beaker with 1.6 L 1X PBS and stir bar. Stir on ice for 2 hours
10. Replace 1X PBS and stir on ice for another 2 hours
11. Replace 1X PBS and stir on ice overnight
12. Transfer lysate containing recombinant protein from dialysis cassette to 1.5 mL microtubes
13. The recombinant protein is stored at -80 °C

Done by Sophie C and Abigail
Supervised by Gabriel

Date: 20260806
Title: Transformation
Purpose: to amplify DNA part

Procedure:
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21
X: ACDI
V: ABDI

1. Set the water bath to 42 °C and prepare ice
2. Get 1 LBA and 2 LBC plates
3. Thaw competent cells until they are 50 % thawed and add 10 µl of plasmid to 100 µl of competent cell
4. Incubate on ice for 20 min
5. Heat shock 42°C for 45 seconds
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium
8. Incubate for 1 hr at 37°C
9. Centrifuge at 5000 g for 5 min
10. Remove 1 mL supernatant and vortex
11. Spread on plates
12. Incubate at 37 °C for 16hours

Done by Sophie C

Date: 20260806
Title: Add CH protectant on horizontal agar plates (exp set #7)
Purpose: test CH protectant
Procedure:

1. Use the following amount to apply for each plates CH biopepitde 1000µL (dillution 1/500-> 6986 µL of sterile water +14µL of peptide)
2. Apply the protectant for the following plants
    - Aftertreatment
        - 100 mM NaCL , CH biopeptide (Day 1)
        - 0,CH biopeptide (Day1)

Done by Sophic C and Naomi
Supervised by Abigail

Date:20260807
Title: Colony PCR
Purpose: Amplify desired DNA sequence
Procedure:
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21:

1. Prepare the following reaction

L *40

	40 reactions	1 reaction
Fw primer Exp1	40 µl	1 µl
RV primer Exp1	40 µl	1 µl
Q amp master mix	400 µl	10 µl
ddH_2O	320 µl	8 µl

2. Picked up colonies with tips and streak on grid
3. Wait 5 minutes and take out tip
4. Put in thermal cycle

95°C	5 min	1 cycle
95°C	20 sec	35 cycles
65°C	20 sec	
72°C	3 min 30 sec	
72°C	5 min	1 cycle

Done by Sarah, Sophie L, Sophia Y, and Sophie H

Date:20260807
Title: Golden Gate Assembly of G, H, L, X, V
Purpose: Assemble level 2 construct
Procedure:

G: PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015
H: 3P-EEBD-RBS0-Csn:LEA-B0015
L: 3P-EEBD-RBS0-GanA-Gala-L3S2P21
X: A+C+D+I (1:1)
V: A+B+D+I (1:1)

1.

G	
Part	**Amount**

Vector 1D (132ng/μL, 3120bp)	1.5μL (1/10)
PcpcG2-veg (168ng/μL, 235bp)	1.175μL (1/50)
EEBD-RBS0 (151ng/μL, 2392bp)	1.04μL (1/5)
Csn:LEA (212ng/μL, 634bp)	0.76μL (1/20)
B0015 (107ng/μL, 2213bp)	1.34μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.4μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

H	
Part	**Amount**
Vector Exp1 (162ng/μL, 6929bp)	1.4μL (1/5)
3P (106ng/μL, 2909bp)	1.43μL (1/9)
EEBD-RBS0 (150ng/μL,2392bp)	1.04μL (1/5)
Csn:LEA (212ng/μL, 634bp)	0.76μL (1/20)
B0015 (107ng/μL, 2213bp)	1.34μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.3μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

L	
Part	**Amount**
Vector Exp1 (162ng/μL, 6929bp)	1.4μL (1/5)
3P (106ng/μL, 2909bp)	1.43μL (1/9)
EEBD-RBS0 (150ng/μL,2392bp)	1.04μL (1/5)

GanA-Gala (177ng/μL, 4198bp)	1.22μL (1/9)
L3S2P21 (170ng/μL, 2195bp)	1.64μL (1/5)
T4 DNA ligase buffer	2μL
Sterile ddH_2O	10.3μL
T4 DNA ligase	1μL
Bsa1-HFv2	1μL

X	
Part	**Amount**
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
C(98ng/μL,3828bp)	1.27μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3757bp)	0.74μL
T4 DNA ligase buffer	2μL
Sterile ddH_2O	11.06μL
T4 DNA ligase	1μL
BsmBI	1μL

V	
Part	**Amount**
Vector Exp2 (163ng/μL, 5104bp)	1.017μL
A (109ng/μL, 3754bp)	1.119μL
C(98ng/μL,3828bp)	1.484μL
D (134ng/μL, 3243bp)	0.786μL
I (165ng/μL, 3757bp)	0.74μL
T4 DNA ligase buffer	2μL

Sterile ddH_2O	10.85μL
T4 DNA ligase	1μL
BsmBI	1μL

Done by Sophie C, Chloe

Date: 20260807
Title: Liquid culture of ABDI
Purpose: bacteria culture from successful colony PCR product
Procedure:

1. Label liquid culture tube
2. Add 5mL LB medium and 25μL chloramphenicol to the tube
3. Dip a pipette on a colony
4. Dispense the tip into tube
5. Place the tube into the incubator at an angle

Done by Sarah, Sophie L

Date: 20260807
Title: Liquid culture of B-sub WBS00
Purpose: bacteria culture from successful colony PCR product
Procedure:

1. Label liquid culture tube
2. Add 5mL LB medium to the tube
3. Dip a pipette on a colony
4. Dispense the tip into tube
5. Place the tube into the incubator at an angle

Done by Sophie L, Sarah

Date: 20260807
Title: Make 250mL LBC with glycerol agar
Purpose: Make LBC agar plate
Procedure:

1. Weight 250mL dd H_2O and 9.25g LB agar powder
2. Add the ingredient above in 500mL flask and pu ta stirring bar
3. Stir the mixture until all the powder were fully dissolved
4. Cover the flask with double layer aluminum foil and put a autoclave sticker on it as well
5. Autoclave for 20 minutes at 127°C.
6. Put the flask into BSC after incubating (or put in 55 ℃ water bath if not pour shortly after incubating)
7. Add 250μL chloramphenicol into the liquid gel and gently mix it
8. Pour 20mL of liquid agar into each plate
9. Wait for it to solidify (about 40min later)

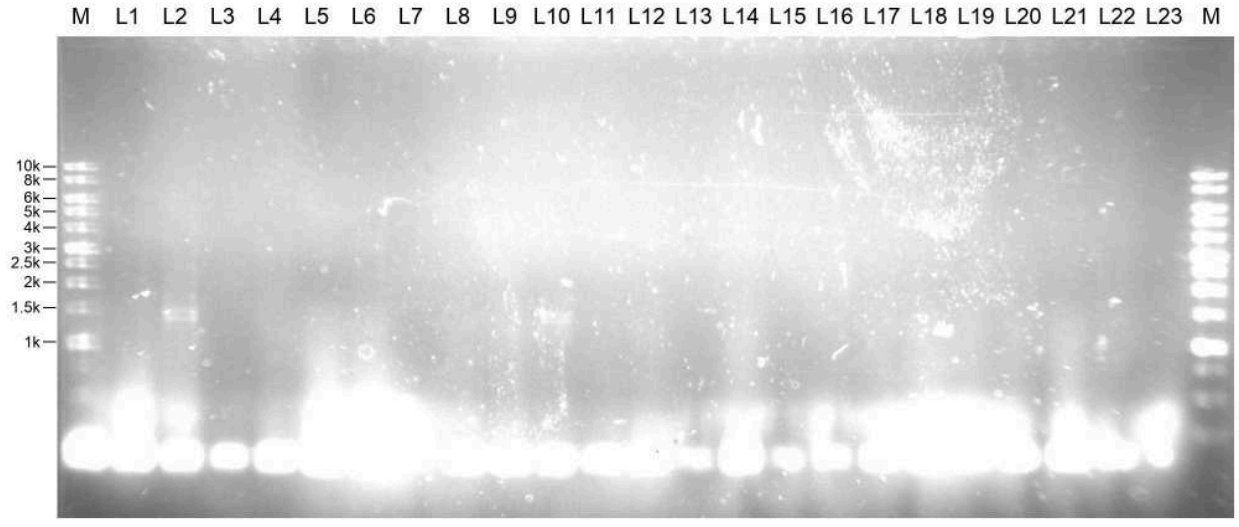
Done by Sarah, Sophie L, Sophie H.

Date: 20260807
Title: Make 0.8 % Agarose gel for two 25well big tray
Purpose: Make gel for electrophoresis
Procedure:

1. Weight 0.64 g agarose powder and 80 mL TAE buffer
2. Add the ingredient above in 250 mL flask and cover the flask with double layer plastic wrap
3. Poke some holes on plastic wrap with pipette tip
4. Microwave the flask until all the powder were fully dissolved
5. Take the plastic wrap off and wait till it stop smoking
6. Add 4μL of diluted safety dye and mix it
7. Pour the gel into 2 assembled tray evenly
8. Use pipette tip to press the air out and pop bubbles
9. Wait for it to solidify
   Done by Sarah, Sophie H, Sophie L.

Date: 20260807
Title: Electrophoresis of L
Purpose: Check the bp of colony PCR product
Procedure:

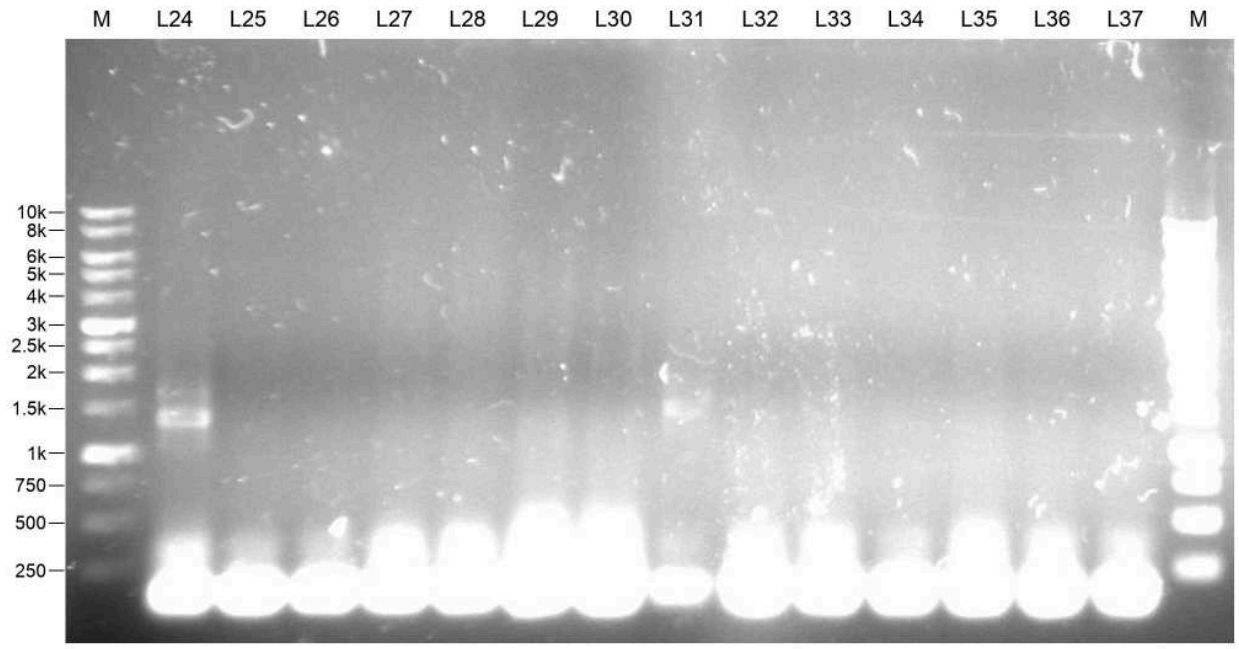
1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1Kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

L: 3P-EEBD-RBS0-GanA-Gala-L3 S2P21	Exp-sq Fw	Exp-sq Rv	3516

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1Kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L: 3P-EEBD-RBS0-GanA-Gala-L3 S2P21	Exp-sq Fw	Exp-sq Rv	3516

Done by Sarah, SophieL, Sophie H

Date:20260808
Title: Miniprep
Purpose: Isolate and purify Level 2 GGA product
Procedure:
ABDI:
PtagC-RBS-Ho1-pcyA-L3S2P21+Pliag-RBS0-ccaSm3-B0015+Pveg-RBS0-ccaR-B0015+PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Transfer 1000 µL liquid culture to 1.5 ml microtube
2. Centrifuge at 16,000g for 1min, discard supernatant
3. Repeat step 1&2 until all medium are collected
4. Add 200 µL of MX1 buffer
5. Resuspend by pipetting or vortexting
6. Add 250 µL of MX2 buffer
7. Invert slowly 8-10 times (don't pipette or vortex!), and incubate at RT for 2min
8. Add 350 µL of MX3 buffer
9. Centrifuge at 23444 g for 5 min
10. Transfer 800 µL of supernatant to column
11. Centrifuge at 7000 g for 30 sec and discard the flow through
12. Add 500 µL of WN buffer, centrifuge at 7000 g for 30 sec
13. Centrifuge at 7000 g for 30 sec and discard the flow through
14. Centrifuge empty column for 8 min at 10000 g and place column into 1.5 mL tube
15. Add 50 µL EB buffer at the center of the column
16. Centrifuge at 23444 for 5 min

	Concentration (ng/µL)	A260/230	A260/280
ABDI	66.00	1.222	1.886

Done by Sarah, Sophie H.

Date:20260808
Title: B Sub electrocompetent cells prep(WN800)
Purpose: electroporation of foreign DNA into a protease-deficient strain to safely express and secrete recombinant proteins.
Procedure:

1. Dilute overnight culture of B Sub WN800 (312.5 µl) in 16 fold in growth medium(5mL) and grow at 37 °C to an OD600 of
2. Cool the cells on ice cold water and put on shaker for 10 minutes
3. Centrifuge at 4 °C at 5000g for 5 minutes to harvest cells
4. Add 1 mL ice cold electroporation medium to resuspend cells in 1.5mL microtube centrifuge 5000g for 1 minute and discard supernatant. Repeat 4 times to wash cells

5. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.125 mL) with a cell concentration of 1-1.3x10^{10} cfu/mL
6. Store competent cell at -80 °C

Done by Chloe, Abigail, Sophie C

Date:20260808
Title: PCR
Purpose: rapidly copy and amplify a specific DNA sequence
Procedure:

1. Prepare the following reactions

Qx2

reagent	1rxn	2rxn
10x ExTAQ buffer	5μL	10μL
2.5uM DNTP mix	4μL	8μL
Fw primer(CcaSm3)	1μL	2μL
Rv primer(CcaR)	1μL	2μL
DNA temp(V)	1μL	2μL
PCR-grade H_2O	37.75μL	75.5μL
Ex-taq poly	0.25μL	0.5μL

R

reagent	1rxn
10x ExTAQ buffer	5μL
2.5uM DNTP mix	4μL
Fw primer(Csn)	1μL
Rv primer(AccD)	1μL
DNA temp(V)	1μL
PCR-grade H_2O	37.75μL
Ex-taq poly	0.25μL

S

reagent	1rxn

10x ExTAQ buffer	5μL
2.5uM DNTP mix	4μL
Fw primer(Ho1)	1μL
Rv primer(pcyA)	1μL
DNA temp(V)	1μL
PCR-grade H_2O	37.75μL
Ex-taq poly	0.25μL

2. Cycle

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	2.5 min	
72°C	1 min	1 cycle

Done by Sophie C, Chloe

Date:20260808
Title: make 1 % agarose gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray, mold, comb
2. Weigh 0.2 g agarose powder and measure 20 mL 1xTAE buffer
3. Add all to a conical flask
4. Cover with plastic wrap&puncture holes
5. Microwave for 30 sec Agitate. Repeat till no sediments
6. When no smoke, add 1μL safety dye
7. Pour into mole
8. Let solidify at RT

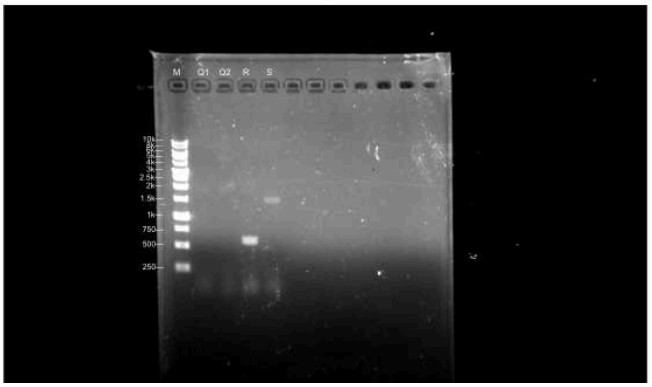
Done by Sophie C

Date:20260808
Title: Gel electrophoresis of PCR

Purpose: check the PCR result
Procedure:

1. Load samples in the following order
   M Q1 Q2 R S
   5 5 5 5 5
2. Resolve the gel at 100V for 30min

0808 Level 2 V PCR



M: 1kb DNA ladder

Q1-2: ABDI (CcaSm3 F, CcaR R)
- bp: 2626

R: ABDI (Csn F, AccD R)
- bp: 1195

S: ABDI (Ho1 F, pcyA R)
- bp: 1549

Done by Sophie C

Date:20260808
Title: Transformation (G,H,L,X,V)
Purpose: amplification DNA part
Procedure:

1. Set up water bath at 42 °C and prepare ice
2. Get 3 LBA and 2 LBC plates
3. Thaw competent cells till they're 50 % thawed and add 10 µL plasmid to 100 µL competent cell
4. incubate on ice for 20 min
5. Heat shock at 42 °C for 45 sec
6. Incubate on ice for 3 min
7. Add 1mL SOC medium
8. Incubate for 1 hr at 37 °C
9. Centrifuge at 5000 g for 5min
10. Remove 1 mL of supernatant and vortex
11. Spread on their respective plates
12. Incubate 37 °C for 16 hr

Done by Sophie C, Ryan W, Chloe, Abby T, Ryan Y

Date:20260808
Title: Colony PCR
Purpose: Amplify the number of desired plasmid

Procedure:

pSTK-Exp1-P3p:RBS0-EEBD-GanA

1. Prepare the Master mix

	1 rxn	25 rxn
pSTK-Exp-seq fw	1 µL	25 µL
pSTK-Exp-seq rv	1 µL	25 µL
ddH_2O	8 µL	200 µL
Q-amp master mix	10 µL	250 µL

2. Thermal cycle

95°C	5min	1 cycle
95°C	30 sec	30 cycles
65°C	40 sec	
72°C	3min 30sec	
72°C	5 min	1 cycle

Done by Chars

Date: 20260809
Title: Colony PCR of G,H,X,V
Purpose: Amplify the number of desired plasmid
Procedure:
G:PcpcG2-veg-EEBD-RBS0-csn:LEA-B0015
H:3P-EEBD-RBS0-csn:LEA-B0015
X:ACDI
v:ABDI

1. Prepare the master mix

X:ACDI

	1 rxn	14 rxn
CcaR fw	1 µL	14 µL
PcpcG2-veg rv	1 µL	14 µL
Q-amp master mix	10 µL	140 µL
ddH_2O	8 µL	112 µL

v:ABDI

	1 rxn	14r xn
CcaR fw	1 µL	14 µL
PcpcG2-veg rv	1 µL	14 µL
Q-amp master mix	10 µL	140 µL
ddH_2O	8 µL	112 µL

G

	1 rxn	8 rxn
Csnfw	1 µL	8 µL
LEA rv	1 µL	8 µL
Q-amp master mix	10 µL	80 µL
ddH_2O	8 µL	64 µL

H

	1 rxn	2 rxn
Csnfw	1 µL	2 µL
LEA rv	1 µL	2 µL
Q-amp master mix	10 µL	20 µL
ddH_2O	8 µL	16 µL

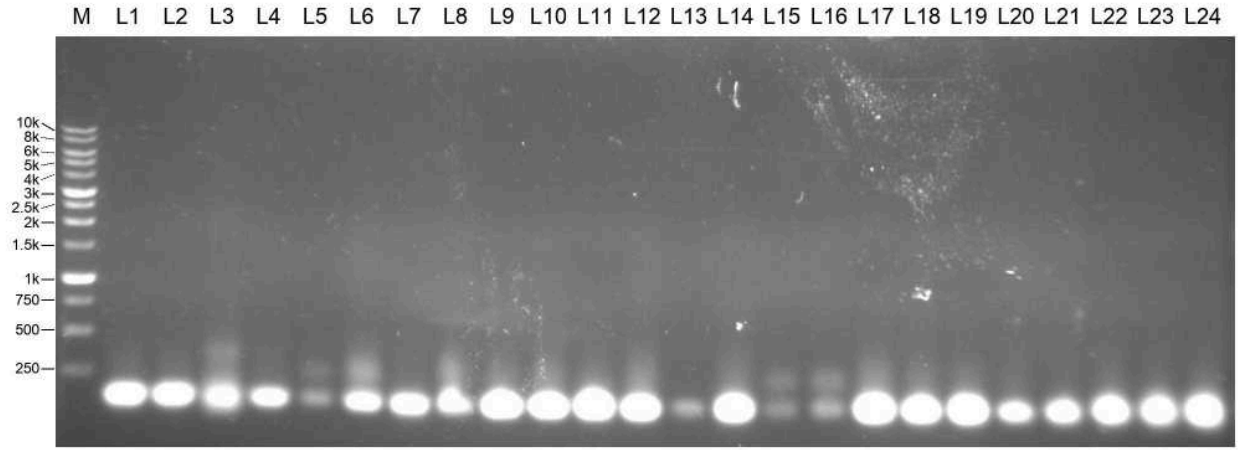
2. Thermal cycle

95 °C	5 min	1 cycle
95 °C	20 sec	35 cycles
65 °C	20 sec	
72 °C	1 min 30 sec	
72 °C	5 min	1 cycle

Done by Abigail, Chloe W
Supervised by Chars

Date: 20260809
Title: Gel electrophoresis of Colony PCR
Purpose: check the PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L1-L24: pSTK-EXP1-P3p:RBS0-EEBD-GanA-Gala	pSTK-EXP-seq fw	pSTK-EXP-seq rv	3516

Date: 20260809

Title: Measure Arabidopsis fresh weight

Purpose: To determine which experimental set is best for Arabidopsis growing

Procedure:

1. Prepare seven 1.5 mL tube and label information on the agar plate (vernalized, incubated, planted, and the date of measuring fresh weight)
2. Open the electronic scale and tare the scale after putting a labeled 1.5 mL tube inside.
3. Take out the tube
4. Carefully take out all the Arabidopsis from the corresponding agar plate (careful not to break the root) and place them inside the tube
5. Put the tube back onto the scale and measure fresh weight
6. Wait until the weighting value stops changing
7. Took the tube out and label the fresh weight on the tube
8. Repeat step 2~7 until all agar plate sets were finished measuring
9. Store the tubes in -80 °C and wait for extraction

Done by Naomi, Sophia L, Sophie H

Results:

Group	Freshweight (in grams)	Number of arabidopsis
Control (0,0)	0.1537	23

Pretreatment (w/ 100mM NaCl)	0.0203	24
Pretreatment (no salinity)	0.1013	23
Cotreatment (w/ 100mM NaCl)	0.0168	25
Cotreatment (no salinity)	0.0406	25
Aftertreatment (w/ 100mM NaCl)	0.0234	25
AFtertreatment (no salinity)	0.1830	24

Date: 20260809

Title: Measure OD of hay infusion

Purpose: To observe if there is bacillus subtilis growth in hay

Procedure:

1. Take out the incubated centrifuge tubes(incubated overnight)
2. Disinfect all equipment and move it into the BSC for aseptic operation
3. Transfer 200 µl from each centrifuge tube into a separate well of a 96-well plate
4. Transfer 200 µl of LB(blank) and filtered hay infusion(blank) into another two wells
5. Place the 96-well plate into the OD microplate reader(Elisa) and record the OD600 result
6. Repeat steps 1-5 for each hour for a total of 7 hours

Hour	LB	Hay infusion
1	0.083	0.080
2	0.090	0.081
3	0.163	0.079
4	0.366	0.081
5	0.353	0.081
6	0.469	0.079
7	0.355	0.081
Blank	0.038	0.065

Done by Sophia L and Abigail

Date: 20260809

Title: Vernalize exp#7 trial 2 seeds

Purpose: Vernalize new seeds for the second trial.

Procedure:

1. Open the BSC
2. Use the seed template (0.8 % agar circular plates) to plant 40 seeds on each agar plates
3. Use breathable tape to seal each of the agar plates
4. Label the vernalization date, incubator date and seed type
5. Wrap the agar plates with aluminum foil
6. Vernalize in the 4°C refrigerator for 4 days

Date: 20260809

Title: agar experiment set 7 fresh weight

Purpose: measure the arabidopsis weight

Procedure:

1. Set the scale by weighing the empty tube
2. Use a tweezers to pick Arabidopsis from each agar plate
3. Slightly press the plants on tissue paper to remove agar
4. Put all arabidopsis into a tube labelled with their experiment group
5. Weigh the tube with collected arabidopsis and record the fresh weight
6. Repeat steps 1-5 until all experiment groups are collected

Result:

group	freshweight(g)	# of arabidopsis
Control (0,0)	0.1587	23
Pretreatment (without 100mM NaCl)	0.0203	24
pretreatment	0.1013	23
Contreatment (without 100mM NaCk)	0.0168	25
cotreatment	0.0406	25
After treatment (without 100mM NaCl)	0.0234	25
After treatment	0.1830	24

Done by Naomi, Sophia L, Sophie H

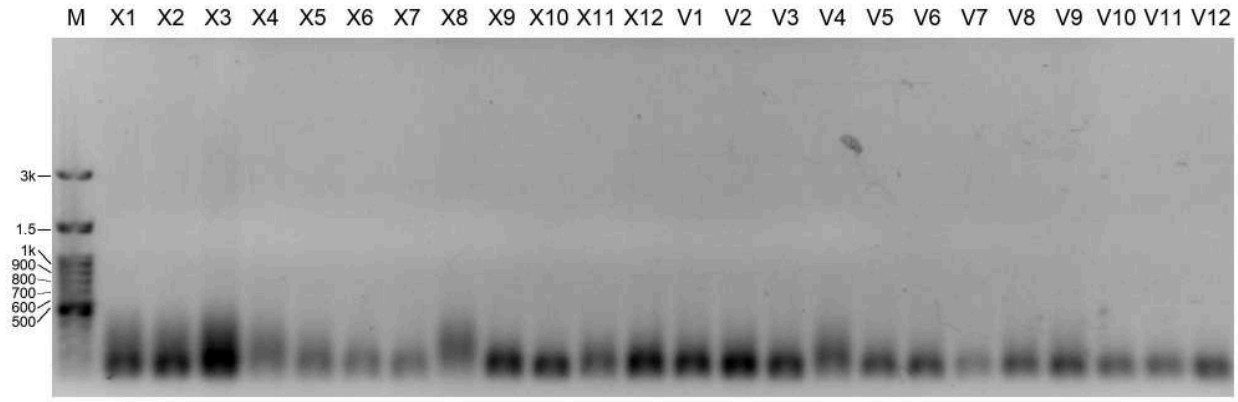
Date: 20260810
Title: Make 1 % gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray,comb,mold
2. Weigh 0.8 g agarose powder& 80 mL 1XTAE buffer
3. Add all in conical flask
4. Wrap with plastic wrap and puncture holes
5. Microwave for 30 sec Agitate.Repeat till no sediments
6. When there is no smoke, add 4 µL safety dye. Agitate
7. Pour melted gel into mold
8. Let it solidify at RT

Done by Sophie C

Date: 20260810
Title: Electrophoresis
Purpose:Check PCR result
Procedure:

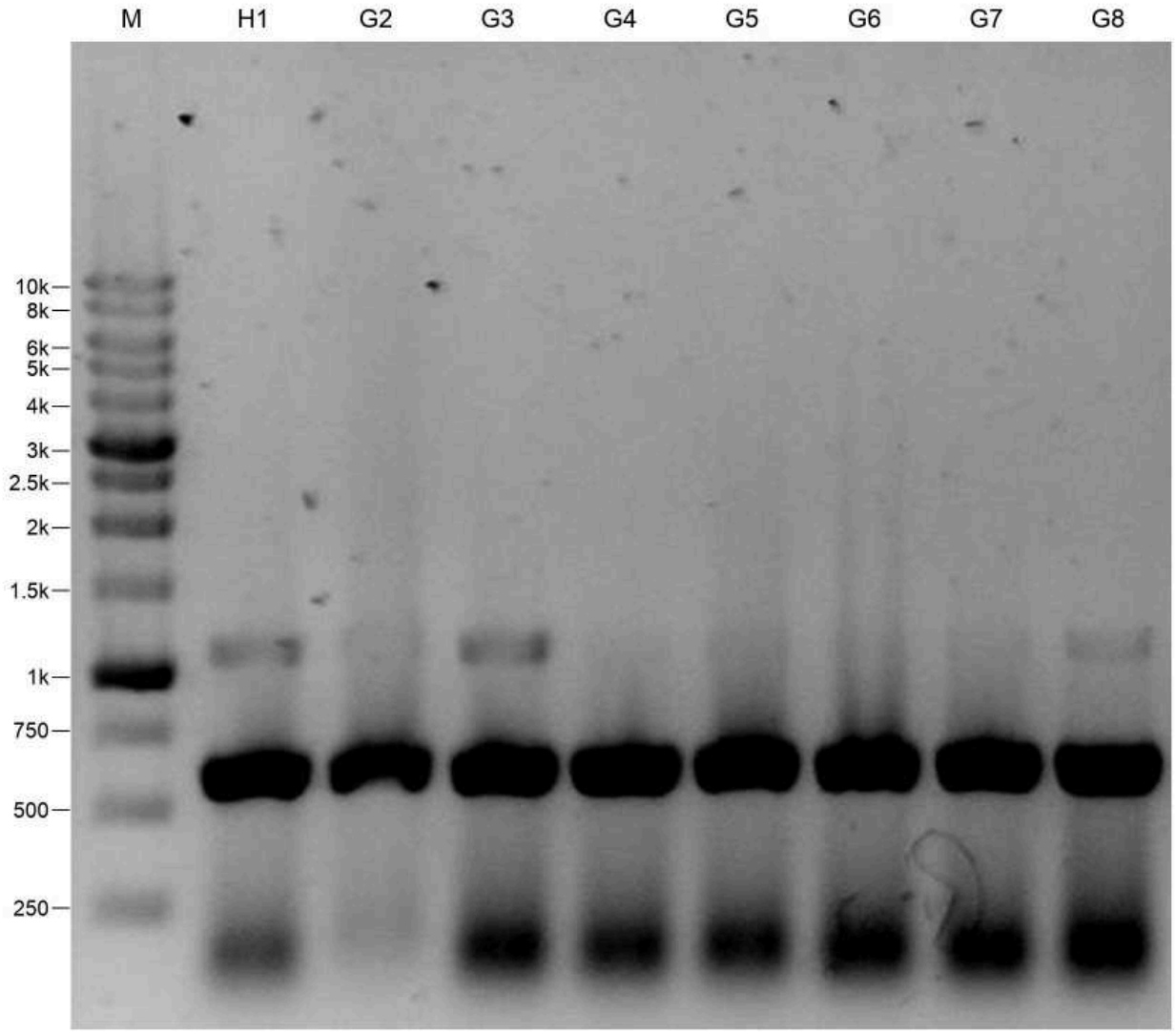
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X: ACDI	CcaR fw	PcpcG2-veg rv	1032
V: ABDI	CcaR fw	PcpcG2-veg rv	1032
Note: All fail -> RE-GGA			

3. Load the gel in the layout shown below (5 µL each):

4. Resolve at 100 v for 30 min



M: 10k DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G: PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015	Csn fw	Lea rv	584
H: 3P-EEBD-RBS0-Csn:Lea-B0015	Csn fw	Lea rv	584

Note: Liquid culture 8/10: H1, G4, G6

Done by Sophie C

Date: 20260810
Title: Colony PCR of L
Purpose:Amplify the number of desired plasmid
Procedure:

1. Prepare the following master mix

Lx25

Exp-seq fw	25μL
Exp-seq rv	25μL
Q-amp master mix	250μL
ddH_2O	200μL

2. Pick up colonies with tips& streak on grids
3. Wait 5 min , take out tips
4. Thermal cycle

95°C	5min	1cycle
95°C	20sec	35cycle
60°C	20sec	
72°C	3min 30 sec	
72°C	5min	1cycle

Done by Sophie C, Chloe

Date: 20260810
Title: Electroporation
Purpose:allowing foreign molecules like DNA to enter the cells
Procedure:
J:pSTK-Exp1-Pveg-EEBD-RBS0-csn:AccD-B0015
X:pSTK-Exp2-Ptag+RBS-H01:PcyA-L3S2P21-PrepA-RBS0-CcaSm3-B0015-Pveg-RBS0-CcaR-B0015-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015

1. Prewarm the sOC medium at 37°C and pre-cool a 0.2cm electroporation curette
2. Mix 15.2μL plasmid in 100μL electrocompetent cell in a 1.5mL microtube, and mix 2.4μL plasmid (400ng/μL)J in 100μL electrocompetent cell
3. Set up electroporator and voltage resistance
    - V:2.3Kv
    - Resistance :300Ω
    - Capitance: 35F
4. Transfer mixture of cells &DNA to electroporation curette&tap suspension to bottom
5. Place in shock pad, close lid&wait until beep
6. Immediately remove form shock pad and add 1mL SOC medium
7. Transfer to liquid culture tube and incubate at 37°C for 3 hours, shaking at 200rpm
8. check&record parameters
9. Centrifuge at 5000g for 5min
10. Spread LB plate with erithmycin

J

TC	6.3
V	2284

X

TC	5.5
V	2282

Done by Chloe, Sophie C

Date: 20260810
Title: Colony PCR
Purpose:Amplify the number of desired plasmid
Procedure:
pSTK-Exp1-3P-EEBD-RBS0-GanA-Gala-L3S2P21

1. Prepare the following mixture

	1 rxn	24 rxn
pSTK-Exp-seq fw	1 μL	24 μL
pSTK-Exp-seq rv	1 μL	24 μL
Q-amp master mix	10 μL	240 μL
ddH_2O	8 μL	192 μL

2. Apply the following thermal cycle

95 °C	5 min	1 cycle
95 °C	20 sec	35 cycles
60 °C	20 sec	
72 °C	3 min 30 sec	
72 °C	5 min	1 cycle

Done by Chloe, Sophie C

Date: 20260810
Title: Golden Gate Assembly
Purpose: Assemble level 1 parts to form level 2 parts
Procedure:
X:ACDI
V:ABDI

1. Prepare the following reaction

X

Part	amount
Exp2 (163 ng/μL,5104 bp)	1.017 μL
A:PtagC-RBS-H01:pcyA-I3S2P21 (109 ng/μL,3754 bp)	1.119 μL
C:PlepA-RBS0-CcaSm3-B0015 (98 ng/μL,3828 bp)	1.27 μL
D:Pveg-RBS0-CcaR-B0015 (134 ng/μL,3243 bp)	0.786 μL
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165 ng/μL,3757 bp)	0.74 μL
T4 DNA ligase buffer	2 μL
Sterile water	11.06 μL

T4 DNA ligase	1 μL
BsmBI	1 μL

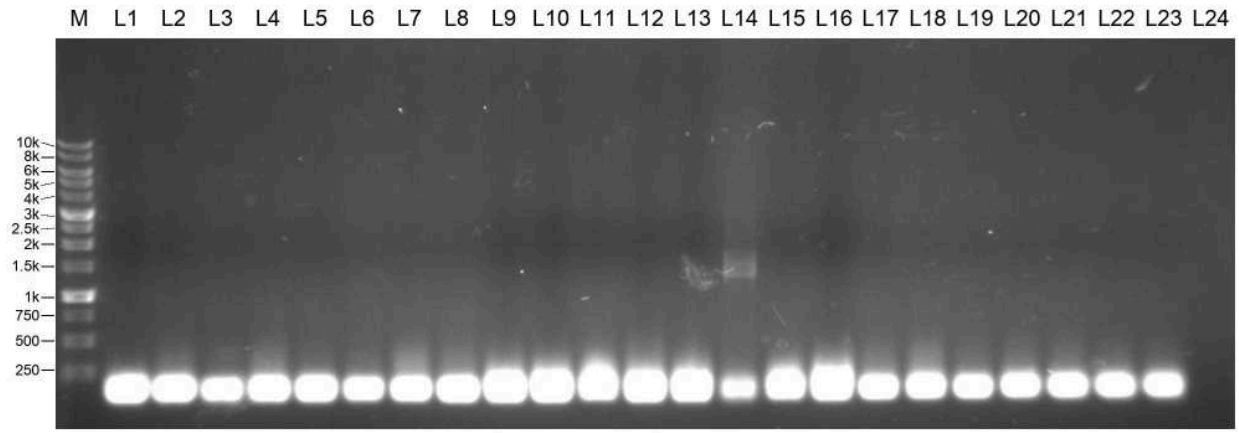
V

Part	amount
Exp2 (163 ng/μL,5104 bp)	1.017 μL
A:PtagC-RBS-H01:pcyA-l3S2P21 (109 ng/μL,3754 bp)	1.119 μL
B:PliaG-RBS0-CcaSm3-B0015 (83 ng/μL,3792 bp)	1.484 μL
D:Pveg-RBS0-CcaR-B0015 (134 ng/μL,3243 bp)	0.786 μL
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165 ng/μL,3757 bp)	0.74 μL
T4 DNA ligase buffer	2 μL
Sterile water	10.85 μL
T4 DNA ligase	1 μL
BsmBI	1 μL

Done by Chloe

Date: 20260810
Title: Gel electrophoresis
Purpose: Checking PCR result
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1k ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L: 3P-EEBD-RBS0-GanA-Gala-L3 P2S21	pSTK-EXP-seq-fw	pSTK-EXP-seq-rv	3516

Done by Chloe

Date: 20260810
Title: Liquid culture
Purpose: bacteria culture form colony PCR product
Procedure:

1. Label 6 liquid culture tubes
2. Add 3 mL TB medium and 3 μL of ampicillin in culture tubes
3. Dip a pipette on a colony
4. dispense the tip into tube
5. Place the tube into the incubator (slanted angle)

Done by Sophie C

Date: 20260810
Title: 1/10 MS medium preparation (1000mL)
Purpose: bacteria culture form colony PCR product
Procedure:

1. Weight the ingredient below
2. Add the ingredient above in 800mL distilled water
3. Add a stirring bar and stir it
4. Add 200mL distilled water
5. Adjust the pH of the medium until it reach 5.7~5.8 (ori: 3.75)
6. Separate the medium into 2 500mL flask

7. Cover flask with double layer aluminum foil and stick autoclave tape
8. Autoclave for 20 minutes at 127° C.

Done by Sophie H

Date: 20260810

Title: agar (exp set #7 trial 1) chlorophyll extraction

Purpose: to isolate green pigment from plant

Procedure:

1. Take the microtubes form 0809
2. Flute acetone to 80 % add 4mL to each tube
3. Add to 10 metal bits into each tube
4. Wait for chlorophyll to extract, and use a vortex to crush the plants to speed up the process
5. Add liquid to 96 wells, 200 µL in each well. Three wells for each sample
6. ELISA reader, Run results through equation:

	F4	F5	F6	F8	F9	F10	G5	G6	G7	G8	G9	G10	H5	H6	H7	H8	H9	H10
C	P100	P0	CO	A100	AO	P100	P0	C100	CO	A100	A0	P100	PO	C100	CO	A100	AO	

7. Correct wavelength

A663
A645

Done by Sophie C, Audrey H
Supervised by Gabriel

Date: 20260810

Title:Arabidopsis soil planting (tray 5 & tray6)

Purpose: planting arabidopsis

Procedure:

1. Autoclave a bag of soil
2. Seak 8 flower pot in 1/10 bleach and wait for 20 min
3. Rinse the put with water for 5 min (until there is no bleach smell)
4. Fill the put with full soil and slightly press it down
5. Gently shake it vertically to soften the soil
6. Water the soil until the upper part of the soil become fully wet
7. Use tip to pick up a seed from preservation solution
8. Dispense the seed with solution on the soil in fixed position (6 seeds in a pot)
    - Make sure not to cover the seed or move it
9. After all the pots are done, cover the pot with plastic , wrap and place them in incubator.

Done by Sophie H
Supervised by Gina

Date: 20260811
Title: Colony PCR (of J in *Bacillus subtilis*)
Purpose: Amplify the number of desired plasmid
Procedure:

1. Prepare master mix as follow X2:

	1 reactions	2 reaction
Fw primer (Exp-seq-fw)	1 µl	2 µl
RV primer Exp1 (Rv primer)	1 µl	2 µl
Qamp master mix	10 µl	20 µl
ddH_2O	8 µl	16 µl

2. Distribute 20µL each into each PCR tube
3. Pick up colonies with tip and streak grid
4. Dispense in PCR tube of master mix
5. Set up thermal cycle and run the machine:

95°C	5 min	1cycle
95°C	20 sec	
55°C	20 sec	35 cycles
72°C	2min 20sec	
72°C	5 min	1 cycle

Done by Sophie C.

Date: 20260811
Title: Miniprep
Purpose: Collect concentrated liquid culture
Procedure:

1. Transfer 1000 µL liquid culture in 1.5 mL microtube
2. Centrifuge for 30 sec at 16000 g , discard the supernatant
3. Repeat step 1&2 until all the liquid were collected
4. Add 200 µL of MX1 buffer
5. Resuspend the pellet by pipetting
6. Add 250 µL of MX2 buffer
7. Invert 8-10 times to mix slowly
8. Incubate at RT for 2 min
9. Add 350 µL of MX3 buffer

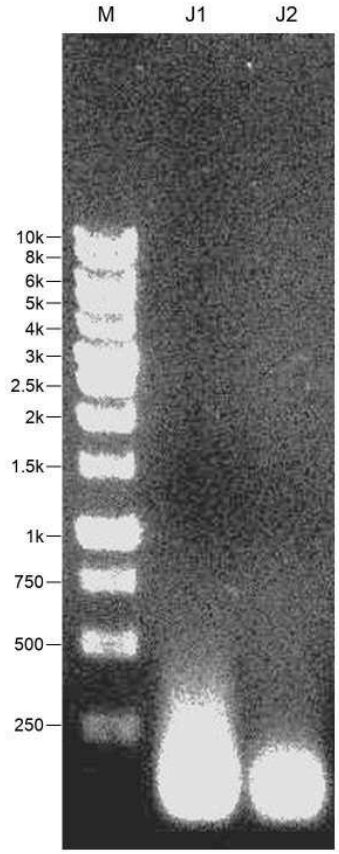
10. Invert 4-5 times to mix gently
11. Centrifuge at 23444 g for 5 minute
12. Assemble the column and tube
13. Transfer the supernatant (~800µL) to the column
14. Centrifuge the column at 7000 g for 30 sec
15. Discard the flow-through
16. Add 500 µL of WNfuvver
17. Centrifuge the column at 7000 g for 30 sec
18. Add 700 µL of WS buffer
19. Discard the flow-through
20. Centrifuge empty column at 10000 g for 3 min
21. Place column into 1.5 mL tube and add 50 µL of EB buffer to center of column membrane
22. Incubate at RT for 3 mins
23. Centrifuge at 10000 g for 3 min

	concentration(ng/µL)	A260/230	A260/180
G4	295	1.724	2.095
G6	187	1.558	2.033
H2	198	1.503	2.010

Done by Sophia Y, Sophie C

Date: 20260811
Title: Gel Electrophoresis
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
J1, J2: Pveg-EEBD-RBS0-Csn:ACCD-B0015	Exp-sq-Fw	ACCD Rv	1868

Date: 20260811
Title: Transformation
Purpose: to cultivate and amplify DNA parts
Procedure:

1. Set up water bath at 42 ℃
2. Thaw competent cells until 50 % melted, add 10 μL plasmid in 100 μL competent cell
3. Ice bath for 20 min
4. Heat shock for 45 sec
5. Ice bath for 3 min
6. Add 1 ml soc medium
7. Incubate 1 hr at 37 ℃
8. Centrifuge 5000 g for 5 min

9. Take out 900 µL supernatant
10. Resuspend pellet
11. Spread on LBC plate
12. Incubate for 16 hr

Done by Sophie C.

Date: 20260811
Title: Electroporation
Purpose: Introduce DNA sequence into competent cells
Procedure:

1. Prewarm the SOC medium at 37 ℃ and pre-cool a 0.2cm electroporation cuvette
2. Mix 15.2µL plasmid in 100µL electroporation cell in a 1.5mL microtube, and mix 2.4µL plasmid (410ng/µL0 J in 100µL electrocompetent cell
3. Set up electroporator and voltage resistance:

V: 2.3kv
Resistance: 300Ω
Capitance: 35F

4. Transfer mixture of cells & DNA to electroporation cuvette & tap suspension to bottom
5. Place in shock pad, close lid and wait until you hear the beeping sound
6. Immediately remove from shock pad and add 1mL SOC medium
7. Transfer to liquid culture tube and incubate at 37 ℃ for 3 hr, shaking at 200rpm
8. Check and record parameters
9. Centrifuge at 5000g for 5min
10. Spread LB plate with erithmycin

Done by Sophie C
Recorded by Sophie H.

Date: 20260812
Title: Hay Infusion
Purpose: Observe the growth of *Bacillus subtilis* in different hay infusion lake water
Procedure:

1. Use an electronic pipette controller with a serological pipette to draw 15mL of hay infusion
2. Transfer the 15 mL into a syringe fitted with a filter.
3. Filter the hay infusion and collect it in a 15 mL tube.
4. Replace the filter with a new one if it becomes clogged and difficult to press down.
5. Repeat step 1 to 3 for the three hay infusions with different lake water.
6. Disinfect all equipment and move it into the BSC for aseptic operation.
7. Use a new serological pipette to transfer 5 mL of the filtered water into a new syringe fitted with a sterile filter.
8. Filter the filtered hay infusion into a centrifuge tube.
9. Use a pipette to add 200 µL of *Bacillus subtilis* into the centrifuge tube
10. Repeat step 6 and 7 for the three filtered hay infusions with different lake water.
11. Place the three centrifuge tubes in the centrifuge overnight.

Results:

Source of hay infusion water	OD of blank(without *Bacillus subtilis*)	OD of *Bacillus subtilis* after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Done by Sophia

Date: 20260812
Title: Colony PCR
Purpose: Amplify engineered DNA plasmid
Procedure:
J: Pveg-EEBD-RBS0-Csn:ACCD-B0015(*Bacillus subtilis*)
V: ABDI
X: ACDI

1. Make the following reactions:

J

	1 reactions	8 reaction
Fw primer (CsnFw)	1 µl	8 µl

RV primer Exp1 (ACCD Rv)	1 µl	8 µl
Qamp master mix	10 µl	80 µl
ddH_2O	8 µl	64 µl

X,V

	1 reactions	27 reaction
Fw primer (CcaR Fw)	1 µl	27 µl
RV primer Exp1 (pcpcG2-veg Rv)	1 µl	27 µl
Qamp master mix	10 µl	270 µl
ddH_2O	8 µl	216 µl

2. Distribute in PCR tubes
3. Dip pipette in colonies
4. Streak grid
5. Dispense tip in PCR tubes
6. Close lid
7. Run through thermal cycle:

95°C	5 min	1cycle
95°C	20 sec	
55°C	20 sec	35 cycle
72°C	2min	
25°C	5 min	1 cycle

Done by Sophie C, Abby T

Date: 20260812
Title: Make 1 % Gel
Purpose: Make Gel for electrophoresis of colony PCR product
Procedure:

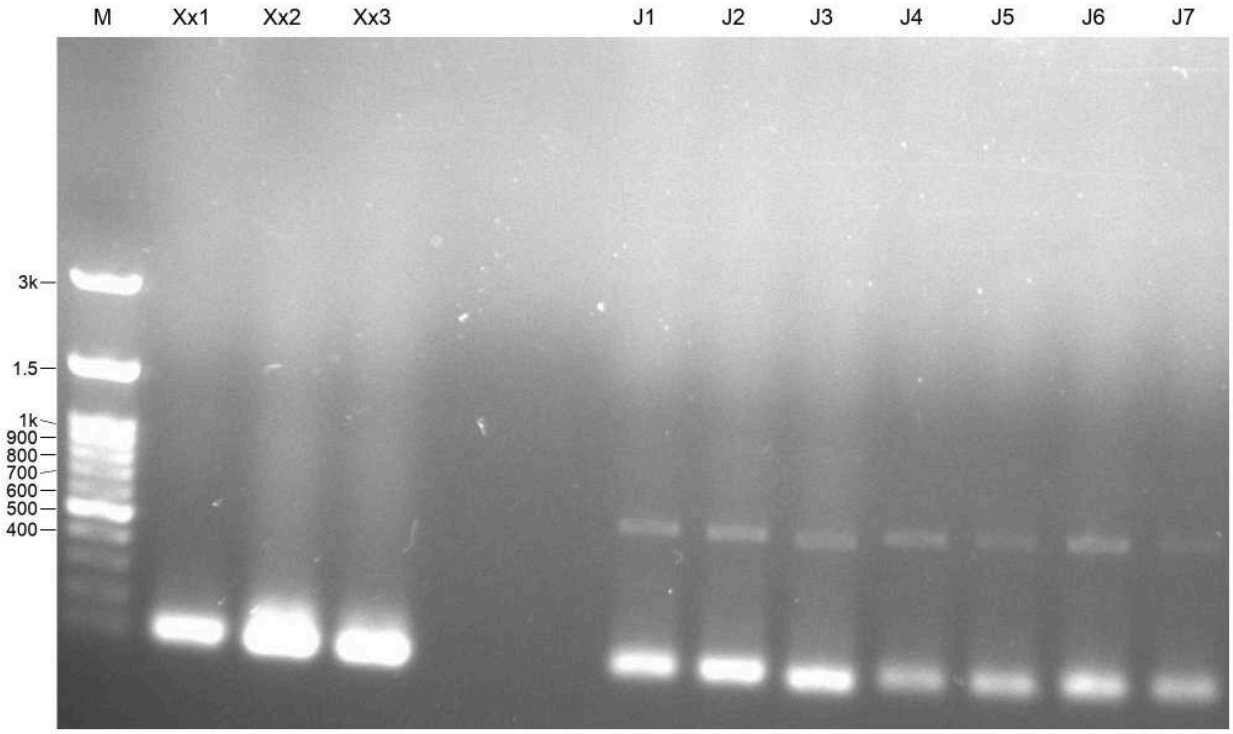
1. Assemble tray, comb, and mold
2. Weight 0.8 g agarose powder and 80 mL 1X TAE buffer
3. Add in 250 mL conical flask
4. Wrap the flask with plastic warp and puncture holes on it with tip
5. Microwave until the powder fully dissolved

6. When there's no smoke, add 4μL safety dye and agitate
7. Pour the melted gel in the mold
8. Let it solidify at room temperature

Done by Chloe, Abby T

Date: 20260812
Title: Gel Electrophoresis
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min

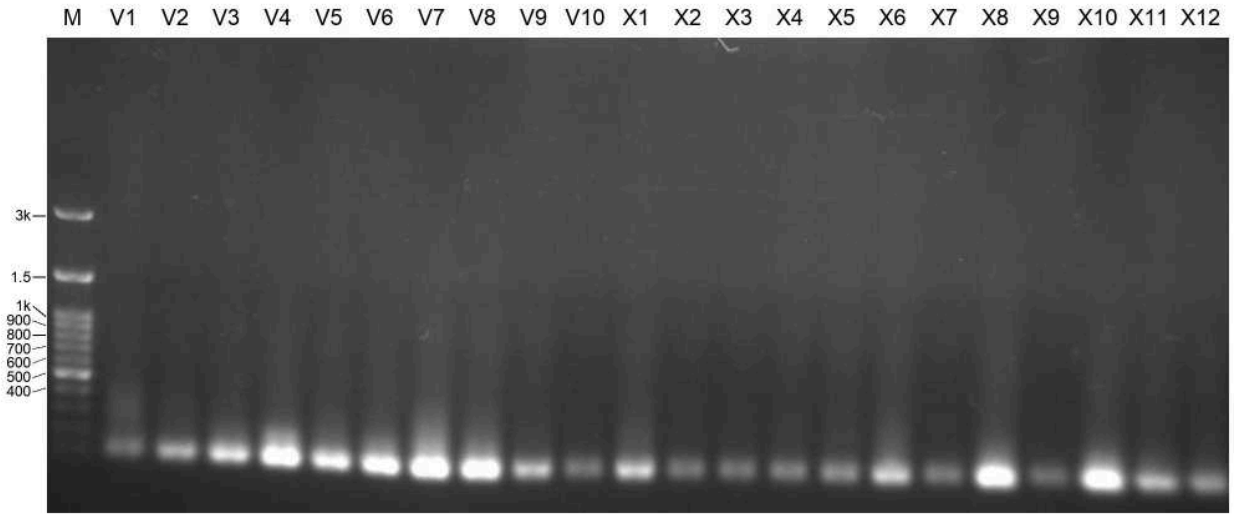


M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X: ACDI (plepA, accd)(for amplification)	ccar fw	pcpcG2-veg rv	1032
J: Pveg-EEBD-RBS0-csn:ACCD-B0015 (electroporation)	csn fw	ACCD reverse	around 615

Done by Abby T

Date: 20260812
Title: Gel electrophoresis
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X: ACDI (plepA, accd)	ccar fw	pcpcG2-veg rv	1032
V: ABDI (pliag, accd)	ccar fw	pcpcG2-veg rv	1032

Done by Chloe

Date: 20260812
Title: Liquid Culture
Purpose: Transfer colonies on agar to medium
Procedure:

1. Prepare liquid culture by the following table:

Bacteria	Liquid Culture	Antibiotics

WB800	5mL LB	None
pSTK-0 CX-Csnfrom glycerol stock	5mL LB	chloramphenicol 5μL
pSTK-Exp1-Csn:ACCD #J1 in *Bacillus subtilis* WB800	5mL LB	Erytheromycin 25μL
pSTK-Exp1-Csn:ACCD #J2	5mL LB	Erytheromycin 25μL
pSTK-Exp1-Csn:ACCD #J6	5mL LB	Erytheromycin 25μL

2. Incubate in 37 ℃ at 200rpm for 16 hr

Done by Chars

Date: 20260812
Title: L0 Golden Gate Assembly
Purpose: Assemble parts
Procedure:

1. Prepare the following reaction:

1. pSTK-0C-SamyQ-Bopep4-2.1

Part	amount
STK001-pSTK-0-sfGFP (346ng/μL, 3090bp)	1.16μL (1/20)
SamyQ-Bopep4-2.1 (10ng/μL, 180bp)	1.17μL (1/5)
T4 DNA ligase buffer	2μL
Sterile water	13.67μL
T4 DNA ligase	1μL
Bbs1-HF	1μL
Total:	20μL

2.pSTK-0C-SamyQ-Bopep4-2.2

Part	amount
STK001-pSTK-0-sfGFP (346ng/μL, 3090bp)	1.16 μL (1/20)
SamyQ-Bopep4-2.1 (10ng/μL, 180bp)	1.17 μL (1/5)

T4 DNA ligase buffer	2 µL
Sterile water	13.67 µL
T4 DNA ligase	1 µL
Bbs1-HF	1 µL
Total:	20 µL

3.pSTK-0C-SamyQ-Csn-Lea

Part	amount
STK001-pSTK-0-sfGFP (346ng/µL, 3090bp)	1.16 µL (1/20)
Csn-Lea (PCR product,212ng/µL, 634bp)	0.97 µL (1/50)
T4 DNA ligase buffer	2 µL
Sterile water	13.87 µL
T4 DNA ligase	1 µL
Bbs1-HF	1 µL
Total:	20 µL

Done by Sophie C.

Date: 20260812

Title: plant new set of seeds for hydroponic system

Purpose: prepare new seed for hydroponic system

Procedure:

1. Prepare 0.8 agar 1/10 MS medium & microtube holder
2. Spray the tip holder with ethanol & shine under UV light for 30 min
3. Fill 48 wrack of the tip holder with 2.2 mL of 1/10 MS medium
4. Microwave the agar until it melts, and pour 250µL agar into each 48 eppenthrf lids
5. Wait for them to solidify
6. Move the lids into respective wrack
7. Use autoclaved tooth pick to pick up one seed and add the seed onto the lid
8. Repeat step 7 until all the lid have the seed
9. Wrap the tube which alumium foil
10. Place the tube wrack into 4 ℃ refrigerator
11. Wait for 4 days for seeds to vernalize

Done by Sophie H

Date: 20260812

Title:Hay infusion

Purpose: observe the growth of *Bacillus subtilis* in different lake water

Procedure:

1. Use an electronic pipette controller with a serological pipette to draw 15mL of hay infusion
2. Transfer the 15 mL into a syringe fitted with a filter
3. Filter the hay infusion and collect it in a 15mL tube
4. Replace the filter with a new one if it become clogged and difficult to press down
5. Repeat step 1 to 3 for the three hay infusions with different lake water
6. Disinfect all equipment and more it into BSC for aseptic operation
7. Use a new serological pipette to transfer 5mL of the filtered water into a new syringe filtered with a sterile filter
8. Filter the filtered hay infusion into a centrifuge tube
9. Use a pipette to add 200μL of *Bacillus subtilis* into the centrifuge tube
10. Repeat step 6 and 7 for the tree filtered hay infusions with different lake water
11. Place the three centrifuge tubes in the centrifuge overnight, start from 18:00

Result:

Source of hay infusion water	OD of blank (without *Bacillus subtilis*)	OD of B.sub after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Done by Sophia L, Abigail

Date: 20260812

Title:fortilize plants

Purpose: fortilize plants

Procedure:

1. Add 0.5g of fertilizer into 500mL of water(1g/1000mL)
2. Add the solution to the plant trays

Done by Alex

Date: 20260813
Title: Hay Infusion
Purpose: Compare the OD of hay infusion with and without *Bacillus subtilis*
Procedure:

1. Take out the three centrifuge tubes from the centrifuge
2. Transfer 200 μL from each centrifuge tube into a separate well of a 96-well plate.
3. Disinfect all equipment and move it into the BSC for aseptic operation.
4. Transfer 200 μL from the 0812 filtered hay infusion into a separate well of a 96-well plate.
5. Place the 96-well plate into the OD microplate reader and record the result.

Results**:**

Source of hay infusion water	OD of blank(without *Bacillus subtilis*)	OD of *Bacillus subtilis* after one day
淡水	0.054	0.042
新竹	0.043	0.055
大湖	0.064	0.064

Done by Sophia L, Abigail

Date: 20260813
Title: PCR
Purpose: Amplify plasmid number
Procedure:
H: Pveg-Csn:LEA

1. Prepare the following reaction

	1 reaction
Forward primer (EEBD F)	1 μL
Reverse primer (LEA R)	1 μL
Q-amp master mix	10 μL
ddH_2O	7 μL
DNA template	1 μL

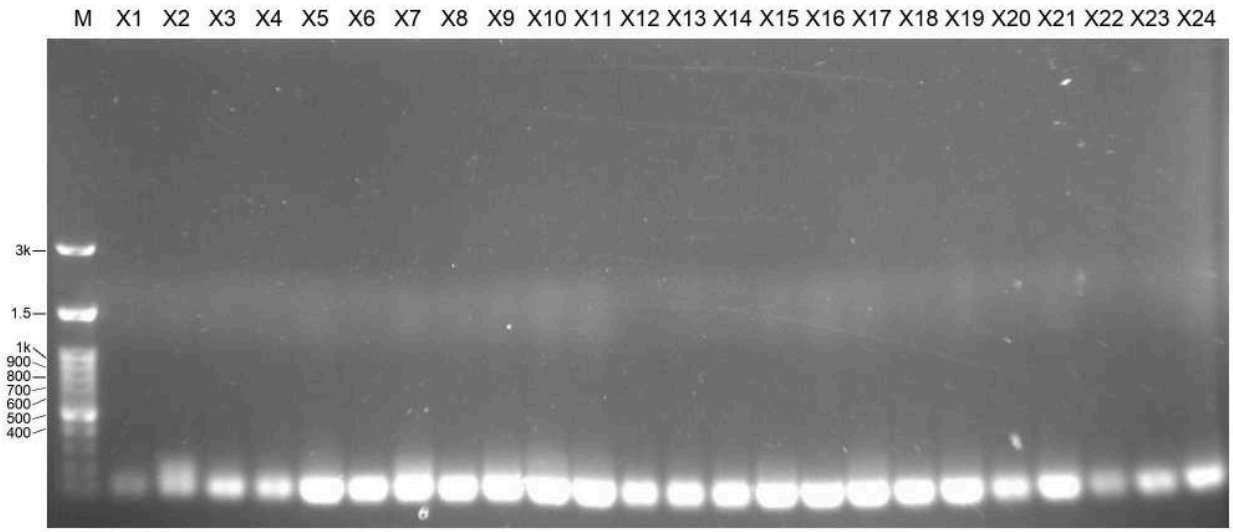
2. Put in PCR machine for thermal cycle

95°C	2 min	1 cycle
95°C	20 sec	35 cycles
60°C	20 sec	
72°C	1 min 20 sec	
25°C	5 min	1 cycle

Done by Sophie C.

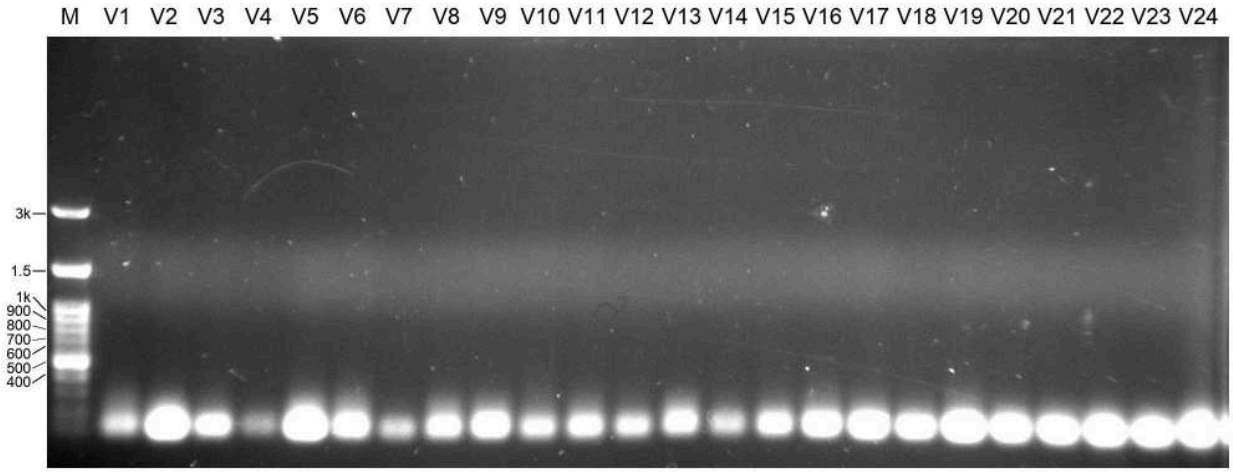
Date: 20260813
Title: Gel electrophoresis (colony pcr)
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X1-24: ACDI	CcaR F	PcpcG2-veg R	1032
Note: -> all failed(re-gga)			

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min

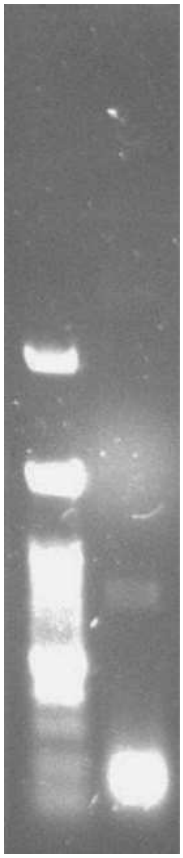


M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
V1-24: ABDI	CcaR F	PcpcG2-veg R	1032
Note: -> all failed(re-gga)			

Done by Sophie C.

Date: 20260813
Title: Gel electrophoresis (PCR)
Purpose: Check the approximate length of PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
H1: Pveg-EEBD-RBS0-Csn:Lea-B0015	EEBD F	LEA R	735

Done by Sophie C.

Date: 20260813
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:

Samples:
F10: Pveg-EEBD-RBS0-Csn:AmilCP-B0015

1. Transfer 1000 µL of liquid culture to 1.5 mL microtube.
2. Put in a centrifuge and centrifuge at 16000 g for 1 minute.
3. Discard the supernatant.

4. Repeat steps 1–3 until all cultures are collected.
5. Add 200 µL of MX1 buffer.
6. Resuspend by pipetting and vortexing.
7. Add 250 µL of MX2 buffer.
8. Invert slowly for 8–10 times, incubate at RT for 2 mins.
9. Add 350 µL of MX3 buffer and invert slowly for 4–5 times.
10. Centrifuge at 23444 g for 5 minutes.
11. Transfer 880 µL of supernatant to column.
12. Centrifuge at 7000 g for 30 seconds.
13. Discard the flow through.
14. Add 500 µL of WN buffer to the column.
15. Centrifuge at 7000 g for 30 seconds.
16. Discard the flow through.
17. Add 700 µL of WS buffer to the column.
18. Centrifuge at 7000 g for 30 seconds.
19. Discard the flow through.
20. Centrifuge the empty column for 8 mins. at 10000 g.
21. Place column into 1.5 mL tube.
22. Add 50 µL EB buffer to the center of the column membrane.
23. Incubate at RT for 3 mins.
24. Centrifuge at 10000 g for 3 mins.
25. Measure the concentration by using Biodrop

Done by Ryan W.

BioDrop Results:

Sample	Concentration (ng/µL)	A260/A280	A260/A230
F10	81	1.929	1.227

Date: 20260813
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Set the water bath to 42 °C and prepare ice
2. Thaw 100 µL competent cells until they’re 50 % thawed and add 10µL (L0 GGA 20260812) of plasmid
3. Incubate on ice for 20 min
4. Heat shock at 42 °C for 45 seconds
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37 °C
8. Centrifuge at 5000 g for 5 min
9. Remove supernatant, leave around <100µL

10. Resuspend pellet
11. Spread on LBC plate
12. Incubate at 37 °C for 16 hr

Done by Sophie C.

Date: 20260813
Title: Golden Gate Assembly (L)
Purpose: To assemble the parts for L
Procedure:

Sample: pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21

1. Add the following

Part	Original	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP concentration: 162 ng/µl 6929 bp	0.2780154321	1.4 µl (dilute 1/5)
Insert promoter(3P) concentration: 106 ng/µl 2909bp	0.3567641509 µL	1.43 µL (dilute 1/4)
Insert RBS (EEBD-RBS0) concentration: 151 2392 bp	0.2059337748	1.03 (dilute 1/5)
Insert CDS(GanA-Gala) concentration: 177 4148 bp	0.3046553672	1.22 (dilute 1/4)
Insert Terminator (L3S2P21) concentration: 170 2145 bp	0.1640294118 µL	1.64 µL (dilute 1/10)
STK331_pSTK-0x-Cansecretion tag 2188 bp 81 ng/µL	0.3511604938 µL	1.07 µL (dilute 1/10)
Histag (pSTK-OCZ-HisGX) ● Concentration (ng/µL): 135	0.2028962963 µL	1 µL (dilute 1/5)

• bp: 2107		
T4 DNA ligase buffer	2 µL	2 µL
Sterile water	7.21 µL	
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Done by Sophie C.

Date: 20260813
Title: Golden Gate Assembly (V)
Purpose: to assemble the parts for V
Procedure:
Sample: V:A+B+D+I

1. Assemble the following

V	
Part	Amount
Vector Exp2 (163ng/µL, 5104bp)	1.017 µL
A (109ng/µL, 3754bp)	1.119 µL
C(98ng/µL,3828bp)	1.484 µL
D (134ng/µL, 3243bp)	0.786 µL
I (165ng/µL, 3757bp)	0.74 µL
T4 DNA ligase buffer	2 µL
Sterile ddH_2O	10.85 µL
T4 DNA ligase	1 µL
BsmB1	1 µL

Done by Sophie C, Sarah, Abigail

Date: 20260813
Title: Add Zhenhan protectant on horizontal agar(trial #2)
Purpose: test zhenhan protectan

Procedure:

1. Use the following amount & apply on 4 pretreatment plant
    - [1mL per plate]
    - [1/500 dilution -> 30 µL of CH protectant with 14.97 mL ddH_2O]
2. Evenly distribute liquid protectant by tilting the plate sideways until all plants are converted with a thin layer of protectant
3. Seal plates with tape and label date of pretreatment

Done by Chloe
Supervised by Abigail

Date: 20260813
Title: hay infusion
Purpose: compare the OD of hay infusion with and without *Bacillus subtilis*
Procedure:

1. Take out the three centrifuge tubes from the incubators
2. Transfer 200µL from each centrifuge tube into a separate well of a 96-well plate
3. Disinfect all equipment and move it into the BSC for aseptic operation
4. Transfer 200µL from the 0812 filtered hay infusion into a separate well of a 96-well plate
5. Place the 96-well plate into the OD microplate reader and record the result

Done by Sophia L, Aibgail

Date: 20260814
Title: Colony PCR (Q1, Q2, S)
Purpose: Amplify engineered DNA plasmid
Procedure:

1. Add the following (2 master mix)

pSTK-0C-SamyQ-BoPep4-2.1
pSTK-0C-SamyQ-BoPep4-2.2

	1 reaction	35 reactions
Forward primer (STK fw)	1μL	35 μL
Reverse primer (STK rv)	1μL	35 μL
Q-amp master mix	10μL	350 μL
ddH_2O	8μL	280 μL

pSTK-0C-Csn:Lea

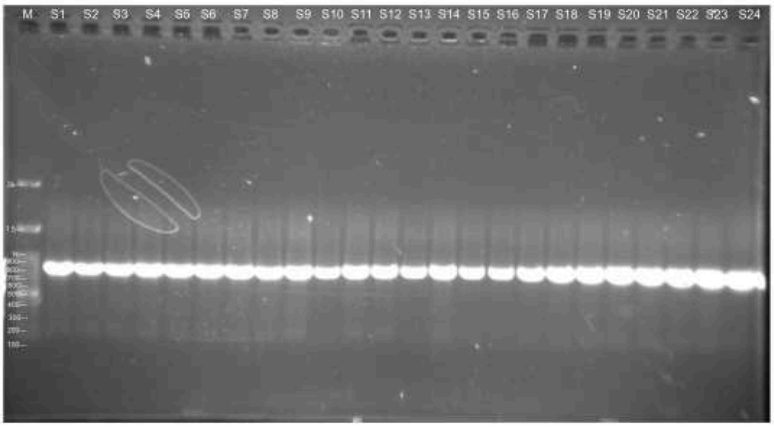
	1 reaction	26 reactions
Forward primer (Csnfw)	1μL	26 μL
Reverse primer (LEA rv)	1μL	26 μL
Q-amp master mix	10μL	260 μL
ddH_2O	8μL	208 μL

2. Circle 12 colonies from plate G and M, 16 colonies from L
3. Pick up colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
65 ℃	40 sec	
72 ℃	3 min	
72 ℃	1 min	1 cycle

Results:

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

S1-S24

pSTK-0C-Csn-Lea

Csn F & STK-seq-rv

Bp: 785 bp

S2, S10, S16

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

Q2(5)-Q2(13)

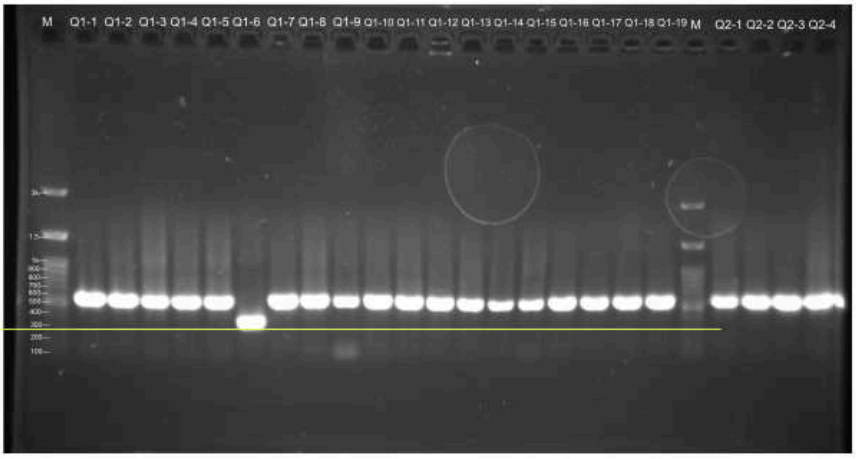
pSTK-0C-SamyQ-Bopep4 -v2t2

STK-seq-fw & STK-seq-rv

Bp: 536 bp

Q2-5, Q2-7, Q2-13

0814 Colony PCR (Q1)



M: M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder

Q1(1)-Q1(19)

pSTK-0C-SamyQ-Bopep4-v2t1

STK-seq-fw & STK-seq-rv

Bp: 536 bp

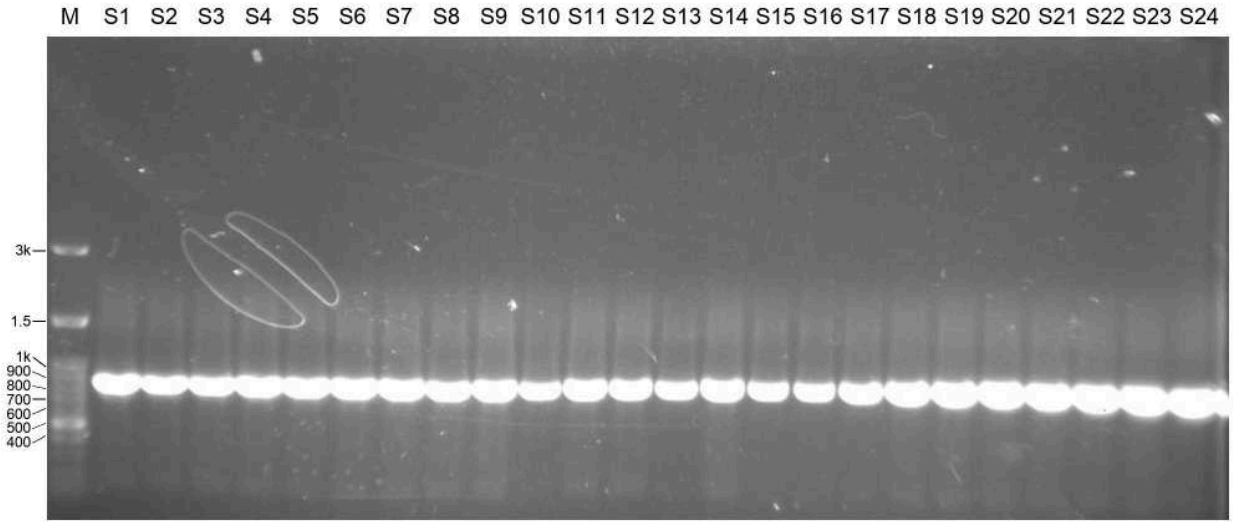
Q2(1)-Q2(4)

pSTK-0C-SamyQ-Bopep4-v2t2

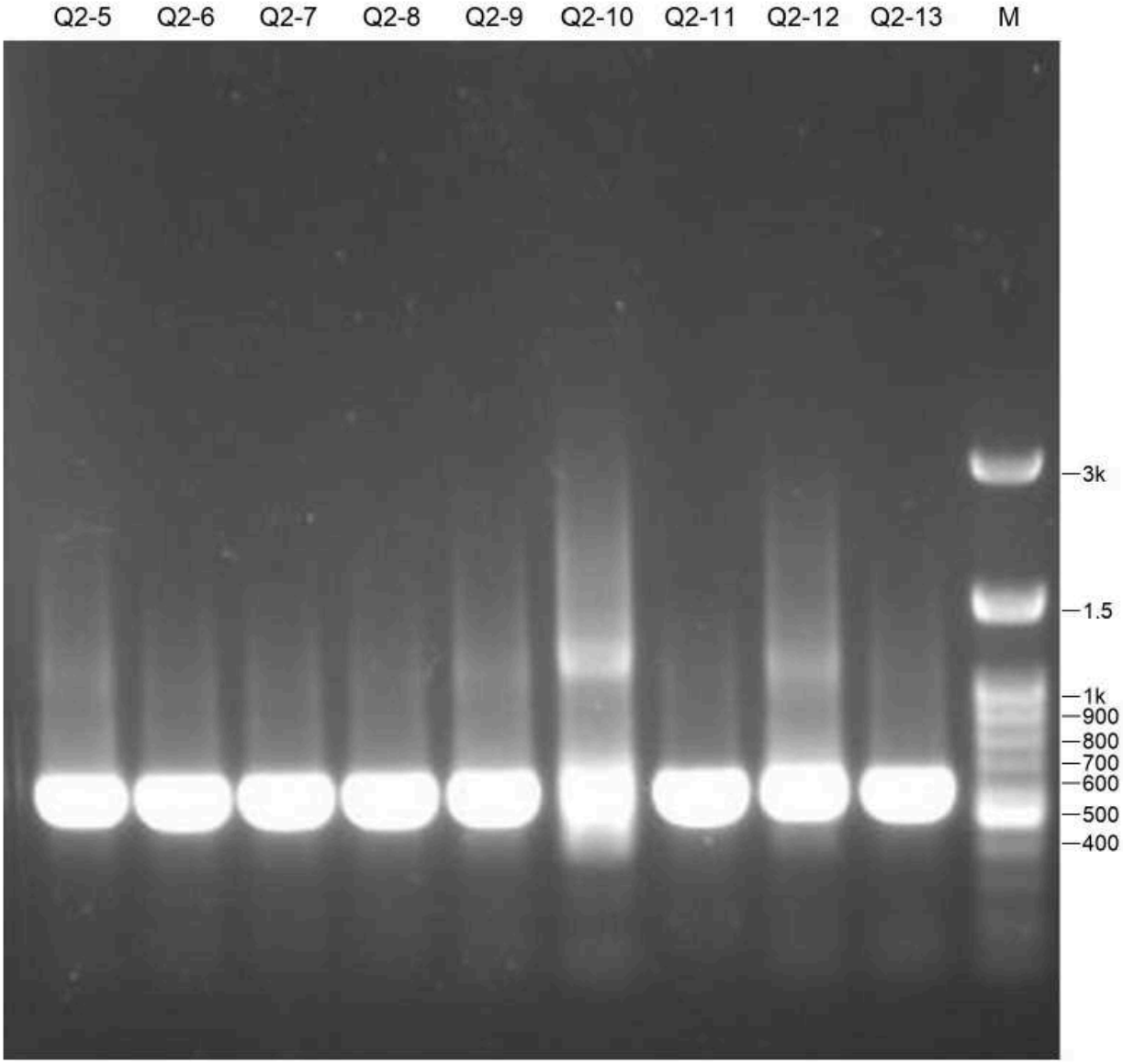
STK-seq-fw & STK-seq-rv

Bp: 536 bp

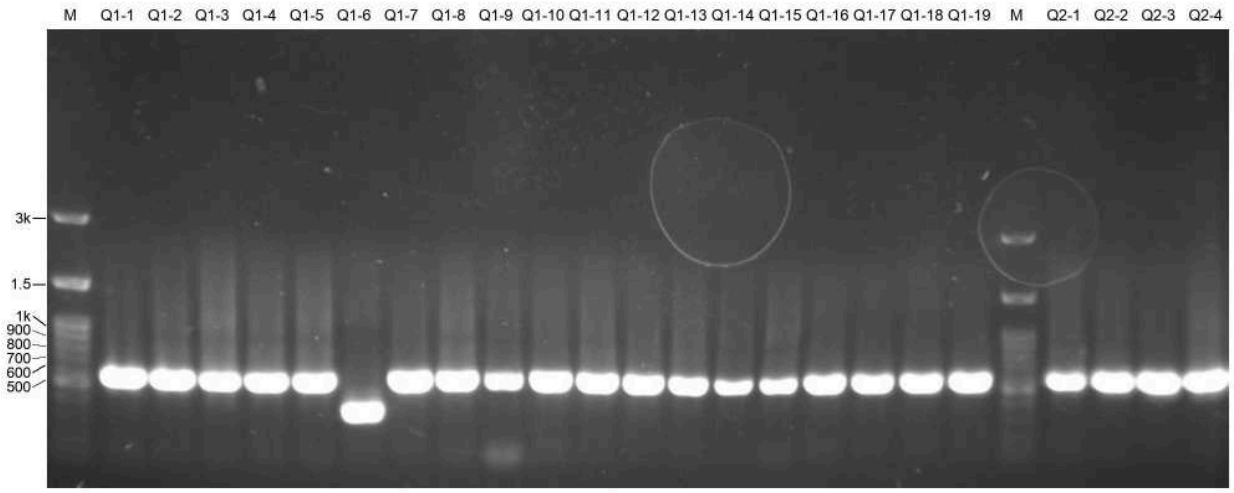
Q1-4, Q1-10, Q1-19 -> liquid culture



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
S1-S24: pSTK-0C-Csn-Lea	Csn F	STK-seq-rv	785
Note: S2, S10, S16			



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Q2(5)-Q2(13): pSTK-0C-SamyQ-Bopep4-2.2	STK-seq-fw	STK-seq-rv	536
Note: Q2-5, Q2-7, Q2-13			



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Q1(1)-Q1(19): pSTK-0C-SamyQ-Bopep4-2.1	STK-seq-fw	STK-seq-rv	536
Q2(1)-Q2(4): pSTK-0C-SamyQ-Bopep4-2.2	STK-seq-fw	STK-seq-rv	536
Note: Q1-4, Q1-10, Q1-19 -> liquid culture			

Done by Sophie C.

Date: 20260814
Title: Make 1 % agarose gel x3
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble tray, mold, comb
2. Weigh 1 g agarose powder & measure 100 mL 1XTAE buffer
3. Add all in conical flask
4. Wrap with plastic wrap (cover) & puncture holes
5. Microwave 30 sec, agitate
6. Microwave 5s ec. Repeat till no sediments
7. When no smoke, add 5 µL safety dye
8. Agitate
9. Pour melted gel into mold
10. Let it solidify at RT

Done by Sophie C

Date: 20260814
Title: Fertilize plants (in soil)
Purpose: Fertilize plants
Procedure:

1. Prepare nutrient solution:
   -Add 0.5 g fertilize powder to 500mL tap water in 500 mL flask (ratio: 1g fertilizer: 1000 mL tap water)
   -Gently mix the solution
2. Add the solution to each plates

Done by Sophie H.

Date: 20260814
Title: Transformation of L and V
Purpose: Transfer
Procedure:

1. Set up water bath at 42 ℃
2. Wait until competent cell melted 50 % , and add 5 μL plasmid to each 100 μL competent cell
3. Incubate on ice for 20 min
4. Heat shock for 45 sec
5. Incubate on ice for 3 min
6. Add 1mL of soc medium to each tubes
7. Incubate for 1 hr in incubator (37 ℃)
8. Centrifuge 5000 g for 5 min
9. Place the centrifuged tube in BSC along with an empty microtube & TBC plates
10. Discard around 900 mL supernatant
11. Repellent the rest
12. Spread the liquid on TBC plate
13. Place in incubator for 16 hr

Done by Naomi, Sophie H

Date: 20260814
Title: Make LBC plate
Purpose: prepare LBC plate
Procedure:

1. Add 250 mL ddH_2O 9.25 LB agar powder, and 2 mL glycerol to a 500 mL flask
2. Add a stirring bar
3. Stir the mixture until all powder are dissolved
4. Cover the flask with aluminum foil and an autoclave indicator sticked on top
5. Autoclave for 20 minutes at 127 °C.

6. Put it in BSC while wait for it to cool down (can also stir before putting in BSC)
7. Add 250 µL of chloramphenicol and gently mix it
8. Distribute 250µL LBC medium in plate (20mL each)
9. Wait for it to cool down

Done by Sophic H, Alex Li, Naomi

Date: 20260814
Title: Make LB medium
Purpose: prepare LB medium
Procedure:

1. Add 11 g of TB powder, 250 mL of ddH_2O and 2 mL glycerol to a 500 mL flask
2. Add a stirring bar to the flask
3. Stir until all powder are dissolved
4. Cover the flask with aluminum foil and an autoclave indicator sticked on top
5. Autoclave for 20 minutes at 127 °C.

Done by Naomi, Sophie H

Date: 20260814
Title: Fertilize plants
Purpose: Fertilize plants
Procedure:

1. Prepare nutrient solution
    - Add 0.5 g fertilize powder to 500 mL tap water(1g fertilizer:1000 mL taps water)
    - Gently mix it
2. Add the solution to each plates

Done by Sophie H

Date: 20260814
Title: Cotreatment(正瀚 protectant+100NaCl)
Purpose: test 正瀚 protectan
Procedure:
Aftertreatment (100NaCl)
pretreatment(100NaCl)

1. Use previous diluted 正瀚 protectant (1/500 dilution) into tube in the BSC
2. Add 1 mL of the diluted 正瀚 protectan into two cotreatment plates
    - Cotreatment (protectant+100 NaCl)
    - Cotreatment (protectant+0 NaCl)
3. Add 1 mL of diluted 100 mM NaCl into two cotreatment +aftertreatment
    - Cotreatment (protectant+100 mM NaCl)
    - aftertreatment(100 NaCl+protectant)
4. Evenly distribute the liquid by lightly shaking left to right

Done by Abigail

Date: 20260814

Title: prepare 0.1 MS medium

Purpose: prepare medium for experiment

Procedure:

1. Prepare 700mL of 0.1 MS medium using the following protocol
2. Add the contents into a beaker
3. Use a magnetic stirring bar to stir until no sediments are left in the solution
4. Use a pH probe to measure the pH and ensure the pH is between 5.7~5.8. If pH is too low slowly add 1M KOH to increase it
5. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
6. Auto clave for 20 min at 127 ℃

   Done by Alex

Date: 20260814

Title: Add CH protectant in Box C2 (pretreatment) for heat stress testing

Purpose: test heat stress

Procedure:

1. Change the medium of Boxes C1,C2,C3 and C4. Add 300mL of 0.1MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are too small
3. Evenly distribute the rest of arabidopsis into boxes C1,C2,C3,C4
4. Add 600 µL of CH protectant into box C2

Done by Alex

Date: 20260814

Title: Hay infusion

Purpose: test which nutrient additive best support *Bacillus subtilis* growth in lake water

Procedure:

1. Take out half of the hay from the 新竹 container
2. Use an electronic pipette controller with 25mL serological to draw all the liquid from the container and pour it into a 100mL beaker
3. Measure the total amount of liquid
4. Take out 5 bottles and label them as following
   - 10 g coffer grounds
   - 10 g sugarose
   - 10 g hay
   - 10 g hay / 5 g sugarose / 10 g coffee grounds
   - 10 g hay / 10 g sugarose / 5 g coffee grounds
5. Take out the scale and measure the additive as labeled on the battle
6. Use the beaker and get 500 mL of water
7. The water into the log sugerose bottle to see if water can cover the sugerose
8. Remove the water from the bottle, make sure not to pour the sugarcane out

9. Pour the liquid in the 1000mL beaker back into the 新竹 container
10. Add 10 g of sugarose and log of coffer grounds into the 新竹 container and close the lid

Done by Sophia L, Sophie H, Olivia Du, Mia

Date: 20260815
Title: Make LBC with glycerol
Purpose: prepare LBC plate with glycerol
Procedure:

1. Follow the recipe
2. Add a stirring bar and stir until no more sediments
3. Cover the flask with aluminium foil and stick indicator tip on top
4. Autoclave at 127 ℃ for 20 min
5. Place new petri dishes and equipment in the BSC
6. Spray equipment with 70 % ethanol
7. Add 500µL chloramphenicol to the mixture
8. Use a pipette gun to add 20 mL of mixture into each petri dishes
9. Solidify at room temperature under the BSC

Done by Abby K/T

Date: 20260815
Title: Colony PCR (L)
Purpose: Amplify engineered DNA plasmid
Procedure:
L:3P-EEBD-RBS0-csn:GanA-Gala-L3S2P21

1. Prepare the following master mix x35

Fw primer EEBD F	35 µL
Rv primer CsnR	35 µL
Q-amp master mix	350 µL
ddH_2O	280 µL

2. Distribute 20 µL to each PCR tube
3. Pick up colony with pipette
4. Dispense pipette in master mix
5. Close the lid
6. Place in PCR machine for thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	35 cycles
60 ℃	20 sec	
72 ℃	30 sec	
25 ℃	5 min	x1

Date: 20260815
Title: Make 2 % Agarose Gel x2
Purpose: prepare gel for gel electrophoresis of colony PCR
Procedure:

1. Assemble mold, tray, comb
2. Weigh 1.2 g agarose powder and measure 60 mL 1XTAE buffer
3. Add all in conical flask
4. Cover with plastic wrap & puncture holes
5. Microwave 30 sec agitate
6. Repeat till no sediments
7. When no smoke, add 3μL safety dye
8. Pour melted gel into mold
9. Let solidify at RT

Done by Sophie C

Date: 20260815
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
S2,S10,S16 : pSTK-OC-csn:LEA
Q1-19,Q2-13,Q2-5,Q1-4,Q2-7,Q2-10:pSTK-OC-SamyQ-Bopep4-2.2

1. Pellet 1.8mL bacterial overnight culture by centrifuge 6800 g for 3 min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250 μL P1 buffer, and resuspend pelleted bacterial
4. Add 250 μL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350 μL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900 g
7. Transfer 800 μL supernatant into new column
8. Centrifuge 1 min for 17900 g
9. Add 500 μL PB buffer
10. Centrifuge 1 min for 17900 g , and discard the flow-through
11. Add 750 μL PE buffer
12. Centrifuge 1 min for 17900 g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900 g , and discard the flow-through
14. Put into new column
15. Add 50 μL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1 min at 17900 g

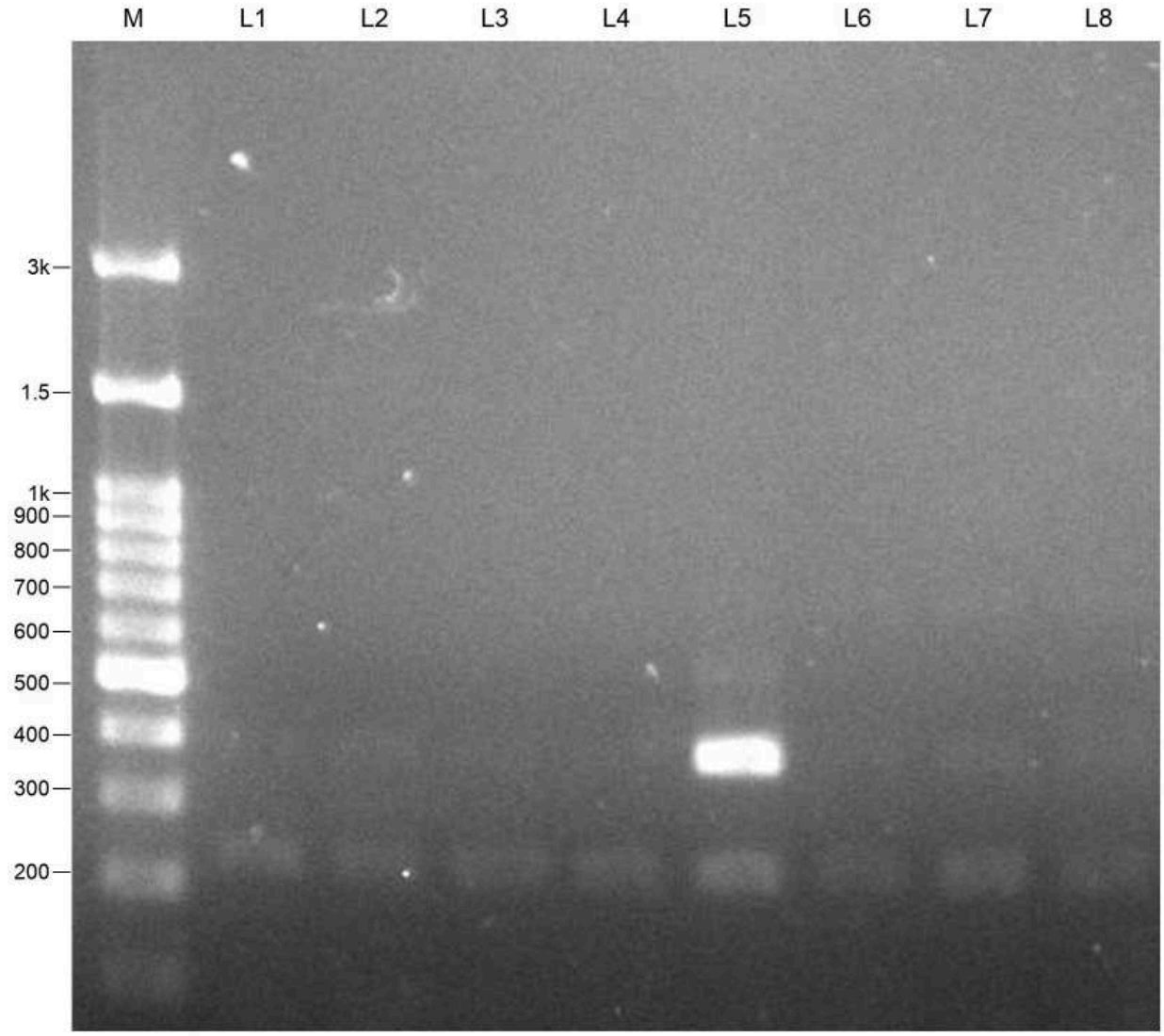
	Concentration(ng/μL)	A260/230	A260/280

S10	48.00	2.000	1.920
S2	56.00	1.931	1.931
S16	55.00	1.774	1.897
Q1-19	61.00	1.906	1.900
Q1-4	77.00	1.400	1.925
Q1-10	52.00	2.364	2.000
Q2-5	50.00	2.000	1.923
Q2-7	57.00	2.036	1.900
Q13	57.00	1.500	1.900

Done by Sarah, Sophie H

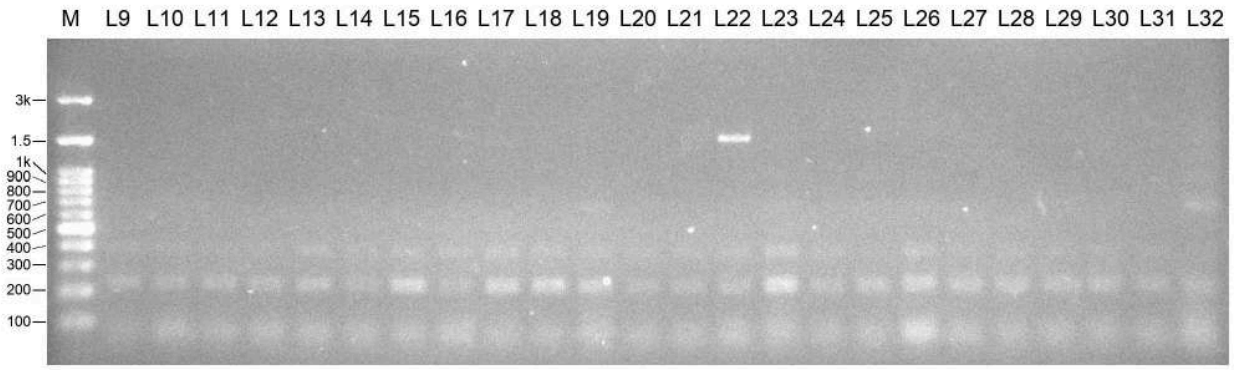
Date: 20260815
Title: Gel electrophoresis (L)
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L1-L8: pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21	EEBD F	Csn R	210

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
L9-L32: pSTK-EXP1-3P-EEBD-RBS0-Csn:GanA-Gala-L3S2P21	EEBD F	Csn R	210/371

Done by Sophic C

Date: 20260815
Title: Colony PCR
Purpose: Amplify engineered DNA plasmid
Procedure:
V1-V24:ABDI

1. Prepare the following master mix x28

Fw primer CcaR	28 µL
Rv primer PcpcG2-veg	28 µL
Q-amp master mix	280 µL
ddH_2O	224 µL

2. Distribute 20µL to each PCR tube
3. Pick up colonies with pipettes
4. Strike the grid three times
5. Dispense pipette tip in master mix
6. Close the lids and spin down
7. Place into PCR machine for the following thermal cycle

95 ℃	5 min	x1

95 ℃	20 sec	35 cycles
60 ℃	20 sec	
72 ℃	1 min	
25 ℃	5 min	x1

Done by AbbyK, Sophie C, Chloe

Date: 20260815
Title: Golden Gate Assembly
Purpose: assemble the parts
Procedure:
pSTK-OC-SamyQ-AccD-2.1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(1/5)
SamyA-AccD-2.1(pTwist) 125ng/μL , 3401bp	1.78μL(1/5)
T4 DNA ligase buffer	2μL
Sterile water	12.78μL
T4 DNA ligase	1μL
BbsI-HF	1μL

pSTK-OC-SamyQ-AccD-v3t1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(1/5)
SamyA-AccD-v3t1(pTwist) 142ng/μL , 3401bp	1.56μL(1/5)
T4 DNA ligase buffer	2μL
Sterile water	12.99μL
T4 DNA ligase	1μL
BbsI-HF	1μL

pSTK-OC-SamyQ-LEA-2.1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(1/5)
SamyA-AccD-2.1(pTwist) 141ng/μL , 2840bp	1.3μL(1/5)
T4 DNA ligase buffer	2μL
Sterile water	13.25μL
T4 DNA ligase	1μL
BbsI-HF	1μL

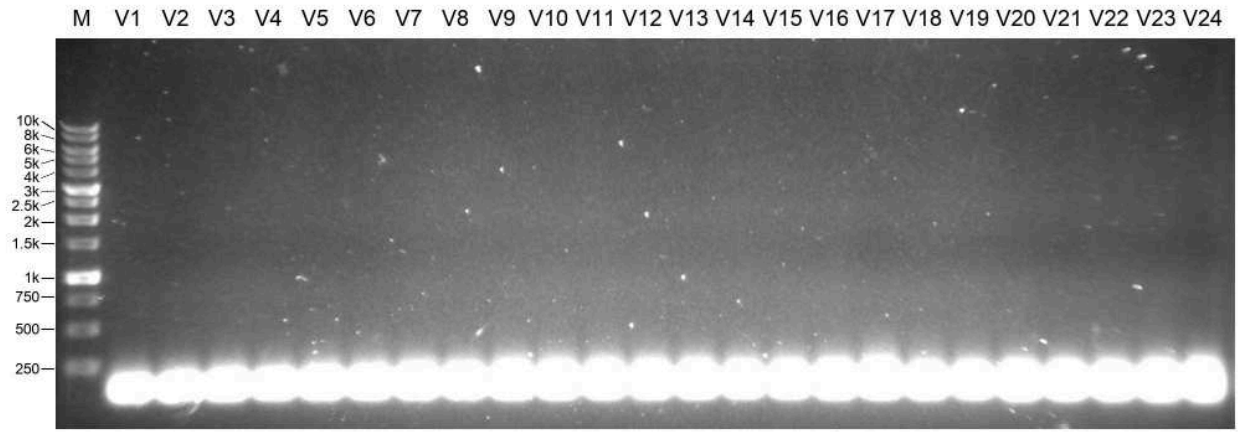
pSTK-OC-SamyQ-LEA-2.1

part	amount
pSTK-001-pSTK-O-sfGFP 346ng/μL , 3090bp	1.45μL(1/5)
SamyA-AccD-2.1(pTwist) 111ng/μL , 2840bp	1.66μL(1/5)
T4 DNA ligase buffer	2μL
Sterile water	12.8μL
T4 DNA ligase	1μL
BbsI-HF	1μL

Done by Sophie C, Abby T/K

Date: 20260815
Title: Gel electrophoresis (V)
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
V1-V24: A: PtagC-RBS-Ho1:pcyA-L3S2P21; B:Pliag-RBS0-ccaSm3-B0015; D: Pveg-RBS0-ccaR-B0015; I: PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015	ccaR F	pcpcG2-veg R	1032

Done by Abby K

Date: 20260815
Title: Make 1 % agarose gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble mold, tray, comb
2. Weigh 0.2g of agarose powder and 20mL of 1XTAE buffer
3. Add into a flask
4. Cover with plastic wrap and poke a few holes
5. Microwave for 20 seconds, then agitate
6. Repeat for 5 seconds each until no sediments
7. When no smoke, add 2mL of safety dye
8. Pour melted gel into mold
9. Solidify at RT

Done by Abby K , Sophie C

Date: 20260815
Title: hydroponic heat stress testing

Purpose: testing heat stress in hydroponic
Procedure:

1. Prepare 4 new boxes and add 300mL of old MS medium into each box. Label them as C1, C2,C3,C4
2. Add 600L CH protectant into C3 (co-treatment)
3. Take another box and add 300mL of water into it
4. Put a thermometer into the water
5. Put all boxes into the incubator(setting 37 ℃, actual temperature 33 ℃)
6. Wait until the thermometer reaches 33 ℃
7. Take out the new C1 box from incubator
8. Transfer the floating board with plants form old C1 box into the new C1 box that is at 33 ℃
9. Put the new C1 box back into the incubator
10. Repeat steps 7~9 until all boxes changed medium

    Done by Alex

Date: 20260815
Title: add zhenhen protectant to heat stress testing agar plates
Purpose: testing zhenhen protectant
Procedure:
pre-treatment 7:00pm

1. Dilute zhenhen protectant by 1/500 (40μL of peptide+19960μL of ddH_2O -> total:20mL)
2. Apply the sprayer to the centrifuge tube
3. Place the plate vertically and spray the plants with protectant with a distance of ~15cm in between (2 sprays)

    Done by Abby T
    Supervised by Gina

Date: 20260816
Title: Protein Extraction
Purpose: purify protein samples
Procedure:

1. Take 24 2mL centrifuge tubes and add 1800 µL of overnight liquid culture (*Bacillus subtilis*)
2. Centrifuge 16000 g for 30 seconds
3. Take tubes out of microcentrifuge and pour the supernatant into a 50 mL centrifuge tube
4. Repeat the steps 1-3 until the 50 mL centrifuge tube is filled
5. Pour the supernatant into a syringe connected to 0.22 µm filter and filter the supernatant into a 50mL centrifuge tube
6. Repeat the previous steps until all culture is filtered
7. Centrifuge the Ni-NTA column at 175 g for 2 min to remove the storage buffer
8. Add 6mL of equilibrate buffer into the column by dispensing the buffer along the walls of the tube
9. Add 237.5 µL of imidazole into supernatant
10. Add all supernatant from 1 tube into the column
11. Add 6mL of wash buffer and remove the flow-through repeat this step 3 times
12. Centrifuge at 175 g for 2 min
13. Collect the flow through from the last wash buffer step
14. Add 3 mL of elute buffer
15. Centrifuge at 175 g for 2min
16. Collect the flow-through as the purified protein

Date: 20260816
Title: Miniprep of H1 (0809 liquid culture)
Purpose: DNA plasmid purification
Procedure:
H1: 3P-EEBD-RBS0-Csn:LEA-B0015

1. Transfer 1.8 ml of overnight liquid culture medium to a microcentrifuge tube. Centrifuge at 8000 g for 3 minutes, and discard the flow-through. Repeat this step until all liquid cultures are used.
2. Resuspend pelleted bacterial cells in 250 µl Buffer P1
3. Add 250 µl Buffer P2 and mix thoroughly by inverting the tube 4-6 times
4. Add 350 µl Buffer N3 and mix immediately and thoroughly by inverting the tube 4-6 times.
5. Centrifuge for 10 min at 17,900 g
6. Transfer the supernatant from to the QIAprep spin column. Centrifuge for 60s and discard the flow-through
7. Wash the spin column by adding 0.5 ml Buffer PB. Centrifuge at 17900g for 60s, and discard the flow-through
8. Wash the spin column by adding 0.75 ml Buffer PE. Centrifuge at 17900g for 60s, and discard the flow-through
9. Centrifuge for 1 min to remove residual wash buffer

10. Add 50 µL of EB buffer to the center of the column membrane, let stand for 1 min, and centrifuge for 1min.

Results:

Concentration (ng/µL)	A260/230	A260/280
178	2.177	1.914

Done by Abby T

Date: 20260816
Title: Level 2 Golden Gate Assembly
Purpose: Assemble level 1 genetic circuits
Procedure:

V: A+B+D+I (1:3)

Part	amount	dilution
Exp2 (163 ng/µL,5104 bp)	0.203	1.015 µL (1/5) → 1µl of plasmid with 4µl of ddH_2O
A:PtagC-RBS-H01:pcyA-I3S2 P21 (109 ng/µL,3 754 bp)	0.690	1.38 µL(1/2) → 1µl of plasmid with 1µl of ddH_2O
B:PliaG-RBS0-CcaSm3-B0015 (83 ng/µL,3792 bp)	0.890	1.78 µL (1/2) → 1µl of plasmid with 1µl of ddH_2O
D:Pveg-RBS0-CcaR-B0015 (134 ng/µL,3243 bp)	0.471	1.413 µL (1/3) → 1µl of plasmid with 2µl of ddH_2O
I:PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015 (165 ng/µL,3757 bp)	0.444	1.332 µL (1/3) → 1µl of plasmid with 2µl of ddH_2O
T4 DNA ligase buffer	2 µL	2 µL
Sterile water	13.3 µL	13.3 µL
T4 DNA ligase	1 µL	1 µL
BsmBI	1 µL	1 µL
Total	20 µL	20 µL

2. Run thermal cycle

55 °C	5 min	45 cycles
16 °C	5 min	
55 °C	10 min	1 cycle
80 °C	20 min	

Done by Abby T, Alex

Date: 20260816
Title: Spray CH bio peptide to co-treatment group, and place all groups into incubator at 37°C for thermo priming
Purpose: apply bio peptide and heat stress
Procedure:

1. Place the co-treatment plate vertically, spray evenly toward the entire plant from leaves to root with around 15cm of distance in between, 1 spray on the left side and 1 spray on the right side. Make sure the entire plant receives even treatments.
2. Place all groups on the rack and put the entire rack into the incubator, under the white light set, set at 37 °C

Done by Abby T

Date: 20260816
Title: Ni-NTA Regenerate Buffer (for protein purification)
Purpose: prepare buffer for protein purification
Procedure:

1. Prepare 0.39 g of MES and 0.584 g of NaCl
2. Dissolve both in 100 mL of ddH_2O in a flask
3. Adjust pH to 5.0 using NaOH (final: 5.0 pH)
4. Transfer all to the jar label (N-NTA regenerate buffer)
   Formula:20mM MEs 0.1 M NaCL 5.0 pH (therme fisher)

Done by Ryan Y

Date: 20260816
Title: Transformation
Purpose: transform and cultivate engineered bacteria on agar plate
Procedure:
pSTK-OC-SamyQ-AccD-2.1
pSTK-OC-SamyQ-AccD-2.1
pSTK-OC-SamyQ-LEA-2.1
pSTK-OC-SamyQ-LEA-2.1

1. Turn on water bath to 42 ℃ prepare ice bucket
2. Prepare 4 LBC plates

3. Thaw competent cells until 50 % thawed and add 10μL plasmid
4. Incubate in ice for 20 min
5. Heat shock at 42 ℃ for 45 seconds
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium to each tube
8. Incubate at 37 ℃ for 1 hour
9. Centrifuge at 5000g for 5 min
10. Take out 1 mL of supernatant
11. Resuspend cell, and spread entirely on LBC plates

Done by Abby T/K

Date: 20260816

Title:Remove protectant from C3 and add protectant in C4

Purpose: testing protectant

Procedure:

1. Prepare 2 new boxes and add 300μL of 0.1X MS medium into both boxes. Label them as C3, C4
2. Add CH protectant into box C4(post-treatment)
3. Take another box and add 300 mL of water into it
4. Put a thermometer into the water
5. Put both boxes into the incubator(setting 37 ℃, actual temp 33 ℃)
6. Wait until thermometer reaches 33 ℃
7. Take out new C3 box from incubator
8. Transfer the floating board with plants form old C3 box into new C3 box that is at 33 ℃
9. Put the new C3 box back into incubator
10. Repeat steps 7~9 to change the medium of C4

Done by Alex

Date: 20260816

Title:Spray CH bio peptide to co-treatment (7:00pm)+place all group into incubator at 37 ℃ for therm priming

Purpose: testing peptide

Procedure:

1. Spray diluted protectants to co-treatment
    - 2 sprays
    - 15 cm distance
2. Place all groups of plates onto a rack , and put in incubator under white light

Done by Abby T

Date: 20260817
Title: LB Medium
Purpose: Prepare medium for bacterial culture
Procedure:

1. Prepare the following

LB broth powder	6.25g
ddH_2O	250mL

2. Add all to a conical flask
3. Add a stirring bar
4. Stir on hot plate till no sediments
5. Remove stirring bar
6. Autoclave
7. Put it in the fridge

Done by Sophie C

Date: 20260817
Title: Colony PCR
Purpose: to screen for successful cloning after transformation of QA, QB, QC, QD
Procedure:
QA:pSTK-OC-SamyQ-LEA 2.1
QB:pSTK-OC-SamyQ-AccD v3t1
QC:pSTK-OC-SamyQ-LEA 2.2
QD:pSTK-OC-SamyQ-AccD 2.1

1. Prepare the following master mix for

QA+QC x26

Fw primer Biobrick F	26 µL
Rv primer Lea R	26 µL
Q0amp master mix	260 µL
ddH_2O	208 µL

QB+QDx26

Fw primer Biobrick F	26 µL

Rv primer AccD R	26 µL
Q0amp master mix	260 µL
ddH_2O	208 µL

2. Distribute 20µL to each PCR tube
3. Dip pipette into colony
4. Dispense the tip into PCR tube
5. Take out tip & close lid
6. Thermal cycle in PCR machine

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	1 min 20 sec	
25 ℃	5 min	x1

Done by Sophie C, Abby K

Date: 20260817
Title: Make 1 % agarose gel x2
Purpose: for gel electrophoresis of colony PCR
Procedure:

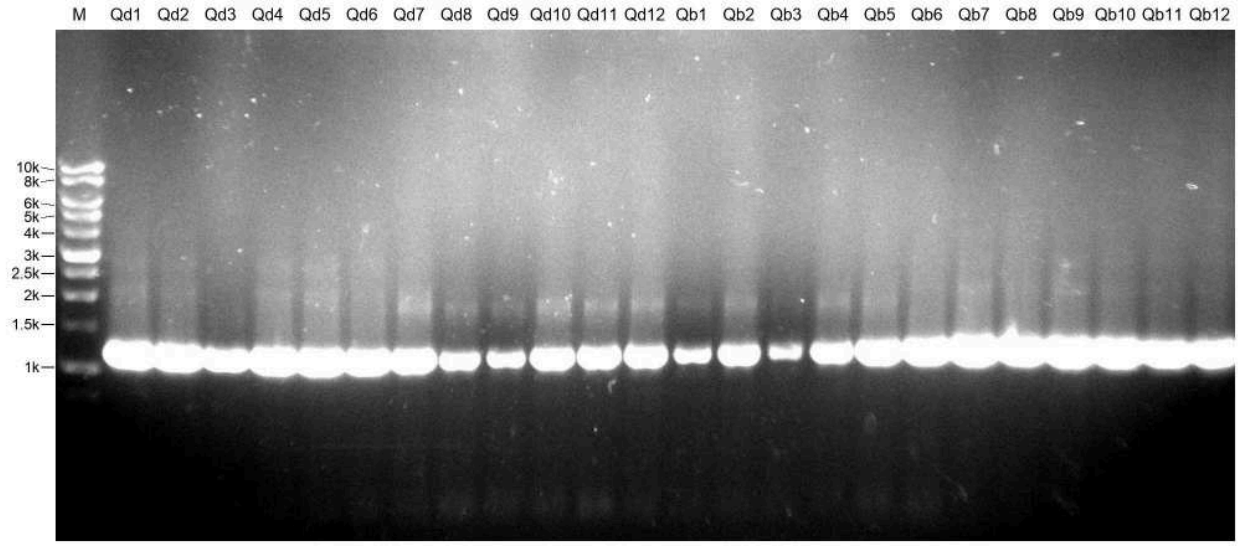
1. Assemble comb, tray, mold
2. Weigh 0.8 g agarose powder & 80 mL 1XTAE buffer
3. Add all in conical flask
4. Cover with plastic wrap & puncture holes
5. Microwave for 30 sec agitate
6. Microwave for 5 sec. Agitate. Repeat until no sediments
7. When no smoke, add 4 µl safety dye
8. Pour melted gel into mold
9. Let solidify at RT

Done by Sophie C

Date: 20260817
Title: Gel Electrophoresis of QD and QB
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):

2. Resolve at 100 v for 30 min

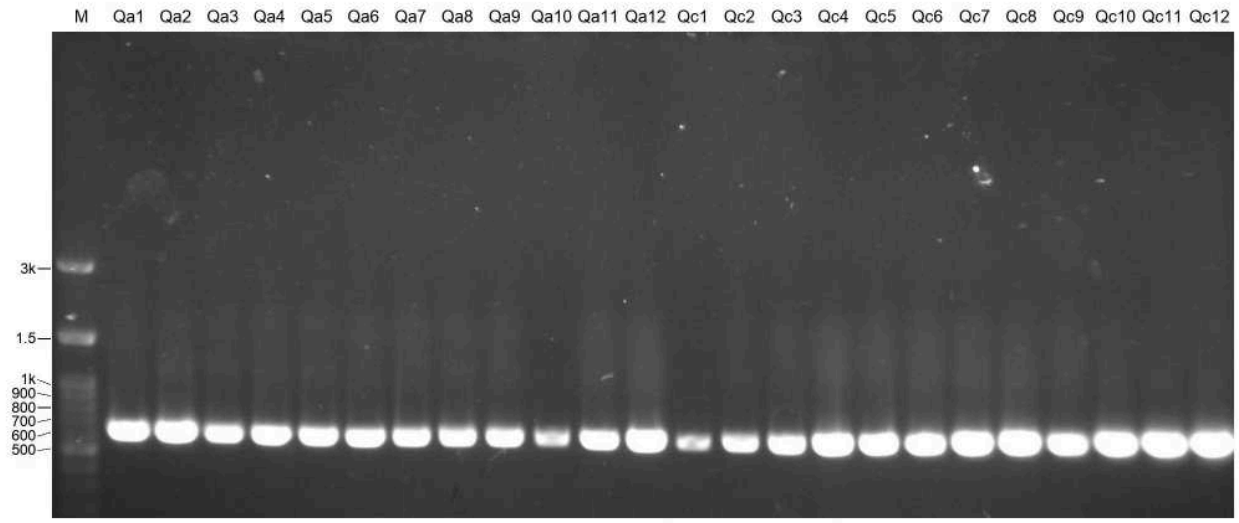


M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Qd1-12 (ACCD 2.1)	Biobrick F	ACCD Rv	1331
Qb1-12 (ACCD v3t1)	Biobrick F	ACCD Rv	1331

Done by Abby K

Date: 20260817
Title: Gel Electrophoresis of QA and QC
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Qa1-12 (LEA 2.1)	Biobrick F	Lea Rv	596
Qc1-12 (LEA 2.2)	Biobrick F	Lea Rv	596

Done by Sophie C

Date: 20260817
Title: Transformation (V)
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Set up water bath to 42 ℃
2. Thaw competent cell till 50 % thawed, add 10 μL plasmid
3. Incubate on ice for 20 min
4. Heat shock 42 ℃ for 45 sec
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37 ℃
8. Centrifuge at 5000 g for 5 min
9. Take out ~1000 μL supernatant
10. Resuspend the pellet
11. Spread on LBC plate
12. Incubate at 37 ℃ for 16 hr

Done by Sophie C

Date: 20260817

Title: Add zhenhen protectant (pre-treatment) to experiment set #7 trial 3 (3:30pm)

Purpose: testing zhenhen protectant

Procedure:

1. Dilute the CH protectant by 1/500
2. Under the BSC, use 200µ L pipette to remove the excess water
3. Add 3 mL of CH protectant to the agar plate
4. Move it around the plate until the entire surface is covered
5. remove 2 mL of the CH protectant na discard it
6. Repeat steps 3-5 for the other two plates
7. Use breathable paper to tape up the edges of the plates
8. Place them back in the growth chamber

Done by Abby K

Supervised by Abby T

Date: 20260817

Title: Plant new seeds on soil pots

Purpose: plant seeds

Procedure:

1. Autoclave a bag of soil
2. Soak 4 flower pots in 1/10 bleach and wait for 20 min
3. Rinse the pot with water for 5 min
4. Fill the pot with soil and slightly press it down
5. Gently shake it vertically to soften the soil
6. Water the soil until the upper part of the soil beamer fully wet
7. Use a tip to get a seed from preservation solution
8. Dispense the seed on the soil in fixed position (4 seeds in a pot)
9. After all the pots are done, cover with petri dish and place them in the incubator

Done by Abigail , Sophie C

Date: 20260817

Title: Spray CH bio peptide to after-treatment (7:00pm)

Purpose: testing peptide

Procedure:

1. Spray diluted protectants to after treatment
    - 2 sprays
    - 15 cm distance
2. Place all groups of plates onto a rack, and put in incubator (22 ℃ for 17 hr)

Done by Sophie C

Date: 20260817

Title: plant new set of seeds (set F) for hydroponic system

Purpose: prepare new set of seed for hydroponic system

Procedure:

1. Prepare 47 eppendorf lids
2. Add 250 µL of 0.8 % agar into each lid
3. Wait until the agar solidifies
4. Clean tube rack with 10 % bleach and rinse with ddH_2O until there are no smell of bleach
5. Add 2,2 mL of 0,1X MS medium into 47 tube rack
6. Use an autoclaved toothpick to pick one seed and add the seed into the hole of the lid
7. Repeat step 6 until all lids have seeds
8. Close the lid of the tube rack
9. Wrap the tube rack with aluminum foil
10. Place the tube rack into the refrigerator at 4 ℃

Done by Alex

Date: 20260819
Title: Colony PCR (X, V, J)
Purpose: screen for successful cloning after transformation
Procedure:
X: ACDI (X1-X8)
V: ABDI (V1-V8)
J: Pveg-EEBD-RBS0-csn::ACCD-B0015 (J1-J8)

1. Prepare the following master mix (x18)

X&V

Fw primer ccar	18 µL
Rv primer csnR	18 µL
Q-amp master mix	180 µL
ddH_2O	144 µL

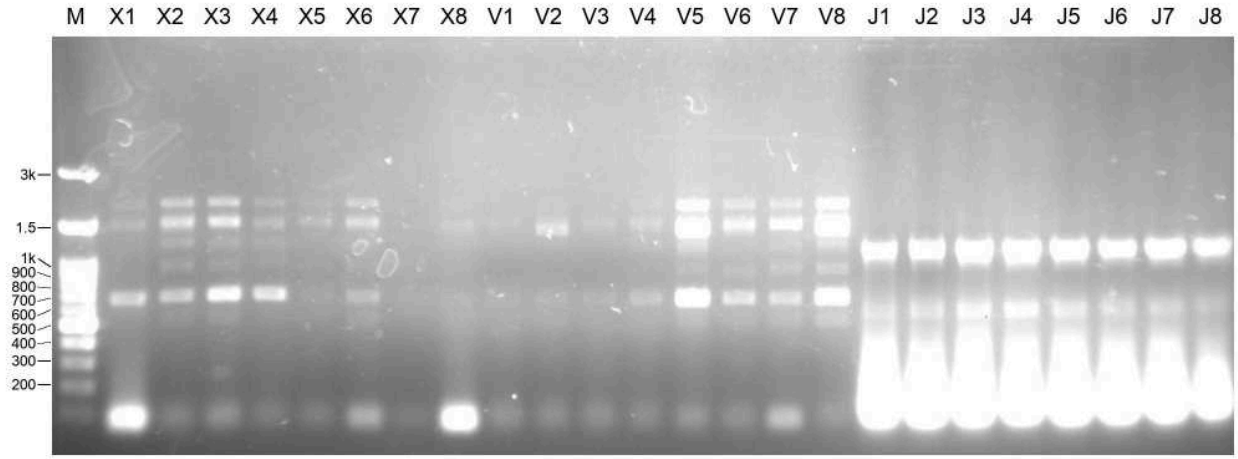
J:

Fw primer Csn	10 µL
Rv primer ACCD	10 µL
Q-amp master mix	100 µL
ddH_2O	80 µL

2. Distribute in PCR tubes (20 µL each)

Done by Abby K, Abigail

Date: 20260819
Title: Gel electrophoresis (1 % gel)
Purpose: to validate PCR product
Procedure:
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X:ACDI	ccar fw	csn rv	1424
V:ABDI	ccar fw	csn rv	1424
J: Pveg-EEBD-RBS0-Csn:ACCD-B0015 (electroporation)	csn F	ACCD R	1195

Done by Sophie C

Date: 20260819
Title: Transformation (L)
Purpose: to validate PCR product
Procedure:
3P-EEBD-RBS0-Csn:GanA-Gala-B0015

1. Set up water bath to 42 ℃
2. Thaw competent cell till 50 % thawed, add 10 μL plasmid
3. Incubate on ice for 20 min
4. Heat shock 42 ℃ for 45 sec
5. Incubate on ice for 3 min
6. Add 1mL SOC medium
7. Incubate for 1 hr at 37 ℃
8. Centrifuge at 5000 g for 5 min
9. Take out ~1000 μL supernatant
10. Resuspend the pellet
11. Spread on LBC plate

12. Incubate at 37 ℃ for 16 hr

Done by Sophie C

Date: 20260819

Title: Prepare 420 mL 0.8 % 1/2 MS agar

Purpose: add agar into lids for new set of seeds

Procedure:

The following protocol is for 420 mL 1/2 MS agar (0.8 %)

1. 1/2 MS powder (w/o sucrose): 0.966 g
2. MES buffer: 0.21 g
3. Distilled water: 420 mL
4. Sucrose: 4.2 g
5. Agar: 3.36 g
6. Add 1/2 MS powder, MES buffer, sucrose, and distilled water into a flask
7. Shake the flask with hands until no sediments are in the solution
8. Use the pH probe to measure the pH of the solution, ensure the pH is between 5.7–5.8
9. If pH is too low, add 0.1 M KOH slowly to increase it
10. Add agar into the flask
11. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
12. Autoclave for 20 min at 127°C

Done by: Abby K

Date: 20260819

Title: Hay Infusion

Purpose: Preparing different ratios of nutrients additive

Procedure:

1. Take out half of the hay from the 新竹 container
2. Use an electronic pipette controller with a 25 mL serological pipette to draw all the liquid from the container and pour it into a 1000 mL beaker
3. Measure the total amount of liquid
4. Take out 5 bottles and label them as follows:

1	10 g coffee grounds
2	10 g sugarcane
3	10 g hay
4	10 g hay 5 g sugarcane 10 g coffee grounds
5	10 g hay 10 g sugarcane 5 g coffee grounds

5. Take out the scale and measure each as labeled on the bottle
6. Use the beaker and get 500 ml of water
7. Pour the water into the 10 g sugarcane bottle to see if the water covers the sugarcane

8. Remove the water from the bottle, make sure not to pour the sugarcane out
9. Pour the liquid in the 1000 mL beaker back into the 新竹 container
10. Add 10 g of sugarcane and 10 g of coffee grounds into the container and close the lid

Done by Sophia L, Sophie H, Olivia Du, Mia

Date: 20260819

Title: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Purpose: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Procedure:

1. Change the medium of boxes D1, D2, D3, and D4. Add 300mL of 0.1x MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are too small
3. Evenly distribute the rest of arabidopsis into boxes D1, D2, D3, D4
4. Add 600 µL of CH protectant into box D2

Done by Alex

Date: 20260819

Title: plant new seeds on agar x12 (on 0.8 % agar)-> vernalize

Purpose: prepare new seeds

Procedure:

1. Open the BSC and turn on fan
2. Place the 0.8 % agar plates on the 3D printed layout (seed template)
3. Take a clean bamboo stick (autoclaved) and dip in sterilized (autoclaved) water
4. Stick arabidopsis (col-o) seeds on the stick and tap lightly on the agar
5. Repeat until all plates are planted
6. Use breathable tape to seal each of the agar plates
7. Label the vernalization date, incubator date and seed type
8. Wrap the agar plates with aluminum foil
9. Vernalize in the 4 ℃ refrigerator for 4 days and avoid light

Done by Abigail

Date: 20260819

Title: Add CH protectant in Box D2 (pre-treatment) for heat stress testing

Purpose: heat stress testing

Procedure:

1. Change the medium of boxes D1, D2, D3 and D4. Add 300 mL of o.1X MS medium when changing medium for each box
2. Take out the plants that aren't growing well or are too small
3. Evenly distribute the rest of arabidopsis into boxes D1,D2,D#,D$
4. Add 600µL of CH protectant into box D2

Done by Alex

Date: 20260820
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
A1: PtagC+RBS-Ho1:PcyA-L3S2P21
B2: Pveg-RBS0-CcaR-B0015
C6: PlepA-RBS0-CcaSm3-B0015
I3: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

#7: L0-Bopep4-2.2 (pSTK-0C)
#19: L0-Bopep4-2.1 (pSTK-0C)

X2: ACDI
X3: ACDI
V2: ABDI
V8: ABDI

1. Pellet 1.8mL bacterial overnight culture by centrifuge 6800g for 3min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250μL P1 buffer, and resuspend pelleted bacterial
4. Add 250μL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350μL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900g
7. Transfer 800μL supernatant into new column
8. Centrifuge 1 min for 17900g
9. Add 500μL PB buffer
10. Centrifuge 1 min for 17900g , and discard the flow-through
11. Add 750μL PE buffer
12. Centrifuge 1 min for 17900g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900g , and discard the flow-through
14. Put into new column
15. Add 50μL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1min at 17900g
18. Measure the concentration in Biodrop

Results

	Concentration (ng/μL)	A260/A280	A260/A230
A1	107	1.945	2.548
V8	212	1.876	2.000

B2	86	1.955	2.457
C6	134	1.887	1.675
X3	131	1.871	1.658
I3	108	1.862	1.385
#19	137	1.877	/
#7	101	1.836	1.063
X2	241	1.898	2.078
V2	217	1.937	2.192

Done by Sophie C., Abigail

Date: 20260820
Title: PCR of V, X
Purpose: to amplify specific segment of DNA
Procedure:

1. Add the following parts

V: ABDI

	1 reaction
Forward primer (EXP F)	1μL
CDS primer (pcyA R)	1μL
DNA (V)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Bp: 1712

	1 reaction
Forward primer (CcaR F)	1μL
CDS primer (PcpcG2 R)	1μL
DNA (V)	1μL
Q-amp master mix	10μL

ddH_2O	7μL

Bp: 1032

X: ACDI

	1 reaction
Forward primer (EXP F)	1μL
CDS primer (pcyA R)	1μL
DNA (X)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Bp: 1712

	1 reaction
Forward primer (CcaR F)	1μL
CDS primer (PcpcG2 R)	1μL
DNA (X)	1μL
Q-amp master mix	10μL
ddH_2O	7μL

Temperature: 65˚C

Done by Sophie C.

Date: 20260820
Title: Enzyme Digestion
Purpose: To confirm plasmid size for each samples before moving on to golden gate assembly
Procedure:
Samples

- #19: Bopep 4 2.1 (pSTK-0C)
- #7: Bopep4 2.1

1. Set up following reaction on ice
   Volume of DNA(μl)= $\frac{desired\ \ DNA\ mass\ (ng)}{DNA\ concentration\ (ng/\mu l\ )}$
   *desired DNA mass = 500ng

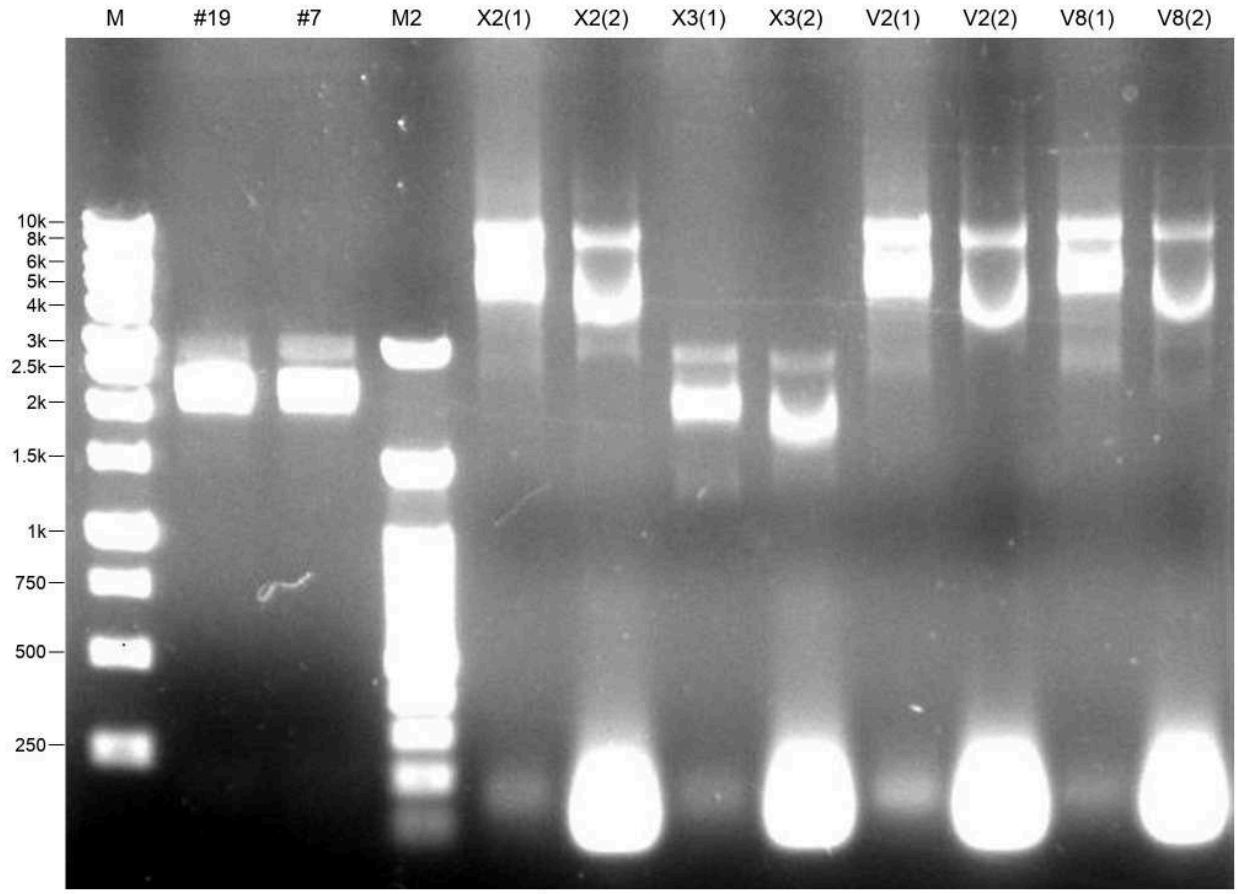
A. Bopep4 2.11
B. Bopep4 2.2

Components	20 µL rxn
DNA - Concentration: 101 ng/µL	4.95 µl
Cutsmart Buffer	2 µL
Restriction Enzyme	0.5 µg
Nucleus free water	12.55 µL

Done by Sophie C., Abby K

Date: 20260820
Title: Gel electrophoresis
Purpose: To validate enzyme digestion and PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder; M2: 3kb DNA ladder			
Sample	Construct	Primers	Expected Size (bp)
#19	Bopep4 2.1 (pSTK-0C) enzyme digestion		183
#7	Bopep4 2.2 (pSTK-0C) enzyme digestion		183
X2(1), X3(1), V2(1), V8(1)	X:ACDI, V:ABDI	EXP-seq-fw, pcyA R	1712
X2(2), X3(2), V2(2), V8(2)	X:ACDI, V:ABDI	CcaR F, PcpcG2-veg R	1032

Done by Sophie C.

Date: 20260820
Title: Prepare 420 mL 0.6 % 1/2 MS agar
Purpose: Add agar into lids for new set of seeds
Procedure:
The following protocol is for 420 mL 1/2 MS agar (0.6 %)

1. 1/2 MS powder (w/o sucrose): 0.966 g
2. MES buffer: 0.21 g
3. Distilled water: 420 mL
4. Sucrose: 4.2 g
5. Agar: 2.52 g
6. Add 1/2 MS powder, MES buffer, sucrose, and distilled water into a flask
7. Shake the flask with hands until no sediments are in the solution
8. Use the pH probe to measure the pH of the solution, ensure the pH is between 5.7–5.8
9. If pH is too low, add 0.1 M KOH slowly to increase it
10. Add agar into the flask
11. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
12. Autoclave for 20 min at 127 °C

Done by: Abby K

Date: 20260820
Title: Level 1 Golden Gate Assembly of Bopep4
Purpose: To assemble level 0 parts to level 1
Procedure:

1. Prepare the following reactions

N: Pveg-EEBD-RBS0-SamyQ:Bopep4(2.1)-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration: 162 ng/ µl • 6929 bp	0.2780154321	1.4 ul (dilute 1/5)
Insert promoter(Pveg) • concentration(ng/µL): 168 • bp: 2361	0.182 uL	1.82 uL (dilute 1/10)
Insert RBS(EEBD-RBS0) • concentration(ng/µL): 151 • bp: 2392	0.2059337748 µL	1.03 µL (Dilute 1/5)

Insert CDS(Samy Q:Bopep4(2.1)) - concentration: 137 - 2263 bp	0.2147372263	1.07 (dilute 1/5)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		9.33 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

O: (STK) Module 4-PcpcG2-veg - EEBD- RBS0 - SamyQ:Bopep4(2.2) - B0015

Vector (1D) ● concentration(ng/μL): 132 ● bp: 3120	0.15 μL	1.5 μL (dilute 1/1]\\ ]0)
Insert promoter(PcpcG2-veg) ● concentration(ng/μL): 168 ● bp: 235	0.01818452381 μL	0.90922619 μl (dilute 1/50)
Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 151 ● bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS(SamyQ:Bopep4 2.2) ng/uL - concentration: 101 - 2263 bp	0.2912772277	1.17 μL (Dilute 1/4)

Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		10.05 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Done by Sophie C, Abby K

Date: 20260820

Title: hydroponic heat stress testing

Purpose: To test hydroponics plant growth when put under heat stress

Procedure:

1. Prepare 4 new boxes, add 300mL of 0.1X MS medium into each box. Label them as D1, D2, D3, D4
2. Add 600 μL of CH protectant into D3 (co-treatment)
3. Take another box and add 300 mL water into it
4. Put a thermometer into water
5. Place all boxes into the heat box(setting 50 ℃, actual temp 35 ℃)
6. Wait until thermometer reaches 35 ℃
7. Take out new D1 box from heat box
8. Transfer the floating board from old D1 box to the new D1 box
9. Put new D1 box back into the hat box
10. Repeat steps 8~10 until all boxes changed medium

Done by Alex

Date: 20260820

Title: Hay infusion

Purpose: observe the growth of *Bacillus subtilis* in 新竹 hay infusion and LB

Procedure:

1. Take the 0815 新竹 1000 mL hay infusion
2. Take two centrifuge tubes and label them
3. Disinfect all equipment and move it into the BSC for aseptic operation
4. Use and electronic pipetted controller with a serological pipette to draw 15 mL of hay infusion
5. Transfer the 15 mL into a syringe fitted with a sterile filter

6. Filter the hay infusion and collect it into the 15mL tube
7. Use a new serological pipette to transfer 5mL of the filtered water into a centrifuge tube (labeled with “hay infusion”)
8. Use a pipette to add 200 µL of *Bacillus subtilis* into the hay infusion centrifuge tube
9. Use the electronic pipette controller with a new serological pipette and add 5 mL of LB into the LB centrifuge tube
10. Use a pipette to add 200 µL of *Bacillus subtilis* into the LB centrifuge tube
11. Place the centrifuge tubes in the centrifuge overnight
12. Take out the centrifuge tube after a wight
13. Use the pipette to transfer 200 µL of each filtered hay infusion and LB(with no *Bacillus subtilis*) into separate wells of a 96-well plate)
14. Transfer 200 µL of hay infusion and LB (with *Bacillus subtilis*) into separate wells of a 96-wells plate
15. Place the 96-well plate into the OD microplate reader and record the result
16. Put the centrifuge back to the centrifuge
17. Repeat step 14 to step 16 every hour

Result:

Hour	OD of LB	OD of Hay Infusion
18	0.489	0.091
19	0.486	0.095
20	0.486	0.097
21	0.503	0.107
22	0.499	0.124

Done by Sophia L, Olivia Du

Date: 20260821
Title: Miniprep
Purpose: to extract plasmid from liquid culture
Procedure:
L1, L2: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21 / 0820 TBA
#3: pSTK-0C-SamyQ:LEA 2.2 (D-C) / 0817 TBC
#1: pSTK-0C-SamyQ:LEA 2.1 (Q-A) / 0817 TBC
#2: pSTK-0C-SamyQ:ACCD 2.1 (Q-D) / 0820 TBC
#S16: pSTK-0C-csn:LEA / 0814 TBC

1. Pellet 1.8 mL bacterial overnight culture by centrifuge 6800 g for 3 min
2. Discard the supernatant and repeat step 1 until collect all of the liquid culture
3. Add 250 µL P1 buffer, and resuspend pelleted bacterial
4. Add 250 µL P2 buffer, and mix thoroughly by inverting the tube 4-5times until the solution become clear(Do not allow the reaction more than 5 min)
5. Add 350 µL N3 buffer and mix immediately and thoroughly by inverting 4-6 time
6. Centrifuge for 10 min at 17900 g
7. Transfer 800 µL supernatant into new column
8. Centrifuge 1 min for 17900 g
9. Add 500 µL PB buffer
10. Centrifuge 1 min for 17900 g , and discard the flow-through
11. Add 750 µL PE buffer
12. Centrifuge 1 min for 17900 g , and discard the flow-through
13. Centrifuge empty column for 1min at 17900 g , and discard the flow-through
14. Put into new column
15. Add 50 µL EB buffer
16. Stand at RT for 1 min
17. Centrifuge for 1min at 17900 g

Results:

	Concentration	A260/A280	A260/A230
L1	34	2.000	1.789
L2	225	1.907	1.433
#3	74	1.947	2.000
#1	105	1.810	0.991
#2	103	1.839	1.084
#S16	90	1.765	0.909

Done by Sarah, Ryan W.
Supervised by Sophie C.

Date: 20260821
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
Sample: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-2.2-B0015

1. Set the water bath to 42 °C and prepare ice
2. Thaw competent cells until they're 50 % thawed and add 10μL of plasmid
3. Incubate on ice for 20 min
4. Heat shock at 42 °C for 45 seconds
5. Incubate on ice for 3 min
6. Add 1 mL SOC medium
7. Incubate for 1 hr at 37°C
8. Centrifuge at 5000 g for 5 min
9. Remove 1 mL of supernatant then vertex
10. Spread on their respective plates
11. Incubate at 37 °C for 16 hr

Done by Sophie C

Date: 20260821
Title: PCR
Purpose: to amplify DNA parts
Procedure:
L2, L1: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21

1. Prepare the following reaction:

	1 rxn	3 rxn
Forward Primer Exp1	1 μL	3 μL
Reverse Primer Csn	1 μL	3 μL
Qamp master mix	10 μL	30 μL
ddH_2O	7 μL	21 μL
DNA	1 μL	1 μL

2. Put it in PCR machine for thermal cycle:

95 ℃	5 min	x1

95 ℃	20 sec	x35
65 ℃	20 sec	
72 ℃	1 min 20 sec	
25 ℃	5 min	x1

Done by Sarah, Ryan W.

Date: 20260821
Title: Prepare 1.0M KOH
Purpose: Prepare KOH for balancing pH levels
Procedure:

1. Prepare the following in a centrifuge tube:
    a. 2.24 g KOH
    b. 40 mL ddH_2O
2. Add the KOH first. Make sure that no KOH remains on the weighing paper/boat
    a. KOH is hygroscopic meaning that it absorbs some water in air so it sticks to the surface
3. Add the ddH_2O slowly in an ice bucket (or ice water)

Done by Sophia Y.
Supervised by Chars

Date: 20260821
Title: Gel electrophoresis
Purpose: To validate PCR products
Procedure:
L2, L1: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21 (PCR)
L20, L10: pSTK-EXP1-P3P-EEBD-RBS0-Csn:GanA-GalA-L3S2P21
C: Water (1 µL) + Qamp master mix (10 µL) + primer (2 µL) + ddH_2O (7 µL)

1. Load the samples in the following order
   M L1 L2 C M L10 L20
   5 5 5 5 5 5 5
2. Resolve the gel at 100V for 30min

Done by Ryan W.

Date: 20260821
Title: Golden Gate Assembly (P, Q, R, S)
Purpose: assemble level 0 parts to level 1
Procedure:
P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015
R: pSTK-EXP1-Pveg-EEBD-RBS0-Csn:LEA-B0015

S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ:LEA-2.1-B0015

1. Assemble the following

P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration: 162 ng/ µl • 6929 bp	0.2780154321	1.4 µl (dilute 1/5)
Insert promoter(Pveg) • concentration(ng/µL): 168 • bp: 2361	0.182 µL	1.82 µL (dilute 1/10)
Insert RBS(EEBD-RBS0) • concentration(ng/µL): 151 • bp: 2392	0.2059337748 µL	1.03 µL (Dilute 1/5)
Insert CDS (SamyQ-ACCD-2.1) - concentration: 103 - 3206 bp	0.4046407767	1.21 µl (1/3)
Insert Terminator (B0015) • concentration(ng/µL): 107 • bp: 2213	0.268869 µL	1.34 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL	2 µL
Sterile water		9.2 µL
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015

Vector (1D) • concentration(ng/µL): 132 • bp: 3120	0.15 µL	1.5 µL (dilute 1/10)
Insert promoter(PcpcG2-veg)	0.01818452381 µL	0.90922619 µl (dilute 1/50)

• concentration(ng/μL): 168 • bp: 235		
Insert RBS(EEBD-RBS0) • concentration(ng/μL): 151 • bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (SamyQ-ACCD-2.1) ng/μL - concentration: 103 - 3206 bp		1.21 μl (1/3)
Insert Terminator (B0015) • concentration(ng/μL): 107 • bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		10.01 μl
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

R: pSTK-EXP1-Pveg-EEBD-RBS0-Csn:LEA-B0015

• Vector (shuttle vector): pSTK-EXP-1-sfGFP • concentration: 162 ng/ μl • 6929 bp	0.2780154321	1.4 μl (dilute 1/5)
Insert promoter(Pveg) • concentration(ng/μL): 168 • bp: 2361	0.182 μL	1.82 μL (dilute 1/10)

Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 151 ● bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (Csn:LEA) - concentration: 105 ng/μl - 2660 bp	0.33 μl	0.99 μl (1/3)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		9.42 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ:LEA-2.1-B0015

● Vector (shuttle vector): pSTK-EXP-1-sfGFP ● concentration: 162 ng/ μl ● 6929 bp	0.2780154321	1.4 μL (dilute 1/5)
Insert promoter(Pveg) ● concentration(ng/μL): 168 ● bp: 2361	0.182 μL	1.82 μL (dilute 1/10)
Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 151 ● bp: 2392	0.2059337748 μL	1.03 μL (Dilute 1/5)
Insert CDS (SamyQ:LEA-2.1)	0.3820555556 μL	1.15 (1/3)

- concentration: 90 ng/μL - 2645 bp		
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		9.26 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Thermal Cycles:

Level 1	Level 2	Time	Cycles
37 ℃	42 ˚C	5 min	45 cycles
16 ℃	16 ˚C	5 min	
37 ℃	55 ˚C	10 min	1 cycle
80 ℃	80 ˚C	20 min	1 cycle
RT			

Done by Chars

Date: 20260821

Title: Prepare 0.1 MS medium

Purpose: prepare medium for hydroponics testing

Procedure:

1. Prepare the following

MS powder without sucrose	0.69 g
MES powder	1.02 g
ddH_2O	1500 mL

2. Place the materials in a 2000mL beaker
3. Use a stirring rod to stir until no sediments remain
4. Calibrate the pH probe
5. Record the initial pH
6. Add 1.0 M KOH in small increments until the pH reaches 5.70~5.80
    - Initial pH: 4.29, final pH: 5.70

Done by Sophia Y.
Supervised by Chars

Date: 20260821
Title: Hay Infusion
Purpose: Prepare the medium for *Bacillus subtilis* to grow
Procedure:

1. Cut the hay into smaller pieces
2. Place the chopped hay into a flask until it reaches the 150 mL mark.
3. Place a container on the scale, and set it to zero
4. Transfer the hay from the flask to measure its weight (6.69g).
5. Return the weighed hay to the flask.
6. Measure 250 mL of distilled water using a beaker and pour it into the flask.
7. Cover the flask with aluminum foil and autoclave for 20 minutes at 127 °C.
8. Blend grapes with water in a blender, then pour the mixture into a cup.
9. Filter the mixture through gauze into another flask to collect the extract.
10. Repeat step 8 to step 9 using blueberries to obtain blueberry extract.

Done by Sophia L

Date: 20260821
Title: Remove D3's protectant and add protectant in D4
Purpose: testing protectant
Procedure:

1. Prepare 2 new boxes and add 300 mL of 0.1 X MS medium into both boxes label them as D3, and D4
2. Add 600 µL CH protectant into box D4 (post-treatment)
3. Take another box and add 300 mL of water into it
4. Put a thermometer in the water
5. Put both boxes into heat box (setting 50 ℃ actual temp 35 ℃)
6. Wait until thermometer reaches 35 ℃
7. Take out new D3 box from heat box
8. Transfer the floating board from old D3 box to new D3 box
9. Put new D3 box back into heat box
10. Repeat steps 7~9 to change the medium of D4

Done by Alex

Date: 20260821

Title: Measure fresh weight of heat stress trial (agar)
Purpose: measuring heat stress pretest freshweight results
Procedure:

1. Use a tweezer, lift and pick up all arabidopsis plants to simultaneously form the agar plate in a single motion
2. Put plants on a tissue paper, lightly press to remove the agar sticking to the plants. (do not press the leaf area too hard)
3. Place all the arabidopsis in a weighting plate, weight the fresh weight (a plate at the time-control , pre-treat, co-treat, post-treat)
4. Pick all the plants up and put them inside a 15mL centrifuge tube, make sure all the plant and the roots goes inside the microtube
5. Label the fresh weight nad the group concentration on the microtube

*problem observed:
Ants climbing inside the incubator resulting in
-> ants climbing into the heat stress pretest agar plates
*possible solution:
Tape should be stuck onto the plate tightly till there's no more gap between the tape and flu plate

Done by Abigail
Recorded by Audrey H

Date: 20260821
Title: planting seeds on 0.6 % agar plates
Purpose: plant Arabidopsis on agar plates
Procedure:

1. Dip an autoclaved tooth pick in deionized water(autoclaved)
2. Reach into the eppendorf with seeds and pick up one at time
3. Plant 40 seeds on each plate according to the grid
4. Repeat with 13 plates
5. Wrap with aluminium foil and place in the refrigerator

Done by Abby K, Abigail, Sophie C

fDate: 20260822
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
N: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-2.1-B0015
X: ACDI

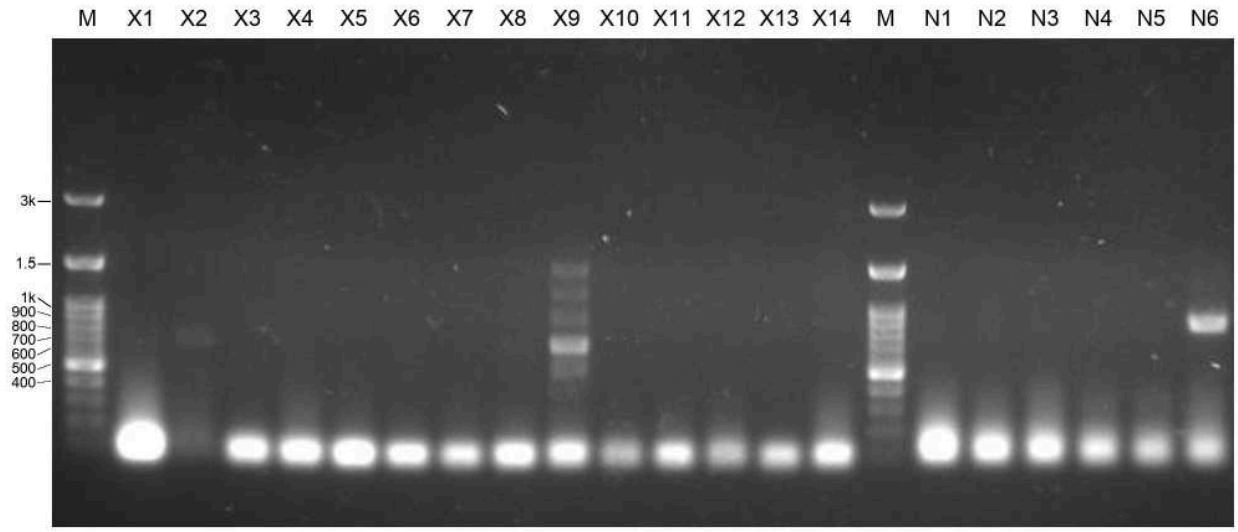
1. Prepare the following reactions
2. Distribute into PCR tubes
3. Dispense pipette into colonies, streak gild
4. Dispense in PCR tube
5. Close lid & spin down
6. Place in PCR machine for thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	1 min 30 sec	
25 ℃	5 min	x1

Done by Sophie C, Abby K

Date: 20260822
Title: Gel electrophoresis
Purpose: check PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X: ACDI	CcaR F	Csn R	1424
N: pSTK-EXP1-Pveg-EEBD-RBS0 -Bopep4-2.1-B0015	pSTK-EXP-seq fw	Bopep4 R	678

Done by Sophie C

Date: 20260822

Title: Measure chlorophyll of heat stress trial (agar)-chlorophyll extraction

Purpose: measure the chlorophyll for heat stress pretest

Procedure:

1. Take the microtubes from 0821 from measured freshweight at 0821
2. Add 4mL to each tube
3. Add 10 metal bits (403) into each microtubes
4. Wait for the chlorophyll to extract and use a vortex to crush the plants to spread up the process(vortex till the plants are no longer green)
5. Add the liquid to the 96 well, 200µL in each well. 3 well for each sample
6. Make sure to select the correct wavelength
    - 663
    - 645

	2	3	4	5	6

F	Control 1	pre–treatment	co-treat	after-treat	80 % acetone
G	Control 1	pre-treatment	co-treat	after-treat	80 % acetone
H	Control 1	pre-treatment	co-treat	after-treat	80 % acetone

Done by Audrey H
Supervised by Gabreil

Date:20260822
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
X1:ACDI(1:1) / X2:ACDI(1:2) / X3:ACDI(1:3)
V1: ABDI(1:1)/ V2: ABDI(1:2)/ V3: ABDI(1:3)
R:pSTK-EXP1-Pvef-EEBD-RBS0-csn:LEA-B0015
S:pSTK-EXP1-Pveg-EEBD-RBS0-Samy Q-LEA-2.1-B0015

1. Set up water bath at 42 ℃
2. Thaw competent cell fill 50 % thawed, add 5 µL plasmid to 100 µL competent cell
3. Incubate on ice for 20 min
4. Heat shack at 42 ℃ for 45 sec
5. Incubate on ice for 3min
6. Add 1mL SOC medium
7. Incubate at 37 ℃ for 1 hr
8. Centrifuge for 5 min at 5000g
9. Remove ~1 mL supernatant
10. Resuspend pellet
11. Spread on respective plates (LBC,LBA)
12. Incubate at 37 ℃ for 16 hr

Done by Sophie C, Abby K, Chloe

Date: 20260823
Title: Make 1 % MS medium without vitamin and 1 % agar
Purpose: prepare MS medium for experiment
Procedure:

1. Measure the following

1/2 MS medium without vitamin

MS powder without vitamin	3.525 g
MES buffer	0.75 g
Distilled water	1500 mL

1 % agar

1/2 MS powder	1.15 g
MES buffers	0.25 g
sucrose	5 g
agar	5 g
Distilled H_2O	500 mL

2. Add them into a flask and add a stirring bar
3. Stir on the hot plate until no sediment
4. Use a pH probe to measure the pH. If too low, add 1M KOH until it reactes 5.7~5.8
5. Wrap both bottle, in aluminium foil and stick a piece of indicator tap on top

Done by Sara, Alex

Date: 20260823
Title: Colony PCR
Purpose: Amplify DNA plasmid sequence
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-2.1-B0015
P:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-AccD-2.1-B0015
R:pSTK-EXP1-Pveg-EEBD-RBS0-can:LEA-B0015
S:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-2.1-B0015

1. Prepare the following reaction

Qx8

PcpcG2-veg fw	9 µL
AccD rv	9 µL

Q-amp master mix	90 μL
ddH_2O	72 μL

Px8 Rx8 Sx8

pSTK-EXP-seq fw	27 μL
pSTK-EXP-seq rv	27 μL
Q-amp master mix	270 μL
ddH_2O	216 μL

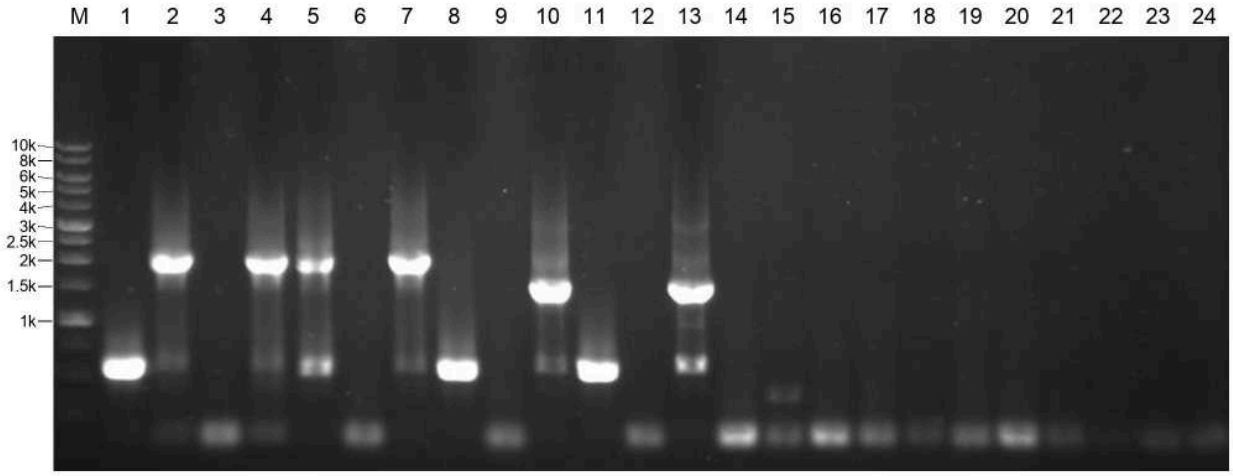
2. Distribute into PCR tube
3. Dip pipette into colonies, streak grid
4. Dispense into PCR tube & wait for 3min
5. Close the lid & spindown
6. Place in machine for thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	2 min	
72 ℃	5 min	
25 ℃	0	x1

Done by Chloe

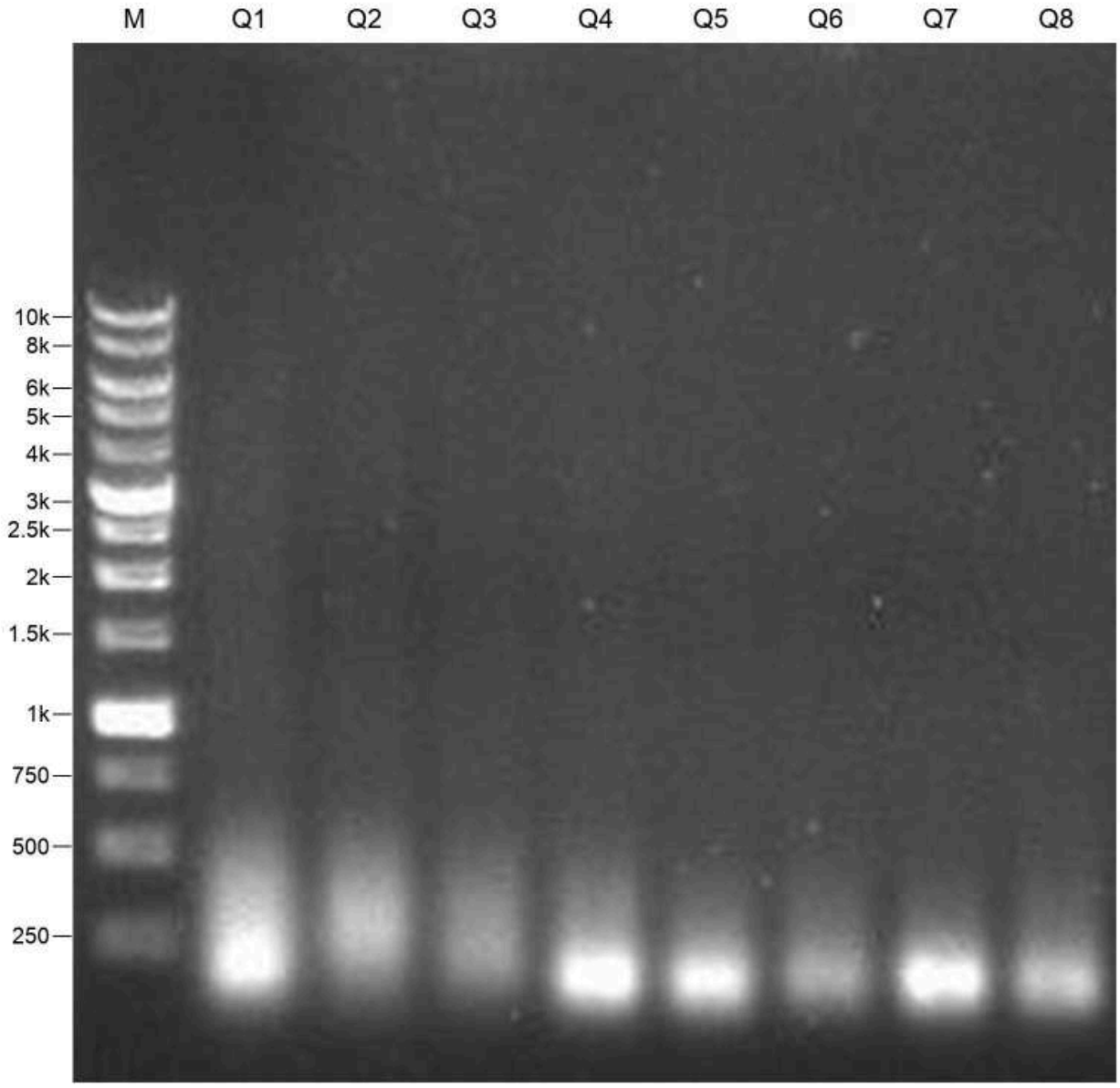
Date: 20260823
Title: gel electrophoresis(1 % gel)
Purpose: check PCR result
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1984
R: pSTK-EXP1-Pveg-EEBD-RBS0-csn:LEA-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1438
S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-2.1-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1423

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	PcpcG2-veg fw	ACCD R	1673

Done by Chloe, Abby K

Date: 20260824
Title: miniprep
Purpose: isolate plasmid of level 2 liquid culture
Procedure:

1. Pellet 1.8mL of liquid culture by centrifuging at 6800 g for 3 min
2. Repeat until collected all medium
3. Add 250 µL of P1 buffer and resuspend the pellet
4. Add 250 µL of P2 buffer and mix by inverting the tube 4-6 times until the solution becomes clear-> do not allow lysis to proceed for more than 5 min
5. Add 350 µL of N3 buffer and mix immediately by inverting the tube 4-6 times
6. Centrifuge for 10 min at 17900 g
7. Transfer ~800µL of supernatant to column and collection tube
8. Centrifuge for 1 min at 17900 g then discard the flow-through
9. Add 500µL PB buffer
10. Centrifuge for 1 min at 17900 g then discard the flow-through
11. Add 750PE buffer
12. Centrifuge for 1 min at 17900 g then discard flow-through
13. Centrifuge for 1 min at 17900 g then discard flow-through
14. Place column on new 1.5 mL microtube
15. Add 50µL EB buffer to the center of the column
16. Let stand at RT for 1 min
17. Centrifuge for 1 min at 17900 g
18. biodrop

	concentration(hg/µL)	A260/280	A260/230
ACDI from 0820	70.00	1.917	1.725
ACDI from 0822	63.00	2.032	1.800

Done by Abby K

Date: 20260825
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
X:ACDI

1. Prepare the following reaction

11 rxn

RV primer PcpcG2-veg	11 µL
Fw primer CcaR-fw	11 µL
Q-ap master mix	110 µL
ddH_2O	88 µL

2. Distribute the master mix to 10 PCR tubes
3. Pick up circled colonies and streak the grid
4. Dispense tips to their tubes for 5 minutes
5. After 5 minutes, remove the tips and spin down
6. Place the tubes in the PCR machine with the following thermal cycle setting

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	1 min 45 sec	
72 ℃	5 min	
25 ℃	0	x1

Done by Naomi, Sarah

Date: 20260825
Title: Colony PCR
Purpose: to screen for successful cloning after transformation
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-2.1-B0015
S:pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-LEA-2.1-B0015

1. Prepare the following reactions

Qx13rxn

RV primer AccD R	13 µL
Fw primer PcpcG2-veg F	13 µL
Q-ap master mix	130 µL
ddH_2O	104 µL

2. Distribute master mixes into respective tubes
3. Pick up colonies and streak the grid
4. Dispense the tips into the corresponding tubes
5. Wait for 5 minutes before removing tips and spinning down
6. Place tubes in the PCR machine with the following thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	1 min 45 sec	
72 ℃	5 min	
25 ℃	0	x1

Done by Sophie C, Abby K

Date: 20260825
Title: LBK, LBC+glycerol, TBC, LBA+glycerol, LB, LBE+glycerol
Purpose: prepare plate
Procedure:

1. Formula as below:

LB

LB broth powder	6.25 g
Agar powder	3.75 g
ddH_2O	250 mL

TB

TB broth powder	6.25 g

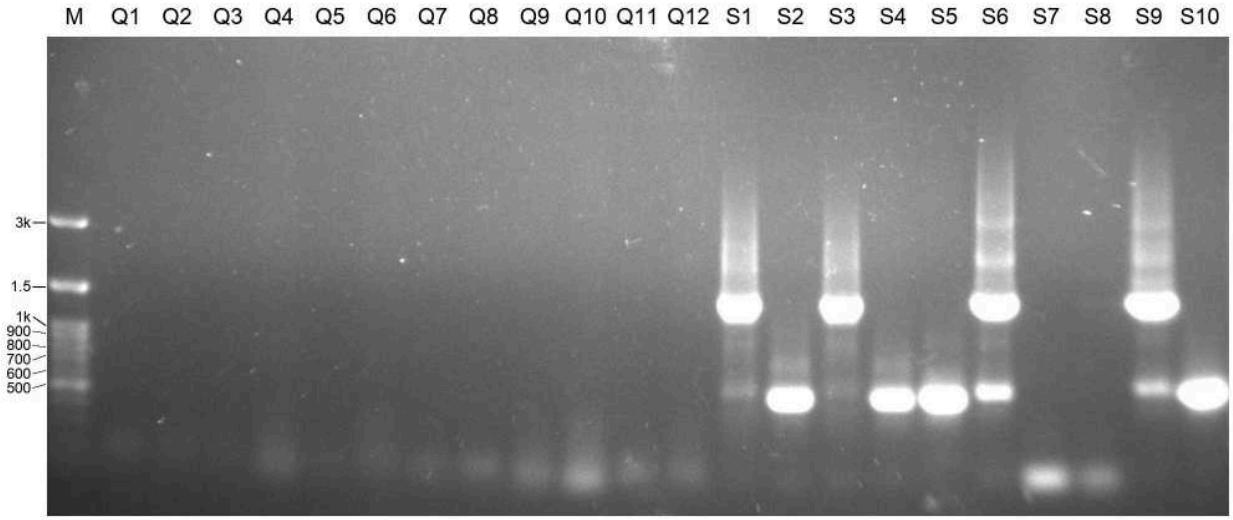
Agar powder	3.75 g
ddH_2O	250 mL

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate, and stir until no sediments
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply on indicator tape to ensure the mixture is successfully autoclave
6. Place the flask in autoclave
7. Place the new petri dishes in BSC and turn on the UVC
8. Set up water bath at 55 ℃
9. Add 500 µL ampicillin/ 25 µL kanamycin / 1250 µL erythromycin / 250 µL chloramphenicol
10. Use serological pipette to add 20 mL of melting agar into the petri dish
11. Let the agar plates dry for at least 1 hr in the BSC

Done by Naomi, Sarah

Date: 20260825
Title: gel electrophoresis
Purpose: checking PCR results
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	PcpcG2-veg fw	ACCD R	1673
S: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-LEA-2.1-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1423

Done by SophieC, Sarah

Date: 20260825
Title: make acrylamide gel
Purpose: Prepare gel
Procedure:

1. Assemble casting fromes with two 1.5mm glass plates
2. Prepare solutions for the stacking (top) and resolving (bottom) gels

component	stacker	resolver
Resolver A	x	4.5 mL
Resolver B	x	4.5 mLx
Stacker A	1.5 mL	x
Stacker B	1.5 ml	x
total	3 ml	9 mL
temed	3 µL	4.5 µL
10 % APS	15 µL	45 µL

3. Transfer resolving gel between glass plates
4. Add stacking gel to the top
5. Insert comb to form wells and allow it to set

Done by Sophie C

Date: 20260825
Title: golden gate assembly of N,O
Purpose:assemble part
Procedure:
N:Pveg-EEBD-RBS0-SamyQ:Bopep4(2.1)-B0015

1. Prepare the following reactions

pSTK-Exp-1-sfGFP	1.4 µL (dill 1/5)
Pveg (168 ng/µL , 2361 bp)	1.82 µL (dill 1/10)
EEBD-RBS0 (151 ng/µL, 2992 bp)	1.03 µL (dill1/5)
B0015 (107 ng/µL, 2213 bp)	1.34 µL (dill(1/5)
SamyQ-Bopep4 (2.1) (137 ng/µL, 2263 bp)	1.07 µL (dill1/5)
T4 DNA liquid buffer	2 µL
Bsa-HFv2	1 µL
T4 DNA lig	1 µL
ddH_2O	9.33 µL

O:Pveg-EEBD-RBS0-SamyQ-Bopep4(2.2)-B0015

pSTK-Exp-1-sfGFP	1.4 µL (dill 1/5)
Pveg	1.82 µL (dill 1/10)
EEBD-RBS0	1.03 µL (dill1/5)
B0015	1.34 µL (dill(1/5)
SamyQ-Bopep4 (2.1) (101 ng/µL, 2263 bp)	1.17 µL (dill1/5)
T4 DNA liquid buffer	2 µL
Bsa-HFv2	1 µL
T4 DNA lig	1 µL
ddH_2O	9.24 µL

2. Put in PCR machine & apply the thermal cycle

37.0 ℃	5 min	x45
16.0 ℃	5 min	

37.0 ℃	10 min	x1
80.0 ℃	20 min	

Done by Chloe, Sophie C

Date: 20260825
Title: Transformation
Purpose:Spread bacterial liquid on agar plate
Procedure:
B:Pliag-RBS0-CcaSm3-B0015

1. Set up water bath to 42 ℃
2. Thaw competent cell until 50 % thawed & add 2 μL plasmid
3. Incubate on ice for 20min
4. Heat shock at 42 ℃ for 45 sec
5. Recover on ice for 3 min
6. Spread on LBA plate in BSC
7. Incubate at 37 ℃ for 16 h (overnight)

Done by Chloe

Date: 20260825
Title: liquid culture
Purpose: Transfer colonies on agar to medium
Procedure:
A:pSTK-1A-PtagC+RBS-H01:pcyA-L3S2P21(#A1)
C:PcepA-RBS0-CcaSm3-B0015(#C6)
D:pSTK-1C-Pveg-RBS0-CcaR-B0015(#B2)
I:pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:AccD-B0015(#I3)
S:pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-LEA-2.1-B0015(#S5, #S11)
P:pSTK-EXp1-Pveg-EEBD-RBS0-SamyQ-AccD-2.1-B0015(#P2,#P7)
R: pSTK-EXp1-Pveg-EEBD-RBS0-csn:LEA-B0015 (#R2,#R5)

1. Add 3mL TB + 3μL ampicillin per tube
2. Pick up one colony in each plate and insert the tip into the medium.

Done by Chloe

Date: 20260826
Title: Miniprep (A, C, D, I, P, R)
Purpose: Collect concentrated bacterial liquid
Procedure:
A1: PtagC+RBS-Ho1:PcyA-L3S2P21
C6: PlepA-RBS0-CcaR-B0015
B2: Pveg-RBS0-CcaR-B0015
R2, R5: Pveg-EEBD-RBS0-Csn::EA:B0015
P2, P7: Pveg-EEBD-RBS0-SamyQ-ACCD
I3: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Pellet 1.8mL of liquid culture by centrifuging at 6800 g for 1 min
2. Repeat until all the medium were collected
3. Add 250 μL of P1 buffer, resuspend the cell pellet by vortexting or pipetting
4. Add 350 μL of P2 buffer, and mix thoroughly by inverting the tube 4~6 times
5. Centrifuge for 10 min at 17900 g
6. Transfer the supernatant to the spin column, centrifuge it for 60sec at 17,900 g, then discard the flow through
7. Add 500 μL PB buffer, and centrifuge for 1min at 17,900 g
8. Discard flow through
9. Add 750 μL PE buffer, and centrifuge for 1min at 17,900 g
10. Discard flow through
11. Centrifuge empty column for 1min at 17,900 g to remove residual wash buffer
12. Place the column to 1.5 mL microcentrifuge tube
13. To elute DNA, add 50μL EB buffer to the center of the membrane
14. Let the tube stand for 1 min, then centrifuge it for 1min at 17,900 g

Biodrop Results:

Concentration	A260/230	A260/280
94ng/μL	1.918	1.918
86ng/μL	1.87	1.911
85ng/μL	1.889	1.932
253ng/μL	2.018	1.888
243ng/μL	2.229	1.898
220ng/μL	2.178	1.897
248ng/μL	2.234	1.893
32ng/μL	1.524	1.882

Done by Abby K, Abby T

Date: 20260726
Title: Colony PCR of B, Q
Purpose:Amplify the number of engineered circuit design
Procedure:
B: PliaG-RBS0-Ccasm3-B0015
Q: PcpcG2-veg-EEBD-RBS0-SamyQ:ACCD-2.1-B0015 (bp: 2041)

1. Prepare the following reactions:

Q (18rxn)

	1 rxn	18 rxn
pSTK-Seq fw	1 μL	18 μL
pSTK-Seq rv	1 μL	18 μL
Qamp master mix	10 μL	180 μL
ddH_2O	8 μL	144 μL

B (5rxn)

	1 rxn	5 rxn
pSTK-Seq fw	1 μL	5 μL
pSTK-Seq rv	1 μL	5 μL
Qamp master mix	10 μL	50 μL
ddH_2O	8 μL	40 μL

2. Distribute the master mix into labeled PCR tubes (20μL each well)
3. Pick up colonies with a tip and streak on the grid
4. Dip the pipette tip into the respective master mix
5. Incubate at Room Temperature for 5min, then discard the pipette tips
6. Put the PCR tubes into PCR machine and run the thermal cycle as below:

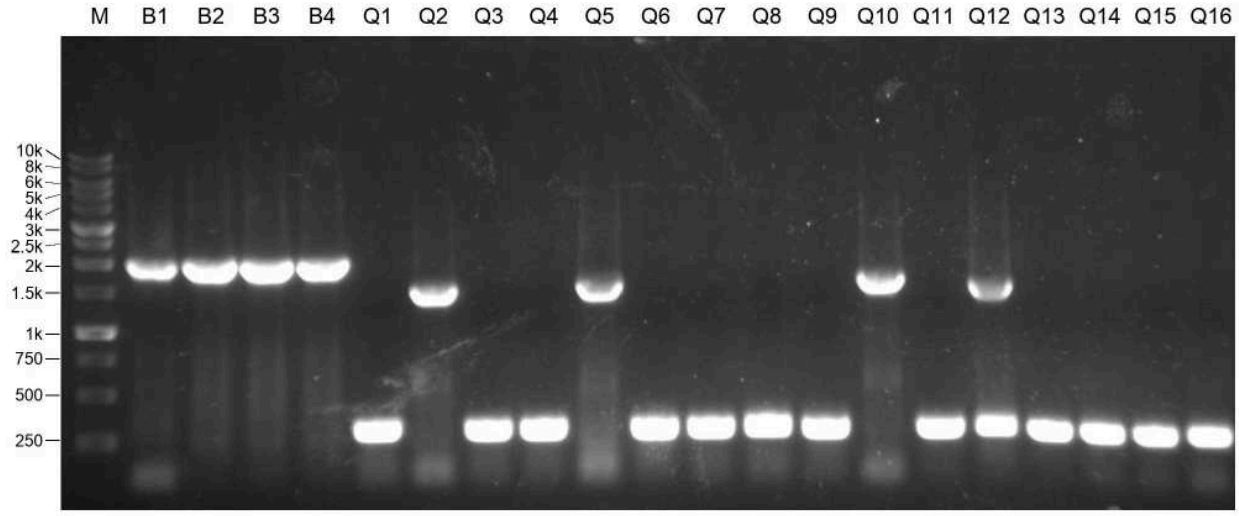
95.0 ℃	5 min	x1
95.0 ℃	20 sec	
60.0 ℃	20 sec	x35
72.0 ℃	5 min	

72 ℃	5 min	
25 ℃	0 min	x1

Done by Abby T

Date:20260826
Title: Gel electrophoresis of Colony PCR (1 % gel)
Purpose: Check the approximate length of PCR product
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
B: pSTK-1A-PliaG-RBS0-Ccasm3-B0015	pSTK_Seq fw	pSTK_Seq rv	1991
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	pSTK_Seq fw	pSTK_Seq rv	2041
Note: → liquid culture B, 8/27 Miniprep			

Date:20260826
Title: Cotreatment + NaCl on soil
Purpose: Add salinity and protectant (co-treatment) on agar grown Arabidopsis
Procedure:

1. Prepare 75mM NaCl and 100mM NaCl
2. Prepare diluted 正翰 protectant by the ratio of 1:500 (1μL of protectant and 499μL of autoclaved ddH_2O)
3. Add 2mL of respective NaCl solution and protectant in the ratio of 1:1 (1mL NacL solution and 1mL diluted protectant) into each respective agar plates with Arabidopsis

Done by Abigail

Date: 20260826
Title: Transformation of N, O
Purpose: Spread bacterial liquid on agar plate
Procedure:
N: Pveg-EEBD-RBS0-SamyQ-Bopep4(2.1)-B0015
O: Pveg-EEBD-RBS0-SamyQ-Bopep4(2.2)-B0015

1. Set up water bath at 42 ℃ and ice tub
2. Wait until the competent cell is 50 % thawed in the ice tub
3. Add 10 μL plasmid
4. Incubate on ice for 20 min
5. Heat shock for 45 sec at 42 ℃ water bath
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium in the microtube in BSC
8. Incubate at 37 ℃ for 1 hr
9. Centrifuge for 5 min at 5000 g
10. Take out about 1 mL of supernatant
11. Resuspend the pellet with pipette
12. Transfer all of the pellet on labeled LBA plate and spread it evenly
13. Place the spreaded agar plate upside down in incubator at 37 ℃ for 16 hr

Done by Sophie C.

Date: 20260826
Title: Golden Gate Assembly of X *3
Purpose: Assemble parts
Procedure:
X: ACDI

part	1:3	1:5	1:7
pSTK-EXP-1-sfGFP Conc: 163 ng/μL 5164 bp	1 μL (dilute 1/5)	1 μL(dilute 1/5)	1 μL(dilute 1/5)

A 107 ng/µL, 3754 bp	0.68 µL	1.14 µL	1.6 µL
C 134 ng/µL, 3828 bp	0.56 µL	0.93 µL	1.3 µL
D 86 ng/µL, 3243 bp	0.74 µL	1.23 µL	1.7 µL
I 108 ng/µL 3757 bp	0.68 µL	1.13 µL	1.58 µL
T4 DNA ligase	0.5 µL	0.5 µL	0.5 µL
T4 DNA ligase buffer	1 µL	1 µL	1 µL
BsmbI	0.5 µL	0.5 µL	0.5 µL
ddH_2O	4.34 µL	2.57 µL	0.82 µL

Done by Sophie C
Supervised by Chloe

Date: 20260827
Title: Miniprep
Purpose: isolate plasmid form liquid culture
Procedure:
#2 PliaG-CcaSm3
#3 PliaG-CcaSm3

1. Pellet 1.8mL of liquid culture by centrifuging at 6800 g for 1 min
2. Repeat until collected all medium
3. Add 250 µL of P1 buffer and resuspend the pellet
4. Add 250 µL of P2 buffer and mix by inverting the tube 4-6 times until the solution become clear
   ->do not allow lysis to proceed for more than 5min
5. Add 350 µL of N3 buffer and mix immediately by inverting the tube 4-6 time
6. Centrifuge for 10min at 17900 g
7. Transfer ~800 µL of supernatant to column and collection tube
8. Centrifuge for 1 min at 17900 g, discard the flow-through
9. Add 500 µL PB buffer
10. Centrifuge for 1 min at 17900 g, then discard the flow-through
11. Add 750 µL PE buffer
12. Centrifuge for 1 min at 17900 g, then discard the flow-through
13. Centrifuge for 1 min at 17900 g, then discard the flow-through
14. Place column on new 1.5mL microtube
15. Add 50 µL EB buffer to the center of the column
16. Let stand at RT for 1min
17. Centrifuge for 1 min at 17900g
18. biodrop

	Concentration(ng/µL)	A260/230	A260/280
#2	93	1.069	1.824
#3	88	1.692	1.913

Done by Sarah, Audrey H

Date: 20260827
Title: Electroporation of P, R
Purpose: Check the approximate length of colony PCR product
Procedure:

1. Prewarm the recovery medium at 37 ℃ and pre-cool a 0.2 cm electroporation cube
2. Hix 403 µL of plasmid P in 100 µL cell in a 1.5 mL microtube(P7:248 ng/µL)
3. Mix 3.95 µL of plasmid R in 100µL cell in a 1.5 mL microtube(R2:253 ng/µL)
4. Set up electroporator and voltage resistance

5. Transfer mixture of cells & DNA to electroporation cuvette & tap suspension to bottom
6. Place in shake pad, close lid&wait until beep (1~2 s)
7. Immediately remove from shock pad and add 1 mL recovery medium
8. Transfer to liquid culture tube & incubate at 37 ℃ for 3hrs, shaking at 200rpm
9. Check&record parameters
10. Centrifuge at 5000 g for 5 min
11. Spread LB plate with erithmycin

Done by Abby T/K
Supervised by Gabriel

Date: 20260827
Title: Transformation
Purpose: Transform bacterial liquid on agar plate
Procedure:
X:ACDI(1:3)
X:ACDI(1:5)
X:ACDI(1:7)

1. Set up water bath at 42 ℃
2. Thaw competent cell till 50 % thawed, add 5 µL plasmid
3. Incubate on ice for 20 min
4. Heat shock for 45 sec
5. Incubate on ice for 3min
6. Add 1mL SOC medium
7. Incubate at 37 ℃ for 1 hr
8. Centrifuge for 5 min at 5000 g
9. Remove ~1 mL supernatant
10. Resuspend the pellet
11. Spread on LBC
12. Incubate 37 ℃ overnight

Done by Sophie C

Date: 20260827
Title: Golden Gate Assembly
Purpose: assemble parts
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-AccD-2.1-B0015

1. Prepare the following

Vector 1D 132 ng/µL,3120 bp	1.5 µL (1/10)
PcpcG2-veg 168 ng/µL,235 bp	0.91 µL (1/50)

EEBD-RBS0 151 ng/μL,2392 bp	1.03 μL (1/5)
pSTK-OC-SamyQ-AccD-2.1 103 ng/μL,2213 bp	1.21 μL (1/3)
B0015 107 ng/μL,2213 bp	1.34 μL (1/5)
T4 DNA ligase buffer	2 μL
Sterile water	10.01 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL

2. Thermal cycle

37 ℃	5 min	x45
16 ℃	5 min	
37 ℃	10 min	x1
80 ℃	20 min	x1

Done by Sophie C

Date: 20260827
Title: protein extraction from cell and medium
Purpose:
Procedure:

1. Centrifuge 250 mL *B. subtilis* culture at 8,000 g for 10 minutes.
2. Carefully collect the supernatant and set aside the cell pellet for lysis (Part B).
3. Immediately add EDTA-free protease inhibitor to the collected supernatant according to manufacturer instructions.
4. Clarify the supernatant by centrifuging a second time at 15,000 × g for 15–20 minutes at 4 °C, then store at 4 °C.
5. Prepare CelLytic B working solution by making a 10-fold dilution of 10X concentrate in the chosen buffer.
6. Prepare:

   -20 mL CelLytic B (1X working solution)

   -400 μL Lysozyme stock (20 μL per 1 mL lysis buffer)

-200 µL Protease Inhibitor stock (10 µL per 1 mL lysis buffer)

7. Add the diluted CelLytic B (pre-mixed with lysozyme and protease inhibitor) directly to the cell pellet.
8. Vortex and mix thoroughly to completely resuspend the pellet.
9. Incubate with shaking at room temperature for 10–15 minutes.
10. Add 250–500 µL DNase I working stock (25 µL per 1 mL lysis buffer) and mix briefly to reduce viscosity.
11. Centrifuge at 16,000 g for 10 minutes to pellet insoluble material.
12. Transfer and collect the supernatant into a fresh tube and retain the pellet for analysis.

Done by Sophie C.

Date: 20260828
Title: Colony PCR
Purpose: amplify engineered DNA plasmid
Procedure:
X:ACD1 0826's ACDE
XI:ACD1(1:7)
XII:ACDT(1:5)

1. Prepare the following reactions

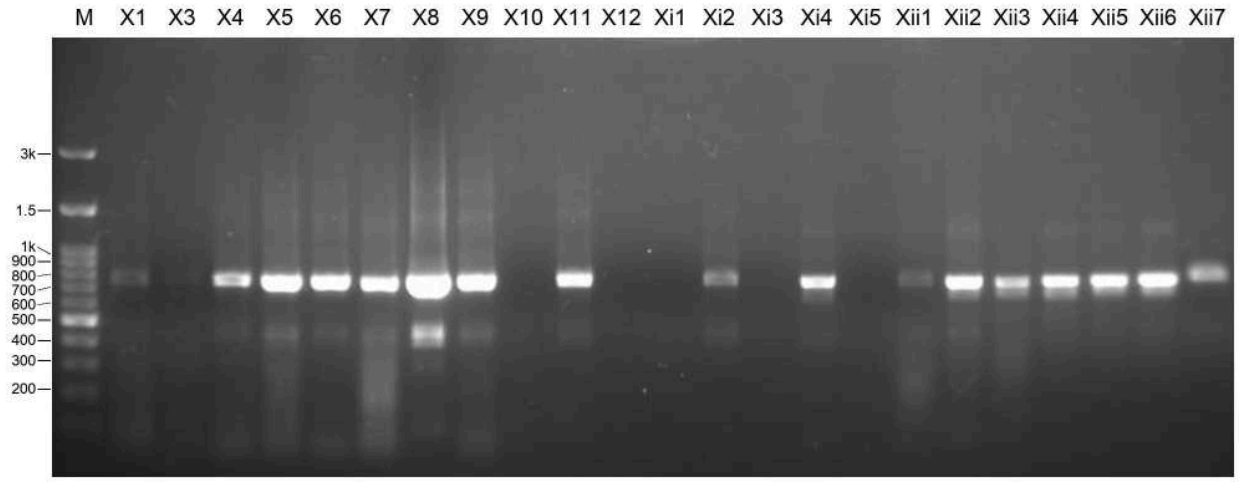
X	1 reaction	25 reaction
CcaR	1 µL	25 µL
Csn	1 µL	25 µL
Q-amp mastermix	10 µL	250 µL
ddH_2O	8 µL	200 µL

2. Distribute 10µL of each tube (24)
3. Pick colonies with a tip and streak on the grid
4. Dip the tip into the mix
5. Discard the tips at the 5mix
6. Run thermal cycle

95 ℃	5 min	1 cycle
95 ℃	20 sec	35 cycle
60 ℃	20 sec	
72 ℃	1 min	
72 ℃	5 min	1 cycle

Done by Olivia

Date: 20260828
Title: Gel electrophoresis
Purpose: checking colony PCR result
Procedure:
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
X: ACDI from 8/28	CCaR fw	csn rv-seq	760
X_I: ACDI (1:7)	CCaR fw	csn rv-seq	760
X_II: ACDI (1:5)	CCaR fw	csn rv-seq	760
Note: X1, X4-9, X11, Xi2, Xi4, Xii1-7 → liquid culture			

Done by Olivia

Date: 20260828
Title: SDS-PAGE
Purpose: separate proteins based strictly on their molecular weight (size) rather than their intrinsic charge or shape
Procedure:

1. Prepare 1.0 % polyacrylamide gel
2. Loading the sample prepare form 260827

M	cell lysort	melium					medium	cell lysort	M
5	50	50	0	0	0	0	50	50	5

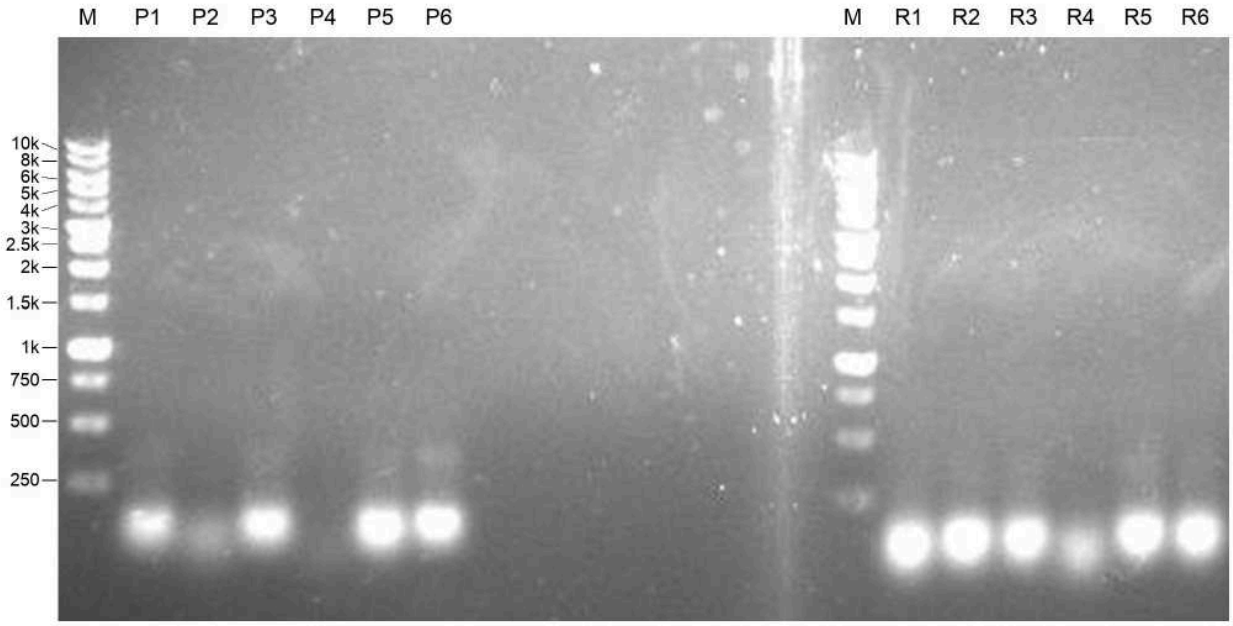
3. Resolve at 200V for 10min and 180V for 40min
4. Cut the gel from the middle. Let part for Western Blot and Right part for Coomassie Blue
5. Transfer the left gel to PVDF for 90min
6. Blocking the PVDP in 20 mL 5 % BSA for 1hr
7. Incubate the PVDP in 1:1000 anti-His-tag mAb (CST) in 2 % BSA overnight
8. Place the right gel in coomassie blue for 45 min
9. Destain the gel increased destain buffer for 30 min

10. Destain the gel in fresh destain buffer overnight

Done by chars

Date: 20260829
Title: Gel electrophoresis
Purpose:checking colony PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
P: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1984
R: pSTK-EXP1-Pveg-EEBD-RBS0-csn:LEA-B0015	pSTK-EXP-seq fw	pSTK-EXP-seq	1438

Done by Abby K

Date:20260829
Title: Colony PCR
Purpose:amplify engineered DNA plasmid
Procedure:
N:pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-Bopep4-2.1-B0015

1. Prepare the following reaction x3 in PCR tubes

Fw primer pSTK-EXP-seq fw	1 µL
Rv primer Bopep 4R	1 µL
Q-amp master mix	10 µL
ddH_2O	8 µL

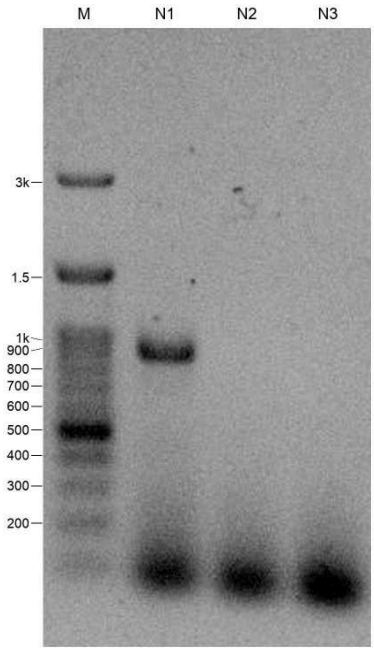
2. Pick up colonies with tip & streak grid
3. Dispense tip in mix
4. Take out tip and close lid
5. PCR machine for thermal cycle

95 ℃	5 min	x1
95 ℃	20 sec	x35
60 ℃	20 sec	
72 ℃	45 sec	
25 ℃	5 min	x1

Done by Sophie C

Date:20260829
Title: Gel electrophoresis
Purpose:checking colony PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 3kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
N: pSTK-EXP1-Pveg-EEBD-RBS0 -SamyQ-Bopep4-2.2-B0015	pSTK-EXP-seq fw	Bopep4 R	678

Done by SophieC

Date:20260829
Title: Miniprep
Purpose:
Procedure:
X6 ACDI
XII4 ACDI

1. Pellet 1-5 mL bacterial overnight culture by centrifuge at 6800 g for 3min
2. Resuspend the pelleted bacterial cells in 250 µL buffer P1, transfer to a micro-centrifuge tube
3. Add 250 µL buffer P2 and mix thoroughly by inverting the tube 4-6 times until the solution becomes clear
4. Add 350 µL of buffer N3, mix immediately and thoroughly by inverting the tube 4-6 times. The solution will turn color less
5. Centrifuge for 10min at 17900 g
6. Apply ~800 µL supernatant to QIA 2.0 spin column by pipetting

7. Centrifuge for 30 sec at 17900 g & discard flow through
8. Wash the column by adding 0.5 mL buffer PB
9. Centrifuge for 30 sec at 17900 g & discard flow through
10. Wash the column by adding 0.75 mL buffer PE
11. Centrifuge for 30 sec at 17900 g & discard flow through
12. Centrifuge for 1 min at 17900 g to remove residual wash buffer
13. Place column in clean 1.5 mL microcentrifuge tube
14. Add 50 µL buffer EB to the center of the column to elute DNA
15. Let stand for 1min
16. Centrifuge for 1min at 17900 g
17. biodrop

	concentration(ng/µL)	A230/260	A260/280
#X6	102.0	2.488	1.902
#XII4	63	2.250	1,969

Done by Sophie C, Abby K

Date: 20260830
Title: Gel electrophoresis of 8/29 PCR product
Purpose: Checking colony PCR result
Procedure:

1. Load the samples in the following format
   M:1kb DNA ladder
   X41, X42, X43, X61, X62, X63: ACDI

Gel1	X41	X42	X43	X61	X62	X63
5	5	5	5	5	5	5 (μL)

2. Resolve the DNA at 100V for 30min

Done by Abby K

Date: 20260830
Title: Colony PCR of Q
Purpose: Amplify engineered DNA plasmid
Procedure:
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ:ACCD 2.1-B0015

1. Prepare the master mix as below:

	1 rxn	20 rxn
Fw primer (SamyQ-fw)	1 μL	20 μL
Rv primer (pSTK-sq-rv)	1 μL	20 μL
dd H_2O	8 μL	160 μL
Qamp master mix	10 μL	200 μL

2. Distribute 20μL master mix into each labeled PCR tube.
3. Pick up a single colony and streak grid.
4. Incubate in RT for 5 min
5. Put in thermal machine and run thermal cycle as below:

90 ℃	5 min	x1
90 ℃	20 sec	x35
65 ℃	20 sec	
72 ℃	1 min 30 sec	

72 ℃	5 min	x1

Done by Sophie H.

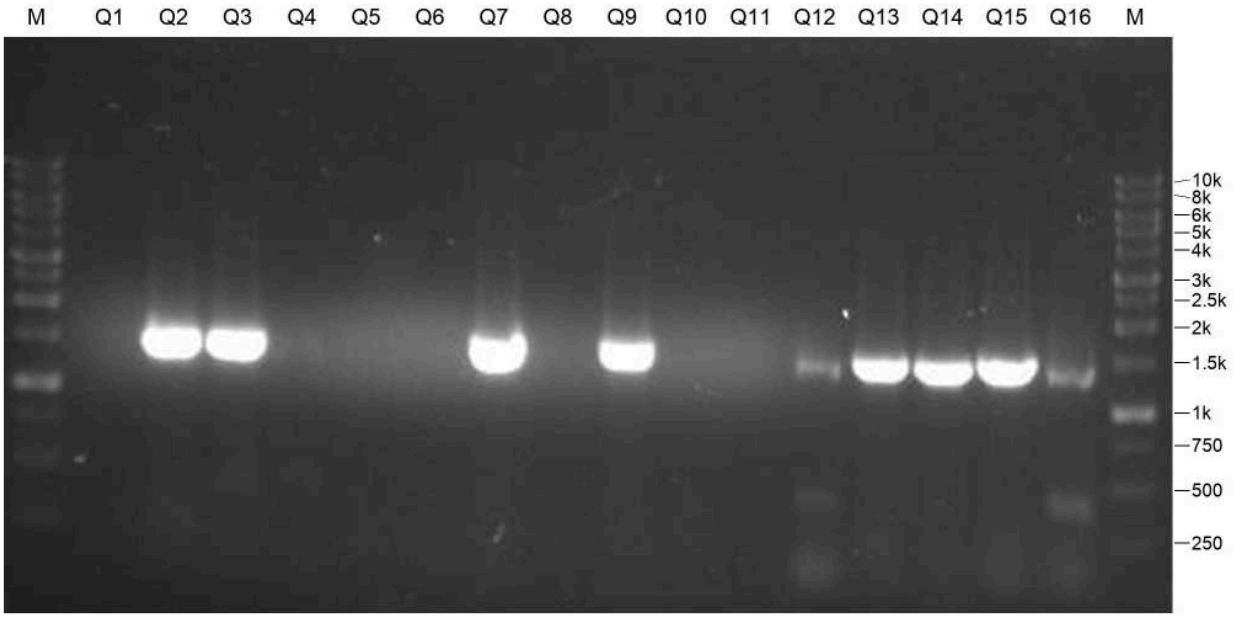
Date: 20260830
Title: Prepare 1 % agarose gel (25 well)
Purpose: Prepare agarose gel for electrophoresis on colony PCR product
Procedure:

1. Add the ingredient below into a 250 mL flask:
   -1X TAe buffer 40 mL
   -Agarose powder 0.4 g
2. Cover the flask with double layer plastic wrap and puncture it with pipette tip
3. Microwave it until all the powder were fully resolved
4. Wait for it to solidify, then add 2 μL of safety dye and agitate it
5. Pour it into assembled tray and comb, use pipette tip to squeeze the bubble out
6. Wait for it to solidify

Done by Sophie H.

Date: 20260830
Title: Electrophoresis of Q
Purpose: Checking colony PCR result
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
Q: pSTK-1D-PcpcG2-veg-EEBD-RBS0-SamyQ-ACCD-2.1-B0015	SamyQ-seq-fw	pSTK-seq-rv	1,420

Done by Sophie H.

Date: 20260830
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
L: pSTK-Exp1-3P-EEBD-RBS0-GanA-Gala-L3S2P21
O: pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-Bopep4-2.1
M: pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn: GanA-Gala-L3S2P21
V(#5, #7): ABDI

1. Set water bath to 42 °C
2. Thaw competent cells until 50 % thawed and add 10μL plasmid
3. Incubate on ice for 20 min
4. Heat shock at 42 °C for 45 sec
5. Incubate on ice for 3 min
6. Add 1 mL SOC medium
7. Incubate at 37 °C for 1 hr
8. Centrifuge at 5000 g for 5 min
9. Take out 1 mL supernatant
10. Resuspend the pellet
11. Spread on LBA and LBC plates
12. Incubate at 37°C for 16 hr

Done by Chloe

Date: 20260830
Title: Miniprep
Purpose: To extract plasmid from liquid culture
Procedure:
X3 XII 1 0829 culture ACDI
X5 XII 2 0829 culture ACDI
X7 XII 3 0829 culture ACDI

1. Pellet 1-5 mL bacterial overnight culture by centrifuging at 6800 g for 3 min at Room Temperature

2. Resuspend the pelleted bacteria cells in 250µL Buffer P1, transfer to a microcentrifuge tube.
3. Add 250 µL Buffer P2 and mix thoroughly by inverting the tube 4-6 times until the solution becomes clear
4. Add 350 µL of buffer N3, mix immediately and thoroughly by inverting the tube 4-6times. The solution will turn color less
5. Centrifuge for 10 min at 17900 g
6. Apply ~800 µL supernatant to QIA 2.0 spin column by pipetting
7. Centrifuge for 30 sec at 17900 g & discard flow through
8. Wash the column by adding 0.5mL buffer PB
9. Centrifuge for 30 sec at 17900 g & discard flow through
10. Wash the column by adding 0.75mL buffer PE
11. Centrifuge for 30 sec at 17900 g & discard flow through
12. Centrifuge for 1 min at 17900 g to remove residual wash buffer
13. Place column in clean 1.5mL microcentrifuge tube
14. Add 50 µL buffer EB to the center of the column to elute DNA
15. Let stand for 1min
16. Centrifuge for 1min at 17900 g
17. Biodrop

	concentration(ng/µL)	A230/260	A260/280
#X8	430	2.287	1.903
#X3	30	1.765	1.875
#X5	53	1.559	1.893
#X7	58	1.349	1.813
#XII 1	81	1.841	1.884
#XII 2	63	1.312	1.853
#XII 3	77	1.638	1.878

Done by Sophie C

Date: 20260831
Title: PCR
Purpose: To check the insert by using PCR
Procedure:
L2 Sample: ACDI
X3, X5, X7, X8, XII 1, XII 2, XII 3 from 8/30 plasmid
Premier: ccaR Forward/ Csn Reverse
ccaSm3 Forward/ ccaR Reverse
pcyA Forward/ ccaSm3 Reverse
1 % Agarose gel
Done by Gabriel

Date:20260831
Title: Electroporation
Purpose:Aim to transfer ACCD and LEA plasmid into B.sub WB700
Procedure:

1. Centrifuge the WB800 in GM medium to O.D 0.91
2. Collect the cell with centrifuge 5000 g for 5min
3. Wash 2 times using EP buffer (100μL per 5000μL origin volume of GM medium)
4. Electroporation the plasmid (Pveg-csn:ACCD, Pveg-SamyQ::ACCD2.2 and Pveg-csb::LEA)
   -Pveg-csn::ACCD: 25 kv, 25 μL, 300 Ω
   -Pveg SamyQ::ACCD 2.2: 2.3 kv, 25 μL, 300 Ω
   -Pveg-csn::LEA: 2.3 kv, 25 μL, 300 Ω
5. Place in recovery medium for 3 hr
6. Spread the medium on agar plate

Done by Gabriel

Date:20260831
Title: Colony PCR
Purpose: To amplify engineered DNA plasmid
Procedure:
L2 V3/ v7 1-4(total 8): ABDI
-Primier: CcaSm3 F/ CcaR R
Pveg-SamyQ: Bopep4 2.2 1-4
-Pveg-GanA 1-4
Premier pExp-Seq Fw/Rv
PcpcG2-GanA 1-3
Primier:
pSTK-seq F/R
Done By Gabriel